Robust Federated Causal Estimation (RoCE) through Calibrated High-Dimensional Models

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Abstract

Integrating evidence across healthcare institutions can improve treatment-effect estimation for patient populations with limited local data. Such integration must address confounding, differences between populations, and restrictions on sharing individual records. Relevant adjustment variables may be hidden among many candidate measurements, and regularized model fitting can introduce bias into subsequent causal inference. We propose robust federated causal estimation (RoCE) for a prespecified target population. RoCE uses outcomes from each source treatment group to augment outcome predictions in the target population. A single joint-weight model adjusts treatment assignment and population shift, allowing paired regularized calibration of the outcome and weighting models. Pivotal selection then penalizes incompatible source treatment groups while jointly optimizing borrowing for the target treatment contrast. With target-based initialization, the procedure uses one batched round of model and summary exchange. Under suitable identification, sparsity, and source-separation conditions, RoCE provides model-doubly-robust inference and has asymptotic variance no greater than calibrated target-only estimation. Simulations demonstrate precision gains with compatible sources and protection against substantial source bias, while revealing undercoverage for small departures from compatibility. An All of Us observational drug comparison with 181 clinical features and a supplementary right heart catheterization analysis illustrate applications to prespecified target populations.

Keywords: causal inference, calibration, federated learning, high-dimensional data, transportability

1 Introduction

Electronic health records (EHRs) collected during routine clinical care provide real-world data for studying treatment effectiveness and safety in the populations receiving those treatments (Sherman et al., 2016; Hernán and Robins, 2016). Longitudinal records of diagnoses, prescriptions, and clinical encounters make it possible to examine treatment use and outcomes beyond the settings of individual clinical trials. Research networks extend this opportunity by connecting institutions whose local samples may be insufficient for a treatment comparison, particularly when eligible patients or outcome events are uncommon. The scientific question, however, often concerns a particular patient population, such as the patients served by one healthcare institution. Additional institutions then provide potential sources of information for that target. When individual records must remain local, this collaboration requires methods that improve target-specific estimation through the exchange of model parameters and summary statistics.

Confounding adjustment is central to such observational comparisons because treatment choices depend on patients' clinical histories. EHRs offer a broad set of candidate adjustment variables, but the measurements that capture these histories may be dispersed among many diagnosis codes, medication records, and utilization features. A small set of prespecified variables may therefore omit information relevant to treatment selection or prognosis. For example, Hou et al. (2021) combined a multiple sclerosis registry with linked EHRs and used data-driven feature selection to identify additional adjustment variables absent from the registry. Regularization makes analysis of these large feature sets feasible, yet the resulting selection and shrinkage errors can enter a treatment-effect estimate at first order (Belloni et al., 2014; Chernozhukov et al., 2018). Valid inference must account for these errors, particularly when the local samples supporting nuisance estimation are much smaller than the full network sample.

A natural starting point for multisite analysis is meta-analysis of local treatment-effect estimates, using sample-size or inverse-variance weights. Each site can use augmented inverse probability weighting (AIPW), which combines an outcome regression with a treatment propensity model and is consistent under the identification conditions if either model is correct (Robins et al., 1994; Bang and Robins, 2005). Federated causal methods extend information sharing through Bayesian synthesis and distributed estimation using locally computed quantities (Vo et al., 2022; Xiong et al., 2023). Such approaches enable collaboration without pooling patient records. The population defining the causal effect nevertheless remains important: when treatment effects vary with patient characteristics, averaging effects from different sites need not recover the target average treatment effect (TATE). Estimating the TATE therefore requires accounting for how source populations differ from the target.

Target-specific borrowing addresses this requirement by adjusting source observations to the target population. Calibration weighting aligns source and target covariate distributions (Josey et al., 2022; Chattopadhyay et al., 2024), while multiply robust federated estimation accommodates covariate shift and differences in the covariates available across sites (Han et al., 2023). Adjustment for covariate shift does not, by itself, ensure that the conditional outcome relationship is shared across populations (Cole and Stuart, 2010; Pearl and Bareinboim, 2014; Dahabreh et al., 2019). Federated adaptive causal estimation (FACE) addresses this additional concern by penalizing source contributions that disagree with target-only estimation; its FACE-X extension uses flexible nuisance estimators with consistency and product-rate conditions (Han et al., 2025). This adaptive borrowing perspective is especially relevant when compatibility is unknown and may differ between treatment arms.

High-dimensional distributed causal estimation addresses a complementary part of the problem. Communication-efficient covariate-balancing methods fit high-dimensional propensity and outcome models using distributed summaries (Wang et al., 2024). DisC2o-

HD incorporates cross-site covariate shift while approximating the estimator available from pooled data (Tong et al., 2025). These developments show how high-dimensional adjustment can be coordinated across institutions. Here we seek to combine this ability to adjust for many covariates with estimation for a prespecified target and adaptation to incompatible source outcomes. The nuisance-fitting step must also support inference when one working model is misspecified. This objective calls for linking adaptive source aggregation to calibration of the target estimating equation.

The single-site literature provides several tools for controlling errors from high-dimensional nuisance estimation, including double selection, approximate residual balancing, and double machine learning (Belloni et al., 2014; Athey et al., 2018; Chernozhukov et al., 2018). These approaches combine variable selection or flexible fitting with estimating procedures that reduce the impact of first-stage error under their respective assumptions. For doubly robust estimators, an additional distinction is consequential: consistency when either model is correct does not automatically imply valid confidence intervals under the same condition. If the other model is misspecified, regularization error in the correctly specified model can contribute at first order. Regularized calibration addresses this problem by choosing fitting losses according to the sensitivity of the effect estimator (Vermeulen and Vansteelandt, 2015; Tan, 2020; Smucler et al., 2019).

For AIPW, calibration exploits the pairing between the outcome regression and propensity model: differentiating the estimating equation with respect to one model motivates how the other should be fitted. Tan (2020) develops penalized calibration with model-assisted confidence intervals that remain valid under a correct propensity model even when the outcome model is misspecified; with a linear outcome model, the intervals are doubly robust. Smucler et al. (2019) combines derivative-matched losses, sample splitting, and cross-fitting to obtain rate and model double robustness for a broader class of causal contrasts. Hou et al. (2025) uses a two-layer cross-fitting construction for sequential calibrated fitting in semi-

supervised treatment-effect estimation with high-dimensional logistic models. Together, these results motivate fitting nuisance models to support inference for the final causal estimand, with the data separation tailored to the sequence of fitting steps.

Extending this paired calibration to transported estimation introduces a third nuisance function: a source-to-target density ratio, in addition to the outcome regression and source treatment propensity. Calibrating the two weighting components separately does not automatically calibrate their product in the transported score. We propose robust federated causal estimation (RoCE) by returning to the role of source observations: outcomes in a source treatment group augment outcome predictions for the entire target population. A single joint weight maps that treatment group to the target, combining treatment and population adjustment. This construction restores an outcome–weight pair and yields paired regularized calibration losses for the source-assisted target mean. Under suitable identification, sparsity, and regularity conditions, calibration supports inference when either of these two working models is correct.

Calibration controls nuisance-estimation error within a source-assisted estimate, while aggregation must address bias from incompatible source outcomes. Building on FACE, RoCE compares each source-assisted estimate with a calibrated target-only estimate on a common standard-error scale and uses a pivotal penalty to limit discrepant contributions. A source can contribute to one treatment arm and receive zero weight in the other. The two arms' weights are optimized jointly for the variance of their difference, accounting for dependence induced by shared target observations. Two-layer cross-fitting accommodates the sequential calibration steps and reserves observations for final evaluation after nuisance fitting and weight learning. With target-based outcome initialization, the procedure requires one batched target broadcast and one source return containing coefficients and empirical summaries. RoCE thus connects calibrated high-dimensional fitting, adaptive borrowing, and federated implementation around a common target treatment contrast.

Our main contributions are:

(i) Joint weighting and paired calibration enable target-specific borrowing with high-dimensional covariates and model-doubly-robust inference under the stated conditions.

(ii) Arm-specific pivotal aggregation limits incompatible borrowing and attains an oracle variance no greater than calibrated target-only estimation under the stated source-selection conditions.

(iii) A federated implementation combines calibrated fitting and joint treatment-effect aggregation in one batched round of communication without transferring individual records.

Section 2 defines the estimand and identification conditions. Section 3 develops source-assisted estimation, calibration, aggregation, and the federated implementation. Section 4 states the corresponding inference guarantees. Section 5 examines precision and coverage under model misspecification and source incompatibility, and Section 6 applies the method to an All of Us drug comparison. Section 7 discusses implications and limitations. The Supplementary Material provides implementation details, proofs, alternative constructions, and a right heart catheterization application.

2 Setting and Identification

Consider one prespecified target site t and K sources $\mathcal{S} = \{s_1, \ldots, s_K\}$. Individual i has observed record $D_i = (R_i, \boldsymbol{X}_i, A_i, Y_i)$, where $R_i \in \{t\} \cup \mathcal{S}$ denotes site membership, $\boldsymbol{X}_i \in \mathbb{R}^p$ baseline covariates, $A_i \in \{0, 1\}$ treatment, and Y_i the outcome. Let $\mathcal{D}_r = \{i : R_i = r\}$ index the observations at site r, with $n_r = |\mathcal{D}_r|$ and $N = n_t + \sum_j n_{s_j}$. Write $\mathbb{E}_r$ and P_r for expectation and probability conditional on $R_i = r$. The sitewise empirical average of h is $\widehat{\mathbb{E}}_r h = n_r^{-1} \sum_{i \in \mathcal{D}_r} h(D_i)$; $I(\cdot)$ denotes an indicator.

Let $Y_i(a)$ be the potential outcome under treatment a. Our estimands are the target potential-outcome means and their contrast,

$$\mu_t^a = \mathbb{E}_t\{Y_i(a)\}, \qquad a \in \{0,1\}, \qquad \tau_t = \mu_t^1 - \mu_t^0.$$

The target population is fixed before borrowing. Source observations add information without redefining the TATE. For binary outcomes, τ_t is the marginal risk difference in the target population.

We distinguish target identification from source eligibility. Let $e_r^{a,\star}(\boldsymbol{x}) = P_r(A_i = a \mid \boldsymbol{X}_i = \boldsymbol{x})$ be the true treatment probability and $f_r^\star$ the covariate density with respect to a common dominating measure. Stars mark true functions; working models follow in Method.

Assumption 1 (Target identification and source eligibility)**.** For each $a \in \{0,1\}$, let $\mathcal{S}_a^* \subseteq \mathcal{S}$ be the possibly empty transportable source set. For constants $\epsilon > 0$ and $C < \infty$:

(a) *Consistency and no interference.* At the target and eligible sources, treatment is well defined with no interference, and $Y_i = Y_i(A_i)$.

(b) *Target identification.* $Y_i(a) \perp A_i \mid \boldsymbol{X}_i, R_i = t$ and $e_t^{a,\star}(\boldsymbol{X}_i) \in [\epsilon, 1-\epsilon]$.

(c) *Source exchangeability and overlap.* For each $s_j \in \mathcal{S}_a^*$, $Y_i(a) \perp A_i \mid \boldsymbol{X}_i, R_i = s_j$ and $e_{s_j}^{a,\star}(\boldsymbol{X}_i) \in [\epsilon, 1-\epsilon]$. In addition,

$$\operatorname{supp}(f_t^\star) = \operatorname{supp}(f_{s_j}^\star), \qquad \|f_t^\star / f_{s_j}^\star\|_\infty \le C.$$

(d) *Conditional mean transportability.* For each $s_j \in \mathcal{S}_a^*$, $\mathbb{E}_{s_j}\{Y_i(a) \mid \boldsymbol{X}_i\} = \mathbb{E}_t\{Y_i(a) \mid \boldsymbol{X}_i\}$ on the common support.

Parts (a) and (b) link observed outcomes to potential outcomes and require the measured covariates to account for target-site confounding, with treatment overlap. They identify

$$\mu_t^a = \mathbb{E}_t\{\mathbb{E}_t(Y_i \mid A_i = a, \boldsymbol{X}_i)\}.$$

Thus target data identify the TATE even when both source sets are empty. Borrowing aims to improve precision under additional source conditions.

Parts (c) and (d) provide those conditions: adjustment for source-site confounding, population overlap, and a shared conditional potential-outcome mean (Cole and Stuart, 2010; Pearl and Bareinboim, 2014; Dahabreh et al., 2019). For an eligible source, they also give $\mu_t^a = \mathbb{E}_t\{\mathbb{E}_{s_j}(Y_i \mid A_i = a, \boldsymbol{X}_i)\}$. The source conditional mean is therefore averaged over the target covariate distribution. Covariate distributions, treatment policies, and residual variation may differ across sites. Eligibility is arm-specific: a source may contribute to one potential-outcome mean without contributing to the other. Equality of conditional treatment effects alone does not justify transporting the two means separately.

Source eligibility is unknown, and measured balance cannot establish either unconfoundedness or outcome compatibility. All sources remain candidates, with discrepancies from target-only estimation guiding adaptive borrowing (Han et al., 2025). The theory distinguishes transportable sources from candidates whose population limits equal the target arm mean. We call the latter *unbiased candidates*, without asserting finite-sample unbiasedness.

Following the stage notation of Hou et al. (2025), hats denote estimates and bars population coefficients; the subscript init identifies initial fits and their population minimizers. Calibrated coefficients omit this subscript, and stars identify true nuisance functions and their coefficients when correctly specified. A *joint weight* $q_{s_j}^a$ adjusts observations from a source treatment group; an *aggregation weight* $\eta_{s_j}^a$ combines the resulting candidate with other estimates. The next section constructs calibrated source-assisted estimates and aggregates them for τ_t.

3 Method

RoCE implements the two tasks described in the Introduction: calibrated estimation of each source-assisted target mean, followed by adaptive borrowing across source treatment groups. We begin with one source and one treatment arm, combining target outcome predictions with a weighted source residual correction. The resulting population mean determines the calibration losses and the fitted candidate estimator. We then combine these candidates with target-only estimation, choosing both arms' weights for the treatment contrast. Section 3.4 completes the construction with cross-fitting and federated communication; fitting and weight learning exclude the observations used for final evaluation.

3.1 Working Models and Source-Assisted Target Means

Fix a source s_j and treatment arm a. Let $m_r^{a,\star}(\boldsymbol{x}) = \mathbb{E}_r(Y_i \mid A_i = a, \boldsymbol{X}_i = \boldsymbol{x})$ denote the true outcome regression at site r. For a transportable source, this conditional mean agrees with $m_t^{a,\star}(\boldsymbol{X})$, and averaging it over target covariates identifies μ_t^a. We approximate the mean using $m(\boldsymbol{x}; \alpha) = g\{\phi(\boldsymbol{x})^T\alpha\}$, where $\alpha \in \mathbb{R}^d$ and g is a canonical generalized linear model (GLM) inverse link. Examples include logistic, linear, and Poisson means; correct specification concerns the conditional mean without requiring the full exponential-family distribution. All sites use the same prespecified feature map ϕ, including an intercept. Its dimension d may differ from the raw covariate dimension p; with main effects only, $\phi(\boldsymbol{X}_i) = (1, \boldsymbol{X}_i^T)^T$ and $d = p + 1$.

The working outcome model may be misspecified. Observed outcomes from source arm a can correct its target predictions if the source residuals are appropriately weighted. Because this correction uses only one source treatment group, its ideal weight is

$$q_{s_j}^{a,\star}(\boldsymbol{x}) = \frac{f_t^\star(\boldsymbol{x})}{f_{s_j}^\star(\boldsymbol{x}) e_{s_j}^{a,\star}(\boldsymbol{x})}.$$

The denominator equals $P_{s_j}(A_i = a) f^{\star}_{s_j}(\boldsymbol{x} \mid A_i = a)$, so the weight maps the source treatment group to the entire target covariate population. The arm-probability factor accounts for averaging over all source observations. In particular, the ideal weight satisfies

$$\mathbb{E}_{s_j}\{I(A = a) q^{a,\star}_{s_j}(\boldsymbol{X}) h(\boldsymbol{X})\} = \mathbb{E}_t\{h(\boldsymbol{X})\} \tag{1}$$

for integrable h. Taking $h = 1$ gives the normalization $\mathbb{E}_{s_j}\{I(A = a) q^{a,\star}_{s_j}(\boldsymbol{X})\} = 1$.

RoCE models this joint adjustment directly through $q^a_{s_j}(\boldsymbol{x}; \gamma) = \exp\{-\phi(\boldsymbol{x})^T \gamma\}$, where $\gamma \in \mathbb{R}^d$ and the intercept accommodates the arm normalization. This formulation combines treatment and population adjustment in one weighting model, paired with $m(\boldsymbol{X}; \alpha)$. A correct weighting model represents the ideal function $q^{a,\star}_{s_j}(\boldsymbol{X})$; balancing a finite set of features alone does not establish that property. The two working models define the source-assisted population mean

$$\mu^a_{s_j}(\alpha, \gamma) = \mathbb{E}_t\{m(\boldsymbol{X}; \alpha)\} + \mathbb{E}_{s_j}\left[I(A = a) q^a_{s_j}(\boldsymbol{X}; \gamma)\{Y - m(\boldsymbol{X}; \alpha)\}\right]. \tag{2}$$

The first term averages outcome predictions over the target population, and the second corrects those predictions using weighted source residuals. Thus the subscript s_j identifies the source supplying the correction, while the estimand remains μ^a_t. The coefficient-dependent quantity $\mu^a_{s_j}(\alpha, \gamma)$ equals that estimand under the conditions below. This transported AIPW construction builds on Josey et al. (2022); Han et al. (2025).

Lemma 1 (Source identification under either correct model)**.** *For a transportable source and coefficient vectors α, γ with finite expectations, Assumption 1 gives*

$$\mu^a_{s_j}(\alpha, \gamma) - \mu^a_t = \mathbb{E}_t\left[\left(1 - \frac{q^a_{s_j}(\boldsymbol{X}; \gamma)}{q^{a,\star}_{s_j}(\boldsymbol{X})}\right)(m(\boldsymbol{X}; \alpha) - m^{a,\star}_t(\boldsymbol{X}))\right].$$

Thus $\mu^a_{s_j}(\alpha, \gamma) = \mu^a_t$ if $m(\boldsymbol{X}; \alpha) = m^{a,\star}_t(\boldsymbol{X})$ or $q^a_{s_j}(\boldsymbol{X}; \gamma) = q^{a,\star}_{s_j}(\boldsymbol{X})$.

Conditioning the source residual on $\boldsymbol{X}$ and applying Eq. (1) gives the identity; see the proof of Lemma 1 in Section S2.3. Inference after regularization additionally requires control of fitting error, which motivates calibration.

3.2 Regularized Calibration

When one working model is misspecified, estimation error in the correctly specified model can affect $\mu^a_{s_j}(\alpha, \gamma)$ at first order. Calibration chooses fitting losses to remove this sensitivity at the population limits (Tan, 2020; Smucler et al., 2019). Continue with the fixed source and arm, and write $\phi = \phi(\boldsymbol{X})$ in the derivatives below. The relevant derivatives are

$$\begin{aligned} \nabla_\alpha \mu^a_{s_j}(\alpha, \gamma) &= \mathbb{E}_t\{g'(\phi^T\alpha)\phi\} - \mathbb{E}_{s_j}\{I(A=a)q^a_{s_j}(\boldsymbol{X};\gamma)g'(\phi^T\alpha)\phi\}, \\ \nabla_\gamma \mu^a_{s_j}(\alpha, \gamma) &= -\mathbb{E}_{s_j}\{I(A=a)q^a_{s_j}(\boldsymbol{X};\gamma)(Y - m(\boldsymbol{X};\alpha))\phi\}. \end{aligned} \tag{3}$$

The first derivative motivates fitting the joint weight to balance derivative-weighted features between target and source. The second motivates fitting the outcome model to balance weighted source residuals. These are the two conditions that the paired losses seek to satisfy at their population limits. If both derivatives vanish and second derivatives are locally bounded, perturbing the fitted coefficients changes the population mean only at second order.

To implement this pairing, let $L(Y, u) = \Psi(u) - Yu$, with $\Psi' = g$, and let $\|v\|_{1,-0}$ denote the sum of absolute slope coefficients, excluding the intercept. We first fit an ℓ_1-penalized outcome GLM in target arm a and an initial exponential-balancing weight at each source. Denote their coefficients by $\widehat{\alpha}_{\text{init}}$ and $\widehat{\gamma}_{\text{init}}$; the target outcome initialization is shared across sources. These initial fits supply fixed multipliers for the subsequent calibrated updates, avoiding alternating optimization of the two models. The initial losses are given in Section S1.1.

For stable multipliers, fix $M > 0$ and clip each initial linear predictor using $\mathcal{T}_M(u) = \text{sign}(u)\min(|u|, 2M)$. The corresponding clipped logistic link is $\sigma_M(u) = \text{expit}\{\mathcal{T}_M(u)\}$. Following the clipping convention of Hou et al. (2025), the theory bounds the relevant population predictors by M, leaving a margin before the numerical radius $2M$; choosing a numerical radius alone does not establish this population bound.

With the initial fits held fixed, use the derivative multiplier $d_{\text{init}}(\boldsymbol{X}) = g'\{\mathcal{T}_M(\phi^T\widehat{\alpha}_{\text{init}})\}$ and the weight multiplier $\omega_{\text{init}}(\boldsymbol{X}) = e^{-\mathcal{T}_M(\phi^T\widehat{\gamma}_{\text{init}})}$. The two calibration losses are

$$\ell_{\text{RCAL}}(\gamma; \widehat{\alpha}_{\text{init}}) = \widehat{\mathbb{E}}_t(d_{\text{init}}\phi)^T\gamma + \widehat{\mathbb{E}}_{s_j}\{I(A=a)e^{-\phi^T\gamma}d_{\text{init}}\} + \lambda_\gamma\|\gamma\|_{1,-0}, \tag{4}$$

$$\ell_{\text{RWL}}(\alpha; \widehat{\gamma}_{\text{init}}) = \widehat{\mathbb{E}}_{s_j}\left[I(A=a)\omega_{\text{init}}(\boldsymbol{X})L(Y, \phi^T\alpha)\right] + \lambda_\alpha\|\alpha\|_{1,-0}. \tag{5}$$

Let $\widehat{\gamma}$ and $\widehat{\alpha}$ minimize these losses, respectively. Each optimization is convex in its coefficient vector when $g' \geq 0$. The joint-weight loss uses the same derivative multiplier in its target and source terms, preserving balance under a correct weight even when the initial outcome fit is misspecified. The outcome loss fits source residuals under the initial joint weight. Only the target moment $\widehat{\mathbb{E}}_t(d_{\text{init}}\phi)$ is needed at the source, which is the basis for the federated implementation below. The empirical averages here represent calibration data; Section 3.4 specifies their folds and the separate data used for initial fitting.

Lemma 2 (Calibration under either correct model)**.** *For a transportable source and one arm, let* $(\bar{\alpha}, \bar{\gamma})$ *be the identifiable calibrated population coefficients satisfying the predictor bound in Section 3.2. If the outcome model is correct at* $\alpha^\star$*,* $\bar{\alpha} = \alpha^\star$*; if the joint weight model is correct at* $\gamma^\star$*,* $\bar{\gamma} = \gamma^\star$*. In either case, the other coefficient minimizes its paired population calibration loss with the correct initial coefficient. Then* $\mu^a_{s_j}(\bar{\alpha}, \bar{\gamma}) = \mu^a_t$*, and both gradients in Eq. (3) vanish at* $(\bar{\alpha}, \bar{\gamma})$*.*

The correctly specified model supplies one zero derivative; stationarity of the paired calibrated loss supplies the other. Initial and calibrated coefficients may differ for the misspecified model. The proof of Lemma 2 is given in Section S2.5; Lemma S1 then controls error propagation from the initial fits.

For a linear outcome mean, $g' = 1$, so the procedure simplifies to exponential feature balancing and penalized weighted least squares; nonlinear means retain the derivative multiplier.

We now return to the source and arm indices. Denote the calibrated functions by $\widehat{m}^a_{s_j}(\boldsymbol{x}) = m(\boldsymbol{x}; \widehat{\alpha}^a_{t,s_j})$ and $\widehat{q}^a_{s_j} = q^a_{s_j}(\cdot; \widehat{\gamma}_{s_j,a})$, without further clipping of these source-assisted fits. Inserting them into the empirical counterpart of Eq. (2) gives

$$\widehat{\mu}^a_{s_j} = \widehat{\mathbb{E}}_t\{\widehat{m}^a_{s_j}(\boldsymbol{X})\} + \widehat{\mathbb{E}}_{s_j}\left[I(A=a)\widehat{q}^a_{s_j}(\boldsymbol{X})\{Y - \widehat{m}^a_{s_j}(\boldsymbol{X})\}\right]. \tag{6}$$

This is the fitted candidate corresponding to $\mu^a_{s_j}(\alpha, \gamma)$. All candidates seek the same target mean μ^a_t, although an incompatible source can have a different population limit. The cross-fitting procedure below specifies when this expression is evaluated on training data for weight learning and when its contributions are evaluated on held-out data for final estimation.

To assess whether borrowing is helpful, we also need a candidate that relies only on the target's own identification conditions. Calibrated target outcome and logistic propensity fits $\widehat{m}^a_t$ and $\widehat{e}^a_t$ give

$$\widehat{\mu}^a_t = \widehat{\mathbb{E}}_t\left[\widehat{m}^a_t(\boldsymbol{X}) + \frac{I(A=a)}{\widehat{e}^a_t(\boldsymbol{X})}\{Y - \widehat{m}^a_t(\boldsymbol{X})\}\right]. \tag{7}$$

We use regularized calibrated target outcome and logistic propensity fits following Tan (2020); Smucler et al. (2019); Hou et al. (2025); the losses and fold updates are given in Section S1.2. The two calibrated arm-specific propensity fits need not sum to one because each is fitted for its own arm score. The true treatment probabilities still satisfy $e^{0,\star}_t = 1 - e^{1,\star}_t$.

The target-only candidate supplies the reference for borrowing. Comparing each source-assisted candidate with this reference connects the calibrated estimation step to adaptive aggregation.

3.3 Joint Aggregation for the Target Treatment Effect

Calibration addresses regularization error within a candidate, while aggregation determines how much to borrow from each source treatment group. We balance the precision available from a source against its discrepancy from target-only estimation. All quantities used to learn these weights are computed outside an outer evaluation fold k_1. Let $n_{r,-k_1}$ be the remaining sample size at site r and $n_{-k_1} = \sum_r n_{r,-k_1}$. In this subsection, the candidate means and fitted functions use only these training observations, and we suppress their fit index $(-k_1)$.

For arm a, let $\eta^a = (\eta^a_{s_1}, \ldots, \eta^a_{s_K})^T$ collect the source aggregation weights. Starting from the target-only reference, the aggregate adds a weighted discrepancy for each source:

$$\widehat{\mu}^a_{\text{agg}}(\eta^a) = \widehat{\mu}^a_t + \sum_{j=1}^{K} \eta^a_{s_j}(\widehat{\mu}^a_{s_j} - \widehat{\mu}^a_t). \tag{8}$$

The corresponding treatment contrast is $\widehat{\tau}_t(\eta^1, \eta^0) = \widehat{\mu}^1_{\text{agg}}(\eta^1) - \widehat{\mu}^0_{\text{agg}}(\eta^0)$. A source can contribute to one arm and receive zero weight in the other. For example, a source with a shifted treated outcome mean but a compatible control mean may still help estimate μ^0_t. Joint optimization allows this distinction while accounting for covariance between the two estimated means. Imposing $\eta^1 = \eta^0$ would instead aggregate assisted treatment effects using a common source-weight vector.

Choosing the two weight vectors jointly requires the variance of their treatment contrast. Stack the weights as $\eta = ((\eta^1)^T, (\eta^0)^T)^T$ and write the aggregate in terms of contributions from individual sites. Using Eqs. (6) and (7), with the arguments $\boldsymbol{X}_i$ suppressed, these contributions are

$$\begin{aligned} \widehat{G}^a_{t,i}(\eta^a) &= \left(1 - \sum_j \eta^a_{s_j}\right)\left[\widehat{m}^a_t + \frac{I(A_i = a)}{\widehat{e}^a_t}(Y_i - \widehat{m}^a_t)\right] + \sum_j \eta^a_{s_j}\widehat{m}^a_{s_j}, \\ \widehat{G}^a_{s_j,i}(\eta^a) &= \eta^a_{s_j} I(A_i = a)\widehat{q}^a_{s_j}(Y_i - \widehat{m}^a_{s_j}). \end{aligned} \tag{9}$$

Averaging within sites and then summing recovers Eq. (8). Under the conditions in Section 4, calibration removes first-order nuisance-estimation terms, leaving a leading variance that

adds across independent sites. The candidates themselves remain correlated because they share target predictions and both arms use the same site records. Computing the variance of the within-site contrast retains these covariances.

Lemma 3 (Sitewise variance and cross-arm covariance)**.** *Independent site samples give additive variances for the first-order site contributions. The joint treatment-effect variance is estimated by*

$$\widehat{V}_{\tau}^{(-k_1)}(\eta) = \sum_{r} \frac{\widehat{\mathrm{Var}}_{r,-k_1}\{\widehat{G}^1_{r,i}(\eta^1) - \widehat{G}^0_{r,i}(\eta^0)\}}{n_{r,-k_1}}. \tag{10}$$

Here each empirical variance centers contributions within the inner calibration folds and pools their second central moments within the site. Under the candidate linear expansions and fitted-score covariance consistency, this expression consistently estimates the treatment-effect variance on the valid candidate subspace.

The proof of Lemma 3 is given in Section S3.2. Its covariance-block construction in Section S1.3 determines the required summaries: prediction and residual means, variances, cross-arm moments, and site counts. Their transmission is described in Section 3.4. Candidate means are pooled within each site's training sample before discrepancies are computed.

Let $\widehat{d}^a_{s_j} = \widehat{\mu}^a_{s_j} - \widehat{\mu}^a_t$ be the training-sample discrepancy and $\widehat{s}^a_{s_j}$ its score-based standard error, including shared-target covariance. Set $\widehat{t}^a_{s_j} = |\widehat{d}^a_{s_j}|/\widehat{s}^a_{s_j}$. This standardization compares discrepancies on a common sampling-error scale. We use the term *pivotal selection* for the resulting penalty within the adaptive aggregation framework (Han et al., 2025). For a positive cutoff c_n, choose both weight vectors in one optimization:

$$\widehat{\eta}^{(-k_1)} \in \arg\min_{\eta\in\mathbb{R}^{2K}} \left\{ n_{-k_1} \widehat{V}_{\tau}^{(-k_1)}(\eta) + \sum_{a=0}^{1}\sum_{j=1}^{K} (\widehat{t}^a_{s_j}/c_{n_{-k_1}} - 1)_+ |\eta^a_{s_j}| \right\}. \tag{11}$$

The variance is quadratic in η, so the objective is a convex quadratic plus a weighted ℓ_1 penalty. The weights may be signed; the target coefficient for arm a is $1 - \sum_j \eta^a_{s_j}$, and

$\eta = 0$ recovers target-only estimation. Positive-definite curvature ensures a finite, unique optimizer. The penalty is zero below the cutoff and grows above it. Selection occurs within the optimization, without a preceding hard-screening step; exceeding the cutoff need not force a weight to zero in finite samples. Penalties remain arm-specific even if the two source-arm biases cancel in their treatment contrast.

Separate armwise variance minimization generally differs from this joint optimization because the cross-arm covariance depends on both weight vectors. If all retained outcome functions equal their correct transportable limits, the cross-arm covariance of the limiting aggregate means is instead $n_{t,-k_1}^{-1}\,\mathrm{Cov}_t\{m_t^{1,\star}(\boldsymbol{X}), m_t^{0,\star}(\boldsymbol{X})\}$, independent of the weights. In this special case, the joint and separate-arm oracle minimizers coincide despite nonzero covariance. Finite-sample optimizers need not coincide, and outcome misspecification generally removes this special structure.

Remark 1 (Distinct roles of the two thresholds). The predictor clipping radius $2M$ belongs to nuisance estimation. The aggregation cutoff c_n, with $n = n_{-k_1}$ during weight learning, controls the arm-specific Wald penalties. For theory, $c_n \to \infty$ and $c_n/\sqrt{n} \to 0$, for example $c_n = \sqrt{\log(n \vee 3)}$. The implementation parameter multiplying $\widehat{t}_{s_j}^{\,a}$ in the penalty is $\lambda_{\mathrm{agg}} = 1/c_n$; a fixed cutoff 2 corresponds to $\lambda_{\mathrm{agg}} = 0.5$. A fixed cutoff is a finite-sample specification, not the growing-cutoff regime of the theorem.

3.4 Two-Layer Cross-Fitting and Communication

The preceding construction has three fitting stages: initial nuisance estimation, calibrated updating, and aggregation-weight learning. We adapt the two-level cross-fitting construction of Hou et al. (2025) to preserve this order. Initial fits exclude the observations used for their calibration updates, and an outer fold is reserved for evaluation after all three stages. Partition each site r into a fixed number $K_f \geq 3$ of balanced folds $\mathcal{I}_{r,1}, \ldots, \mathcal{I}_{r,K_f}$, shared by both arms, with membership fixed in advance or randomized independently of the observed

records. Let $\mathcal{I}_{r,-k_1,-k_2} = \mathcal{D}_r \setminus (\mathcal{I}_{r,k_1} \cup \mathcal{I}_{r,k_2})$, $\mathcal{I}_{r,-k_1} = \mathcal{D}_r \setminus \mathcal{I}_{r,k_1}$, and $n_{r,J} = |\mathcal{I}_{r,J}|$. We write $\widehat{\mathbb{E}}_{r,J} h = n_{r,J}^{-1} \sum_{i \in \mathcal{I}_{r,J}} h(D_i)$ for a fold or fold complement J. For an outer fold k_1, proceed as follows.

1. **Initial fitting and calibration.** For each inner fold $k_2 \neq k_1$, fit the initial models outside both k_1 and k_2. The target outcome initialization is shared across sources. Insert each initial fit into the calibrated loss on fold k_2, then pool the loss blocks over $k_2 \neq k_1$ and minimize once for each calibrated coefficient vector. Each site's blocks are weighted by its own fold fractions, and each slope penalty enters once. The source updates are in Section S1.1; the corresponding local target updates are in Section S1.2.

2. **Joint weight learning.** Evaluate the final calibrated fits on all observations outside k_1. Pool candidate means and the full two-arm covariance blocks within sites, then compute discrepancies and solve Eq. (11). This yields one joint pair $\widehat{\eta}^{(-k_1)}$. The calibration folds contribute fitted training scores here; they are not independent validation data for the final calibrated fits.

3. **Outer evaluation.** Apply these calibrated fits and the learned weight pair to fold k_1. Combine observations using each site's full sample size, retaining the fits and weights specific to that outer fold. All fitting, preprocessing, tuning and weight learning exclude k_1.

Thus each initial fit uses $K_f - 2$ folds, each calibration update uses loss blocks from the $K_f - 1$ folds outside k_1, and k_1 remains untouched until evaluation. For an observation in outer fold $k_1(i)$, evaluate Eq. (9) using nuisance fits and weights trained outside that fold, and write $\widehat{G}^a_{r,i} = \widehat{G}^a_{r,i}(\widehat{\eta}^{a,(-k_1(i))})$ for its held-out contribution. The final estimator, including unequal site and fold sizes, is

$$\widehat{\tau}_t = \frac{1}{n_t} \sum_{i \in \mathcal{D}_t} (\widehat{G}^1_{t,i} - \widehat{G}^0_{t,i}) + \sum_j \frac{1}{n_{s_j}} \sum_{i \in \mathcal{D}_{s_j}} (\widehat{G}^1_{s_j,i} - \widehat{G}^0_{s_j,i}). \tag{12}$$

The two data-separation requirements have different roles. Initial fits exclude their calibration fold, and final evaluation excludes all fitting and weight learning. Step 2 reuses observations that fitted the calibrated models; Assumption 4 states the training moments needed for weight learning. The primary algorithm and reported analyses use this two-layer construction. An alternative reserves independent inner validation folds for weight learning, with $K_f \geq 4$; its algorithm and supporting argument are collected in Section S7 and Lemma S5.

The same fitting order admits a federated implementation. With target outcome initialization, all fold-specific initial coefficients, feature means and derivative moments can be sent to sources in one batch. Sources return calibrated outcome coefficients and training/evaluation residual summaries, including both-arm second moments and counts, in one batch. The target computes prediction covariances and joint aggregation weights locally. Every message identifies its training sample and evaluation fold; outer-evaluation summaries never enter weight learning. This one-round protocol is shown in figure 1.

Here a communication round comprises one target broadcast and one source return. The number of transmitted coefficients and moments grows with the working dimension, sources, and fold-specific tasks, while the message contents do not contain individual records. The round count assumes that the feature definitions and fitting tasks are agreed in advance and that tuning requests no additional target summaries. Cross-validation is confined to the relevant training sample; Section S1.5 gives its loss and stabilization conventions. Preprocessing that requires pooled information must be accounted for separately. Source-based outcome initialization requires one additional exchange of target derivative moments (Section S7.2).

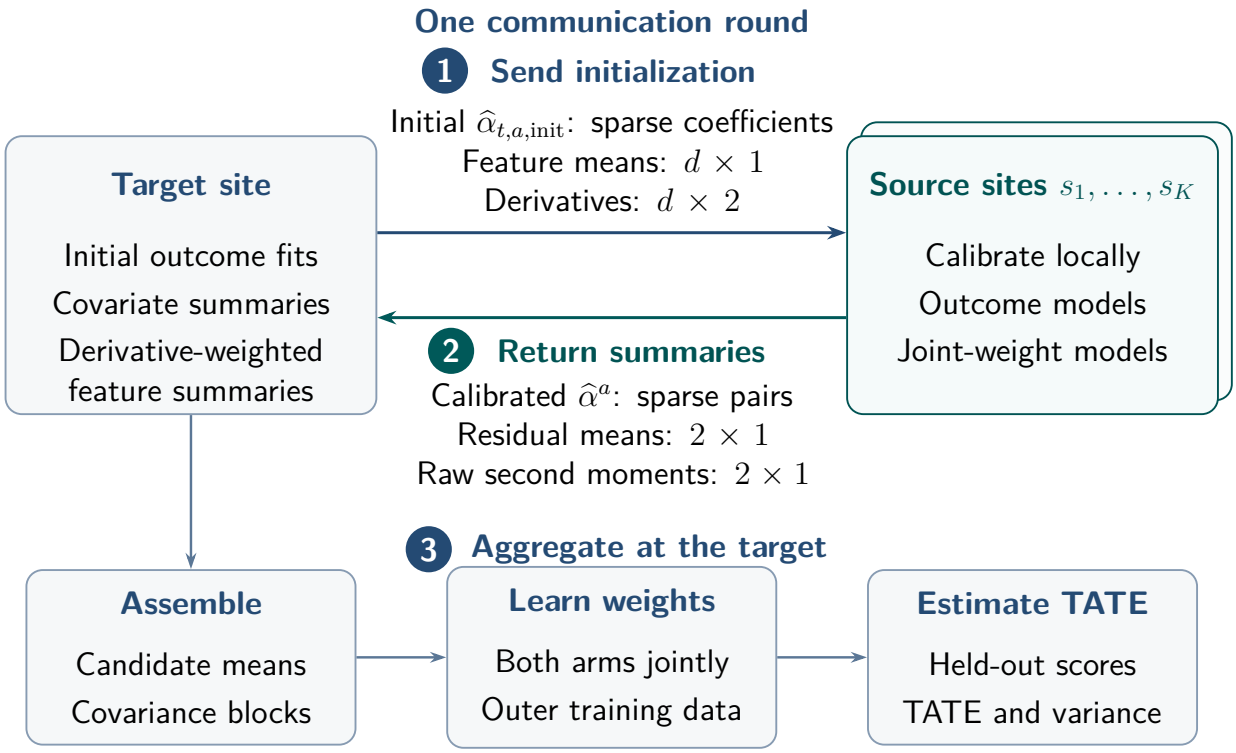


Figure 1: One-round RoCE communication for point estimation and the leading variance, with target outcome initialization and tasks batched over folds. Sources also return sample counts. Patient records and joint-weight coefficients remain local. Primary standard errors use the leading variance; an optional weight-sensitivity diagnostic is described in Section S3.6.

4 Theory

The analysis follows the two stages of RoCE: calibrated candidate estimation and adaptive borrowing. Candidate expansions control nuisance-estimation error. A separate training-moment argument and source-separation conditions then support weight learning. Together, they yield the source-selection and inference theorem for the final contrast in Eq. (12), followed by variance estimation and an efficiency comparison.

Throughout, $N \to \infty$, the numbers of sources K and folds K_f are fixed, and d may grow. Site fractions satisfy $n_r/N \to \rho_r^{\text{all}} > 0$. We study the two-layer procedure in Section 3.4,

building on calibrated estimation (Tan, 2020; Smucler et al., 2019; Hou et al., 2025) and adaptive borrowing (Han et al., 2025). The Supplementary Material follows this order and provides a proof roadmap at the start of Section S2.

4.1 Calibrated Candidate Expansions

We first translate the identification and calibration properties in Lemmas 1 and 2 into sampling limits. Let $\bar{\alpha}_{j,a}, \bar{\gamma}_{j,a}$ be the calibrated source population coefficients, with $\bar{m}^a_{s_j} = g(\phi^T \bar{\alpha}_{j,a})$ and $\bar{q}^a_{s_j} = e^{-\phi^T \bar{\gamma}_{j,a}}$. Let $\bar{m}^a_t, \bar{e}^a_t$ denote the calibrated target outcome and propensity limits. These limits need not equal the true functions under misspecification.

For each arm, define $\mathcal{J}^{a,*} = \{s_j : \operatorname{plim} \widehat{\mu}^a_{s_j} = \mu^a_t\}$, the set of unbiased source-assisted candidates. Membership describes agreement of population means. Inference additionally requires conditional mean transportability and one correct working model for each member, as specified in Assumption 3.

The leading variation comes from centered site scores. The target-only score is $\varphi^a_t = \bar{m}^a_t + I(A=a)(Y - \bar{m}^a_t)/\bar{e}^a_t - \mu^a_t$ (Section S1.2). For a source-assisted candidate, the target score is $\zeta^a_{t,s_j} = \bar{m}^a_{s_j} - \mathbb{E}_t \bar{m}^a_{s_j}$ and the source score is

$$\xi^a_{t,s_j} = I(A=a)\bar{q}^a_{s_j}(Y - \bar{m}^a_{s_j}) - \mathbb{E}_{s_j}\{I(A=a)\bar{q}^a_{s_j}(Y - \bar{m}^a_{s_j})\}.$$

Arguments $\boldsymbol{X}$ are suppressed. The first assumption controls sampling variation and the fitted losses. Constants are independent of d, N; full tail, overlap, and predictor-margin conditions are in Assumptions S1 and S2.

Assumption 2 (Design, link, and moments)**.** For $a \in \{0,1\}$ and sites used by the target-only estimator and candidates in $\mathcal{J}^{a,*}$, there exist $C, \kappa, \delta > 0$ such that:

(a) $D_i \mid R_i = r \overset{\text{i.i.d.}}{\sim} P_r$, $\mathcal{D}_r \perp \mathcal{D}_{r'}$ for $r \neq r'$, and $n_r/N \to \rho^{\text{all}}_r > 0$. Balanced fold memberships are chosen independently of the observed records, as in Section 3.4.

(b) Almost surely $\|\phi\|_\infty \le C$, and

$$\lambda_{\min}\Big[\mathbb{E}_r\{I(A=a)\phi\phi^T\}\Big] \ge \kappa, \qquad \lambda_{\max}\{\mathbb{E}_r(\phi\phi^T)\} \le C.$$

(c) $g \in C^2(\mathbb{R})$, $g' > 0$, and, for $|u| \le 2M$,

$$C^{-1} \le g'(u) \le C, \qquad |g(u)| + |g''(u)| \le C.$$

The conditional-mean and tail bounds, quantitative overlap, and population-predictor margins in Assumption S2(b)–(d) hold. In particular, the ideal source joint weights are bounded above and away from zero.

(d) For each $s_j \in \mathcal{J}^{a,*}$,

$$\mathbb{E}_t|\varphi_t^a|^{2+\delta} \le C, \qquad \mathbb{E}_t|\zeta_{t,s_j}^a|^{2+\delta} + \mathbb{E}_{s_j}|\xi_{t,s_j}^a|^{2+\delta} \le C.$$

The target bound applies also when $\mathcal{J}^{a,*} = \varnothing$.

Positive site fractions keep pairwise and network sample sizes on the same asymptotic scale. The population design bounds allow singular empirical matrices when d exceeds the local sample size. The link and tail bounds control nuisance error; the score moments support the central limit theorem.

For the paired losses in Eqs. (4) and (5), set $n_{t,s_j} = n_t + n_{s_j}$. Effective sparsity $s_{\mathrm{eff},j,a}$ counts both calibrated supports and every correctly specified initial support, including intercepts. Use the population targets in Assumption S3 and penalties satisfying

$$\lambda_\alpha, \lambda_\gamma, \lambda_{\alpha,\mathrm{init}}, \lambda_{\gamma,\mathrm{init}} \asymp \sqrt{\log(2d)/n_{t,s_j}},$$

with the sufficiently large multiplier specified in Lemma S3. That lemma also treats bounded d at the same penalty order. The remaining requirements concern model correctness and sparsity.

Assumption 3 (Source models and sparsity)**.** For every $s_j \in \mathcal{J}^{a,*}$:

(a) $s_j \in \mathcal{S}_a^*$ and $\left(\bar{m}_{s_j}^a = m_t^{a,\star} \text{ or } \bar{q}_{s_j}^a = q_{s_j}^{a,\star}\right)$.

(b) $s_{\text{eff},j,a} \log(2d) = o(\sqrt{n_{t,s_j}})$.

A sparse true model alone need not yield a sparse calibrated limit for the misspecified model; Proposition S1 gives sufficient conditions. Initial and calibrated coefficients remain distinct. The predictor-margin condition places calibrated population predictors and true predictors of correctly specified initial models within $[-M, M]$, leaving room before clipping at $2M$.

Lemma S3 gives coefficient errors of order $s_{\text{eff},j,a}\lambda$ in ℓ_1 and $\sqrt{s_{\text{eff},j,a}}\lambda$ in ℓ_2, where $\lambda \asymp \sqrt{\log(2d)/n_{t,s_j}}$. Calibration makes the population remainder $O_p(s_{\text{eff},j,a}\lambda^2) = o_p(N^{-1/2})$, and outer cross-fitting controls the empirical error. Below, $\widehat{\mu}_{s_j}^a$ pools the candidate's outer-evaluation contributions.

Lemma 4 (Calibrated source expansion)**.** *For the fitting procedure and penalty levels in Section 4.1, under Assumptions 1 to 3, each $s_j \in \mathcal{J}^{a,*}$ satisfies*

$$\widehat{\mu}_{s_j}^a - \mu_t^a = n_t^{-1} \sum_{i \in \mathcal{D}_t} \zeta_{t,s_j,i}^a + n_{s_j}^{-1} \sum_{i \in \mathcal{D}_{s_j}} \xi_{t,s_j,i}^a + o_p(N^{-1/2}),$$

and its fitted site-score covariances are consistent.

The two leading terms describe variation in target predictions and source residuals. Independence across sites makes their variances additive, while shared target observations induce dependence across candidates. This is the structure used in Lemma 3.

The proof of Lemma 4 is given in Section S2.9. It uses the nuisance rates in Lemma S3 and the second-order bounds in Lemma S4, then establishes the cross-fitted expansion and fitted-score covariance consistency. The argument covers logistic, linear, and Poisson working means under the stated predictor and tail conditions.

Aggregation also needs a valid target-only reference in each arm. With a correct target outcome or logistic propensity model and the target regularity, sparsity, and penalty conditions of Proposition S2, calibrated target-only estimation satisfies

$$\widehat{\mu}_t^a - \mu_t^a = n_t^{-1} \sum_{i \in \mathcal{D}_t} \varphi_{t,i}^a + o_p(N^{-1/2}),$$

with consistent fitted-score covariances, including cross-arm covariance. Thus aggregation combines linear expansions evaluated at each candidate's calibrated limits, retaining the corresponding model-correctness conditions.

4.2 Joint Aggregation and Source Selection

Aggregation learns weights on the outer training observations used for calibration. Assumption 4 collects the moment conditions required by the aggregation argument. For the prescribed fitting procedure, Proposition S3 derives these conditions from the source and target candidate conditions in Section 4.1.

Assumption 4 (Training means and score covariances)**.** On each outer training sample, write $n = n_{-k_1}$. The notation $\widehat{\mu}_r^a(\bar{m}, \bar{e}, \bar{q})$ means the same candidate mean evaluated at calibrated population nuisance functions. Hatted scores replace these functions by fitted ones and center within each inner calibration fold. For each a:

(a) For $r \in \{t\} \cup \mathcal{J}^{a,*}$,

$$\sqrt{n}\,|\widehat{\mu}_r^a - \widehat{\mu}_r^a(\bar{m}, \bar{e}, \bar{q})| = o_p(1).$$

(b) For the target and each $s_j \in \mathcal{J}^{a,*}$, the empirical score errors satisfy

$$\widehat{\mathbb{E}}_{t,-k_1}(\widehat{\varphi}_t^a - \varphi_t^a)^2 = o_p(1),$$

$$\widehat{\mathbb{E}}_{t,-k_1}(\widehat{\zeta}_{t,s_j}^a - \zeta_{t,s_j}^a)^2 + \widehat{\mathbb{E}}_{s_j,-k_1}(\widehat{\xi}_{t,s_j}^a - \xi_{t,s_j}^a)^2 = o_p(1).$$

Part (a) makes valid training discrepancies $O_p(n^{-1/2})$; part (b) gives consistent covariance blocks, including shared-target and cross-arm terms. For the alternative with independent inner validation, Lemma S5 supplies the corresponding moment control.

The oracle benchmark minimizes the contrast variance over valid candidates. Let $L_\tau^*(\eta)$ be the limiting quadratic of $n\widehat{V}_\tau^{(-k_1)}(\eta)$ under the convergence conditions below, and define

$$\eta^* = \arg \min_{\eta:\, \eta^a_{s_j}=0 \text{ if } s_j \notin \mathcal{J}^{a,*}} L_\tau^*(\eta), \qquad V_\tau^* = L_\tau^*(\eta^*).$$

The oracle optimizes both arms jointly and may retain a source in only one arm. To recover it without knowing $\mathcal{J}^{a,*}$, the remaining candidates must be distinguishable from the target-only reference.

Assumption 5 (Source separation and aggregation curvature)**.** Uniformly over outer training samples:

(a) For all source arms, $\widehat{d}^a_{s_j} \to_p \bar{d}^a_{s_j}$, with

$$|\bar{d}^a_{s_j}| \geq b_0 > 0 \quad (s_j \notin \mathcal{J}^{a,*}), \qquad \sqrt{n}\,\widehat{s}^a_{s_j} \to_p s^{a,*}_{s_j} \in (0, \infty).$$

(b) In the quadratic representation

$$n\widehat{V}_\tau^{(-k_1)}(\eta) = \widehat{v}_n + 2\widehat{\boldsymbol{l}}_n^T\eta + \eta^T\widehat{Q}_n\eta,$$

the coefficients satisfy $(\widehat{v}_n, \widehat{\boldsymbol{l}}_n, \widehat{Q}_n) \to_p (v^*, \boldsymbol{l}^*, Q^*)$, with $\lambda_{\min}(Q^*) > 0$ and $V_\tau^* > 0$. The limits agree with the population score covariances on the valid subspace.

(c) $c_n \to \infty$ and $c_n/\sqrt{n} \to 0$.

Biased candidates need stable limits and nondegenerate discrepancy scales, without the valid candidates' calibrated expansions. Proposition S4 gives sufficient conditions for this stability. Positive quadratic curvature controls the unconstrained signed weights and makes the oracle unique.

The cutoff links these conditions to the penalty in Eq. (11). For a valid candidate, $\widehat{t}^{a}_{s_j} = O_p(1)$, so its penalty vanishes with probability tending to one as c_n grows. For a candidate with fixed nonzero bias, $\widehat{t}^{a}_{s_j}$ is of order $\sqrt{n}$, so $c_n/\sqrt{n} \to 0$ makes its penalty diverge. These two behaviors yield the following result.

Theorem 1 (Joint treatment-effect aggregation and inference)**.** *For the fitting procedure and penalty levels in Section 4.1, under Assumptions 1 to 5 and the target-only conditions of Proposition S2, the outer-fold weights satisfy* $\widehat{\eta}^{(-k_1)} \to_p \eta^*$ *and*

$$P\{\widehat{\eta}^{a,(-k_1)}_{s_j} = 0 \text{ for all } a = 0, 1,\ s_j \notin \mathcal{J}^{a,*}\} \to 1.$$

Moreover,

$$\sqrt{N}(\widehat{\tau}_t - \tau_t) \xrightarrow{d} \mathcal{N}(0, V^*_\tau), \qquad V^*_\tau \le L^*_\tau(0).$$

If $\mathcal{J}^{0,*} = \mathcal{J}^{1,*} = \varnothing$*, the aggregate equals the target-only contrast with probability tending to one.*

The proof of Theorem 1 in Section S3.5 first bounds the signed weights, then uses the optimality conditions to exclude biased arms and establish oracle-weight convergence. The $o_p(1)$ weight error then multiplies a held-out valid-candidate discrepancy of order $N^{-1/2}$, making the fitted-weight and oracle contrasts first-order equivalent. The calibrated expansions give the normal limit without an additional first-order weight-estimation term.

4.3 Variance Estimation and Efficiency

The limiting variance retains the site and cross-arm structure used to learn weights in Section 3.3. At the oracle weights, write the centered target contribution as $G^a_t = (1 - \sum_j \eta^{a,*}_{s_j})\varphi^a_t + \sum_j \eta^{a,*}_{s_j}\zeta^a_{t,s_j}$ and the source contribution as $G^a_{s_j} = \eta^{a,*}_{s_j}\xi^a_{t,s_j}$. Independence across sites gives

$$V^*_\tau = \sum_{r \in \{t, s_1, \ldots, s_K\}} \frac{\mathrm{Var}_r(G^1_r - G^0_r)}{\rho^{\mathrm{all}}_r}.$$

The within-site difference preserves cross-arm covariance. Whereas V_τ^* is on the $\sqrt{N}$ scale, the variance of $\widehat{\tau}_t$ is estimated using the held-out contributions from Eq. (12):

$$\widehat{V}_\tau^{\text{out}} = \sum_r \frac{\widehat{\text{Var}}_r(\widehat{G}_{r,i}^1 - \widehat{G}_{r,i}^0)}{n_r}, \qquad N\widehat{V}_\tau^{\text{out}} \to_p V_\tau^*. \tag{13}$$

Each empirical second central moment pools outer-evaluation contributions within its site. Variance consistency and the normal limit give the Wald interval $\widehat{\tau}_t \pm z_{1-\alpha/2}\sqrt{\widehat{V}_\tau^{\text{out}}}$.

Because $\eta = 0$ is feasible, the oracle contrast variance cannot exceed that of calibrated target-only estimation. Strict improvement requires a valid source direction that reduces this variance; compatibility alone does not ensure a gain. The comparison concerns the treatment contrast and retains cross-arm covariance, as in the joint optimization in Method.

The simulation and RHC analyses report the leading standard error $\sqrt{\widehat{V}_\tau^{\text{out}}}$ and its Wald interval. No additional weight-learning term is included in these primary intervals. The local sensitivity of learned weights to training moments is retained separately as an optional diagnostic, holding nuisance fits, tuning values, and selected active sets fixed. Section S3.6 distinguishes this diagnostic from the leading variance estimator covered by Eq. (13).

The theorem requires separated nonzero source biases and a growing cutoff; it does not establish uniform protection against biases shrinking on the sampling-error scale. The simulation study next examines precision and coverage for the implemented fixed-cutoff procedure, including weak departures from transportability.

5 Simulation Study

We examine precision and coverage for the TATE under working-model misspecification and source incompatibility. The design varies these two features separately: omitted interactions misspecify the working models, while a shift in one source treatment group violates transportability. Compatible sources provide a baseline for assessing the precision gained through borrowing.

5.1 Data Generation and Fitted Models

Each setting has 200 paired Monte Carlo replicates and $p = 200$ covariates, of which only the first four are active. The target covariates are independent standard normals truncated at ± 2.5 and rescaled to variance one. Write $z_\kappa = X_1(1 + \kappa X_2)/\sqrt{1 + \kappa^2}$. The target treatment logit is $h_A = 0.35 z_{\kappa_A} - 0.25X_2 + 0.125X_3 - 0.0625X_4$, and covariate shift follows $h_X = 0.25 z_{\kappa_A} - 0.20X_2 + 0.15X_3 - 0.10X_4$. Source covariates and treatment are drawn jointly according to $f^\star_{s_j} e^{a,\star}_{s_j} \propto f^\star_t \exp\{(j/K)h_X + (a - \frac{1}{2})h_A\}$, normalized over covariates and treatment arms. Outcomes are Bernoulli with

$$\operatorname{logit} P_r(Y = 1 \mid \boldsymbol{X}, A = a) = a - \tfrac{1}{2} + 0.25 z_{\kappa_Y} + 0.20X_2 + 0.10X_3 + 0.05X_4 + a\rho I(r = s_1).$$

At $\rho = 0$, all sources are transportable. Increasing ρ changes only source s_1's treated-arm outcome, leaving its control arm and all other sources compatible. We consider $\rho \in \{0, 0.5, 1, 1.5, 2\}$ and $K \in \{2, 4, 6, 8\}$ sources, with 1,000 observations per site and total sample size $N = 1000(K + 1)$. This design allows borrowing to differ by treatment arm. We expect compatible sources to improve precision and large source discrepancies to be easier to penalize than small discrepancies. Increasing K increases the total sample size and, when $\rho > 0$, decreases the fraction of sources with an incompatible arm. Numerical integration supplies the true target treatment effect.

Working nuisance models contain an intercept and all p main effects but omit X_1X_2. Scenarios 1–3 set (κ_Y, κ_A) to $(0, 0)$, $(0.75, 0)$, and $(0, 0.75)$, respectively. At $\rho = 0$ these represent both models correct, outcome misspecification only, and treatment/joint-weight misspecification only. Table S1 summarizes which interactions are generated and omitted from the fitted models. The scenarios separate misspecification of a working model from violation of the source transportability condition.

5.2 Implementation and Comparators

RoCE uses two-layer ten-fold cross-fitting, one-round target outcome initialization, 100 nuisance penalty values, a numerical predictor radius of 12, and the fixed aggregation cutoff $c = 2$. Its two arm-specific weight vectors are optimized jointly for the TATE. Benchmarks are calibrated target-only AIPW, sample-size (SS) and inverse-variance (IVW) aggregation of local arm means, density-ratio-weighted source AIPW combined with the target by inverse variance (Federated-DR), and density-ratio-weighted AIPW with common pooled fits (Pooled-DR) (Bang and Robins, 2005; Xiong et al., 2023). The SS and IVW benchmarks aggregate local arm means without transport adjustment; IVW includes a nonnegative between-site variance adjustment. Section S4 describes the roles of these comparators. The methods use the same generated data within each replicate.

We report root mean squared error (RMSE) and coverage of nominal 95% TATE intervals, with pointwise Monte Carlo intervals for RoCE. These comparisons evaluate complete procedures, so they do not isolate the contribution of any single RoCE component. The fixed-cutoff interpretation is given in Remark 1.

5.3 Results

With compatible sources ($\rho = 0$), RoCE reduces RMSE by approximately 35–70% relative to calibrated target-only estimation and gives coverage near the nominal level across the three scenarios (figures 2 and 3). Some nonadaptive comparators have slightly lower RMSE in these settings. The precision gain from borrowing is therefore most clearly distinguished from the additional protection needed when sources become incompatible.

The precision gains at $\rho = 0$ make source compatibility consequential: retaining a biased contribution can trade variance reduction for bias. Weak source deviations remain difficult to detect. At $\rho = 0.5$ and $K = 2$, RoCE coverage is 71.0–74.0% across the three scenarios,

and RMSE is 6.6–7.4% higher than target-only. Undercoverage persists for larger K at this deviation. These results show that finite-sample borrowing can retain enough source bias to outweigh a reduction in variance, even when one working model is correctly specified. Calibration of nuisance fits does not resolve uncertainty about source compatibility.

With larger deviations ($\rho \geq 1$), RoCE generally retains precision gains and near-nominal coverage, whereas nonadaptive borrowing can incur large errors and undercoverage. Larger discrepancies make the incompatible treated arm easier to penalize while compatible arms remain available. Gains tend to increase with K, alongside the larger total sample size and smaller incompatible-source fraction. Taken together, the results identify weak source deviations as the difficult borrowing regime. The pattern is consistent with the separation argument in the theory, but the fixed-cutoff experiments do not establish uniform inference as source biases approach zero.

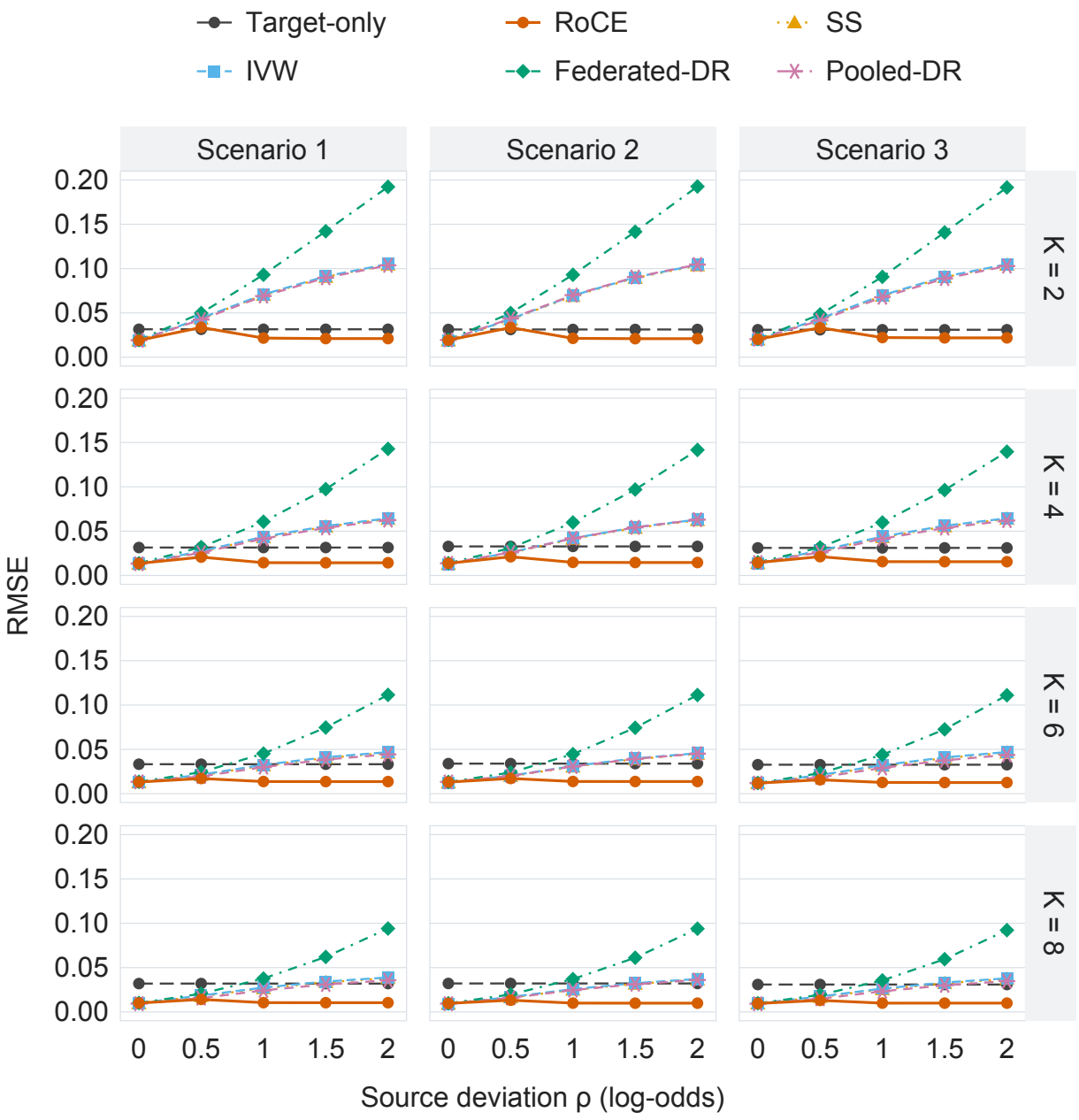


Figure 2: TATE RMSE in the joint-aggregation study at $p = 200$. Columns show Scenarios 1–3 and rows show the number of sources K. Orange bars give pointwise 95% Monte Carlo intervals for RoCE from 200 paired replicates. Target-only denotes calibrated target-site AIPW.

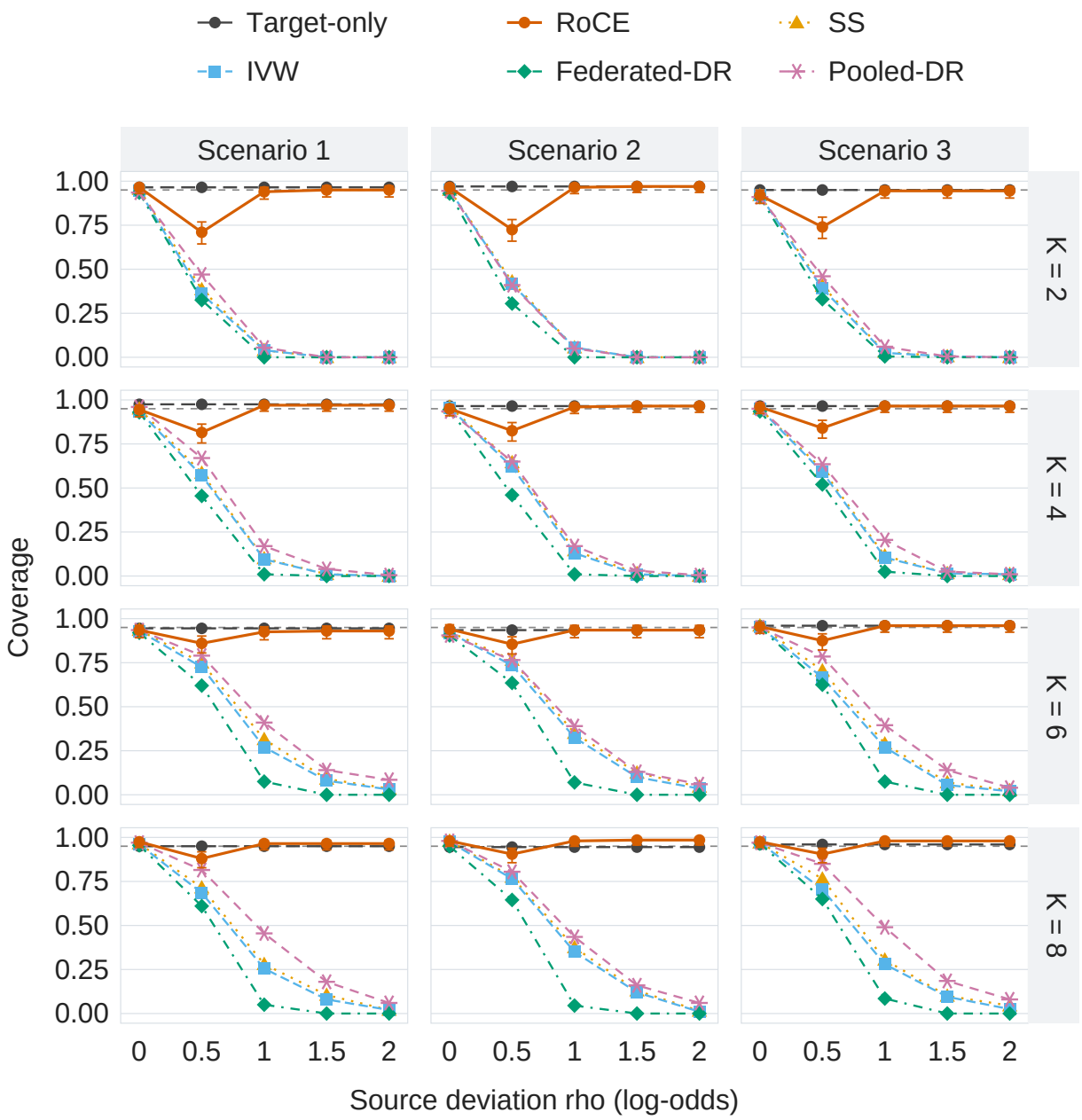


Figure 3: Coverage of nominal 95% TATE intervals in the same study. RoCE intervals use the leading site-centered variance without the optional weight-learning correction. The dashed line marks 0.95; orange bars give pointwise 95% Wilson Monte Carlo intervals for RoCE.

6 All of Us One-Year Observational Application

The simulations separate model misspecification from source incompatibility under known effects. In observational data, compatibility is unknown and must be considered alongside the study design. We examine one-year recorded dermatologic risk after initial exposure to a glucagon-like peptide-1 receptor agonist (GLP-1) or a dipeptidyl peptidase-4 inhibitor (DPP-4) among participants with type 2 diabetes. The statistical question is whether other geographic groups can improve precision for a designated target group while preserving that group's treatment contrast. The analysis uses electronic health records from the All of Us Research Program Controlled Tier version 8, available through the Researcher Workbench (All of Us Research Program Investigators, 2019; All of Us Research Program, 2026). Geographic groups emulate distributed sites within one program, allowing the borrowing procedure to be illustrated on a common data resource.

6.1 Cohort and Target Contrast

Treatment denotes initial recorded exposure to GLP-1 rather than DPP-4, with index defined by the earlier first recorded exposure. We require a type 2 diabetes diagnosis by index and exclude participants with a prior diagnosis from the prespecified composite dermatologic outcome set. Subsequent exposure to the other class does not change the initial assignment. The outcome is a recorded composite dermatologic diagnosis within one year; Section S5 provides the detailed exposure and cohort rules. [To be completed: specify the diseases or diagnostic categories included in the composite dermatologic outcome and provide the corresponding code definitions in the supplement.]

The composite includes hidradenitis suppurativa, inflammatory dermatosis, psoriasis, cutaneous lupus erythematosus, eczema, urticaria, bullous pemphigoid, pemphigus vulgaris,

and alopecia areata, as represented by the selected concepts and their descendants. The concept identifiers and composite outcome definition are provided in Section S5.

An event recorded in the first year establishes a positive outcome; in the absence of an event, a negative outcome requires available follow-up through one year. We use the last condition-occurrence date as a follow-up proxy and exclude unclassifiable outcomes without censoring weighting. Of 15,339 baseline participants with geographic information, 2,074 had an unclassifiable one-year outcome, leaving 13,265 evaluable participants.

The first ZIP-code digit defines Northeast (0–1), South (2, 3, 7), Midwest (4–6), and West (8–9) groups. All four meet the minimum of 25 participants per treatment arm after exclusions. We designate the largest group, Northeast, as the target for this methodological illustration ($n_t = 5{,}040$): 1,082 initially received GLP-1, with 90 recorded events, and 3,958 received DPP-4, with 277 events. The other regions serve as sources. Their treatment-arm sizes and recorded event counts are given in table S3, together with the cohort-definition details in Section S5. The target is held fixed across all methods; changing it would change the estimand.

For this evaluable target cohort, $A = 1$ denotes GLP-1 and $A = 0$ denotes DPP-4. The adjusted risk difference targets $\tau_t = \mu_t^1 - \mu_t^0$, displayed in figure 4 as $100\tau_t$ percentage points. Positive values indicate higher recorded one-year risk under GLP-1.

6.2 Adjustment and Estimation

Models adjust for age, ethnicity, gender, race, sex at birth, and pre-index diagnosis-derived PheCode counts. Counts distinguish the year before index from earlier history; concepts defining diabetes and the dermatologic outcome are excluded. A 5% prevalence screen among participants with mapped PheCodes yields 154 count features among 181 model features. This screen preceded splitting; centering and scaling use outer training samples. RoCE uses five outer folds, one-round source calibration, joint TATE aggregation, numerical predictor

radii of 3 at sources and 5 at the target, and a fixed cutoff of 2. The implementation uses the finite-sample tuning convention described in Remark 1; Section S3.6 gives the RoCE standard-error calculation.

The 181 features provide a broad adjustment set whose relevance can differ across treatment and outcome models. The geographic groups also differ in treatment composition and recorded event frequencies (table S3); these marginal differences motivate transport adjustment but do not establish conditional incompatibility. The same six methods used in the simulation comparison are reported. The target-only estimate supplies a within-population reference, and the borrowing estimators examine the precision available from the other geographic groups under their respective modeling and aggregation choices.

6.3 Results and Interpretation

The supplied All of Us summary gives a RoCE estimate of a 0.79 percentage-point difference (95% CI $[-0.62, 2.20]$), compared with 0.61 points (95% CI $[-1.45, 2.67]$) for target-only estimation. The supplied RoCE interval is about 32% narrower, but all six intervals include zero (figure 4). The analysis therefore illustrates a precision gain from borrowing without resolving the direction of the treatment contrast. Other methods use different borrowing assumptions, so interval width alone cannot establish their bias or coverage.

Reporting update pending: the All of Us intervals above reproduce the original supplied summaries. A matching leading-variance export for this same analysis is required to align them with the simulation and RHC reporting convention; the different Northeast diagnostic analysis must not be substituted.

Interpretation depends on the exposure definition, measured confounding, and outcome ascertainment. Restricting to classifiable one-year outcomes limits the analysis to the evaluable cohort; extending it to the baseline cohort would require additional assumptions or adjustment for follow-up selection. The condition-occurrence proxy does not establish

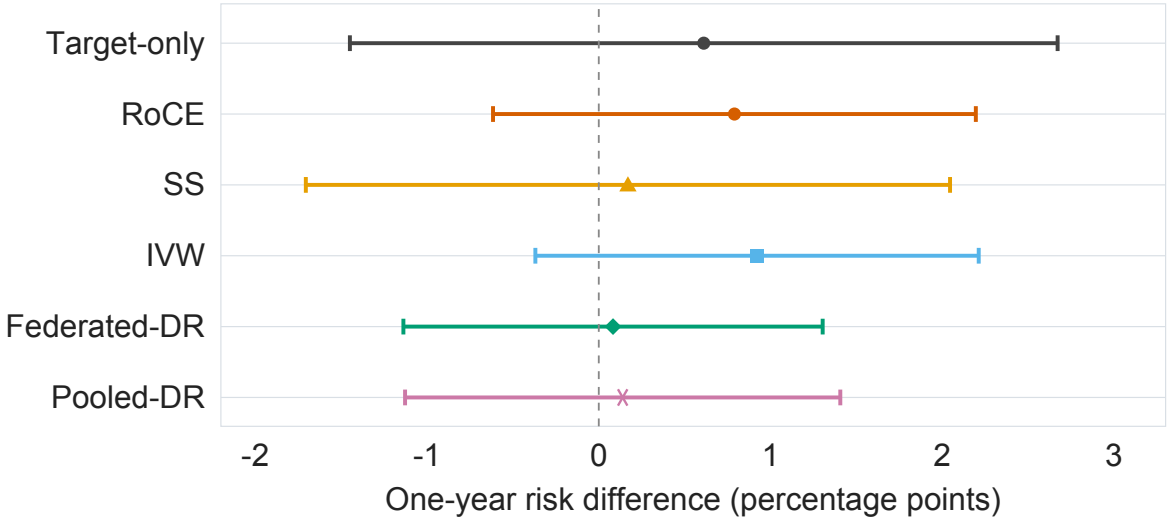


Figure 4: All of Us one-year observational analysis in the evaluable target cohort. Points show adjusted GLP-1 minus DPP-4 risk differences; horizontal bars reproduce the supplied 95% confidence intervals. Their variance-convention update is pending, as noted in the text. Estimates and limits are in percentage points. Method colors and symbols match the supplementary RHC figure; the dashed vertical line marks zero. No censoring weights were used.

continuous observation, and a recorded diagnosis need not coincide with biological onset. These features of the study design are separate from the statistical adjustment for population shift. The supplementary RHC analysis provides a second illustration with a different outcome and target definition (Section S6).

7 Discussion

Target-specific borrowing involves two statistical decisions: how to adjust each source treatment group and how much information to borrow from it. RoCE links these decisions through a source-assisted estimating equation. A joint weight combines treatment and population adjustment, making paired high-dimensional calibration possible. The calibrated candidates then enter an aggregation problem that targets the treatment contrast and retains shared-target and cross-arm covariance. With target initialization, the required fitting tasks can be batched into one communication round.

The separation between calibration and aggregation clarifies the scope of robustness. Calibration controls the first-order effect of nuisance estimation under either correct working model, given transportability and the stated regularity. Aggregation addresses discrepancies that remain because a source outcome relationship is incompatible. Its arm-specific form can preserve useful control outcomes from a source whose treated outcomes are shifted. The oracle variance comparison formalizes the precision available from eligible contributions relative to calibrated target-only estimation; the training-moment and separation conditions are essential to that conclusion.

Weak source deviations are the principal finite-sample difficulty in the reported simulations. A small discrepancy can be hard to distinguish from sampling variation while still contributing enough bias to impair coverage after borrowing. This explains why favorable performance with compatible or clearly incompatible sources does not extend automatically

to intermediate settings. The growing-cutoff theorem concerns separated biases, whereas the simulation and RHC analyses use a fixed cutoff and the leading standard error. The optional local weight-sensitivity diagnostic does not resolve source-selection bias. Inference under local source biases or unstable selection remains an open problem.

The All of Us and RHC analyses illustrate the resulting precision gains for geographic and insurance-defined target populations, respectively. The RHC leading-variance interval includes zero; the All of Us interval convention remains to be reconciled with its supplied summaries. Their substantive interpretation depends on measured confounding, exposure definition, and outcome ascertainment as well as borrowing. In All of Us, restricting to classifiable one-year outcomes makes follow-up selection part of the study population definition. Moreover, both examples emulate distributed sites within a common study; deployment across institutions also requires compatible feature definitions and an explicit accounting of preprocessing communication.

Several extensions follow from these limitations. Growing numbers of sources require simultaneous control of discrepancies and covariance estimation. Weaker sparsity conditions would accommodate broader clinical feature sets, and follow-up adjustment would extend the analysis to censored outcomes. Other effect measures, including parameters of marginal structural models, require their own transport scores and variance objectives. These extensions preserve the central distinction between fitting a reliable source contribution and deciding whether it is informative for the target population.

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Supplementary Material for Robust Federated Causal Estimation (RoCE) through Calibrated High-Dimensional Models

Contents and Reading Guide

The proofs follow the main text's candidate-estimation and joint-aggregation arguments. The roadmap in Section S2 links each main result to its proof; Section S7 collects alternative constructions.

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S1 Method and Implementation Details

We give the fitting, covariance, and communication details for Section 3. Initial fits supply fixed calibration multipliers; calibrated fitting and aggregation-weight learning both exclude the outer evaluation fold.

S1.1 Source Initialization and Fold Updates

We implement the source calibration losses in Section 3.2 using the primary two-layer algorithm in Section 3.4. Let $K_f \geq 3$ and use its sitewise folds $\mathcal{I}_{r,k_1}$, sizes $n_{r,J}$, and averages $\widehat{\mathbb{E}}_{r,J}$. Outer fold k_1 supplies final evaluation; inner fold $k_2 \neq k_1$ supplies a calibration loss block. Initial fitting excludes both folds. The final calibrated fits pool the inner blocks, and their fitted training scores are reused for aggregation-weight learning. Independent inner validation is a separate construction in Section S7.

Write $L(Y,u) = -Yu + \Psi(u)$, where $\Psi' = g$. Following Tan (2020), RML denotes regularized initial outcome fitting, RCAL calibrated weighting, and RWL weighted outcome fitting; IRCAL denotes initial joint-weight fitting. The subscript init marks initial coefficients; calibrated fits omit it. The outcome initialization uses the target arm and is shared across sources. Each source fits its initial joint weight using local observations and a target feature mean. The fold-specific initial fits are

$$\widehat{\alpha}_{t,a,\text{init}}^{(-k_1,-k_2)} \in \arg\min_{\alpha} \left\{ \widehat{\mathbb{E}}_{t,-k_1,-k_2}[I(A=a)L(Y,\phi^T\alpha)] + \lambda_{\alpha,\text{init}}\|\alpha\|_{1,-0} \right\},$$

$$\widehat{\gamma}_{s_j,a,\text{init}}^{(-k_1,-k_2)} \in \arg\min_{\gamma} \left\{ \widehat{\mathbb{E}}_{t,-k_1,-k_2}(\phi)^T\gamma + \widehat{\mathbb{E}}_{s_j,-k_1,-k_2}[I(A=a)e^{-\phi^T\gamma}] + \lambda_{\gamma,\text{init}}\|\gamma\|_{1,-0} \right\}.$$

Within each inner fold k_2, temporarily suppress $(-k_1,-k_2)$ on the initial coefficients and define $d_{k_2}^a(\boldsymbol{X}) = g'\{\mathcal{T}_M(\phi^T\widehat{\alpha}_{t,a,\text{init}})\}$ and $\omega_{j,k_2}^a(\boldsymbol{X}) = e^{-\mathcal{T}_M(\phi^T\widehat{\gamma}_{s_j,a,\text{init}})}$. The source's calibrated

coefficients minimize the sum of losses evaluated on the inner folds, with each site's fold contributions normalized by its own outer-training sample size:

$$\widehat{\gamma}_{s_j,a}^{(-k_1)} \in \arg\min_{\gamma} \left\{ \sum_{k_2 \neq k_1} \frac{n_{t,k_2}}{n_{t,-k_1}} \widehat{\mathbb{E}}_{t,k_2}(d_{k_2}^a \phi)^T \gamma \right.$$
$$\left. + \sum_{k_2 \neq k_1} \frac{n_{s_j,k_2}}{n_{s_j,-k_1}} \widehat{\mathbb{E}}_{s_j,k_2}\{I(A=a) e^{-\phi^T \gamma} d_{k_2}^a\} + \lambda_\gamma \|\gamma\|_{1,-0} \right\},$$
$$\widehat{\alpha}_{t,s_j}^{a,(-k_1)} \in \arg\min_{\alpha} \left\{ \sum_{k_2 \neq k_1} \frac{n_{s_j,k_2}}{n_{s_j,-k_1}} \widehat{\mathbb{E}}_{s_j,k_2}\{I(A=a)\omega_{j,k_2}^a L(Y, \phi^T \alpha)\} + \lambda_\alpha \|\alpha\|_{1,-0} \right\}.$$

The target-only fits use the same inner folds: each initial propensity fit excludes both k_1 and k_2, and each calibrated target update minimizes the fold-size-weighted loss over all $k_2 \neq k_1$; see Section S1.2. Each source initial fit also excludes the fold on which its plug-in is used. Each calibrated update incurs its slope penalty once.

Section S1.3 specifies the covariance summaries from these fits; Section S1.4 describes weight learning and outer evaluation.

S1.2 Target-Only Calibration

We specify the calibrated target-only reference in Eq. (7). Fix an arm $a \in \{0, 1\}$. Retain the main text's outcome function $m(\boldsymbol{x}; \alpha) = g\{\phi(\boldsymbol{x})^T \alpha\}$ and use the logistic propensity model $e_t^a(\boldsymbol{x}; \beta) = \text{expit}\{\phi(\boldsymbol{x})^T \beta\}$. The target-only AIPW contribution is

$$h_0(D; \alpha, \beta) = m(\boldsymbol{X}; \alpha) + \frac{I(A=a)}{e_t^a(\boldsymbol{X}; \beta)}\{Y - m(\boldsymbol{X}; \alpha)\}.$$

Its population mean is $\mu_0(\alpha, \beta) = \mathbb{E}_t\{h_0(D; \alpha, \beta)\}$. The subscript 0 identifies the target-only reference. This smooth score is used in the proof; the implemented final score uses σ_M for the propensity, which is asymptotically equivalent under the established margin and ℓ_1 rate.

Let $(\bar{\alpha}_0, \bar{\beta}_0)$ be the calibrated population coefficients, with $\bar{m}_t^a(\boldsymbol{x}) = m(\boldsymbol{x}; \bar{\alpha}_0)$ and $\bar{e}_t^a(\boldsymbol{x}) = e_t^a(\boldsymbol{x}; \bar{\beta}_0)$. If the outcome model is correct, $\bar{\alpha}_0 = \alpha^\star$ and the target propensity loss with initial $\alpha^\star$ defines $\bar{\beta}_0$. If the propensity model is correct, $\bar{\beta}_0 = \beta^\star$ and the target outcome loss with initial $\beta^\star$ defines $\bar{\alpha}_0$.

Following regularized calibrated target estimation (Tan, 2020; Smucler et al., 2019; Hou et al., 2025), write $u_\alpha = \phi^T\alpha$ and $u_\beta = \phi^T\beta$. Given initial target fits $\widehat{\alpha}_{t,a,\text{init}}$ and $\widehat{\beta}_{t,a,\text{init}}$, the calibrated losses are

$$\begin{aligned}\ell_{\text{RCAL},t}(\beta;\widehat{\alpha}_{t,a,\text{init}}) &= \widehat{\mathbb{E}}_t\left[g'\{\mathcal{T}_M(u_{\widehat{\alpha}_{t,a,\text{init}}})\}\{I(A\neq a)u_\beta + I(A=a)e^{-u_\beta}\}\right] + \lambda_\beta\|\beta\|_{1,-0},\\ \ell_{\text{RWL},t}(\alpha;\widehat{\beta}_{t,a,\text{init}}) &= \widehat{\mathbb{E}}_t\left[I(A=a)e^{-\mathcal{T}_M(u_{\widehat{\beta}_{t,a,\text{init}}})}\{-Yu_\alpha + \Psi(u_\alpha)\}\right] + \lambda_\alpha\|\alpha\|_{1,-0}.\end{aligned} \tag{S1}$$

The exponential multiplier in the outcome loss is propensity odds; the final target AIPW score uses inverse propensity. Target calibration is local and requires no intersite messages.

For the primary two-layer construction, let k_1 be the outer evaluation fold and $k_2 \neq k_1$ an inner calibration fold, with target fold notation as in the main text. The initial target outcome fit uses target arm a outside both folds. The initial propensity fit uses all target observations outside both folds with response $I(A=a)$ and logistic loss $L_{\log}(B,u) = -Bu + \log(1+e^u)$:

$$\widehat{\beta}^{(-k_1,-k_2)}_{t,a,\text{init}} \in \arg\min_\beta \left\{\widehat{\mathbb{E}}_{t,-k_1,-k_2}[L_{\log}\{I(A=a),u_\beta\}] + \lambda_{\beta,\text{init}}\|\beta\|_{1,-0}\right\}.$$

For each k_2, suppress $(-k_1,-k_2)$ on the initial fits and put $d^a_{k_2} = g'\{\mathcal{T}_M(u_{\widehat{\alpha}_{t,a,\text{init}}})\}$ and $\omega^a_{t,k_2} = e^{-\mathcal{T}_M(u_{\widehat{\beta}_{t,a,\text{init}}})}$. The calibrated target fits use only target inner folds and weight each loss block by its share of the outer-training sample:

$$\begin{aligned}\widehat{\beta}^{a,(-k_1)}_t &\in \arg\min_\beta \left\{\sum_{k_2\neq k_1}\frac{n_{t,k_2}}{n_{t,-k_1}}\widehat{\mathbb{E}}_{t,k_2}\left[d^a_{k_2}\{I(A\neq a)u_\beta + I(A=a)e^{-u_\beta}\}\right] + \lambda_\beta\|\beta\|_{1,-0}\right\},\\ \widehat{\alpha}^{a,(-k_1)}_t &\in \arg\min_\alpha \left\{\sum_{k_2\neq k_1}\frac{n_{t,k_2}}{n_{t,-k_1}}\widehat{\mathbb{E}}_{t,k_2}\{I(A=a)\omega^a_{t,k_2}L(Y,u_\alpha)\} + \lambda_\alpha\|\alpha\|_{1,-0}\right\}.\end{aligned}$$

Each plug-in is fitted outside its own calibration fold, and each penalty enters the final update once.

For candidate target functions m, e, consistency and exchangeability give

$$\mathbb{E}_t\{m + I(A=a)(Y-m)/e\} - \mu^a_t = \mathbb{E}_t\left[\frac{e-e^{a,\star}_t}{e}(m-m^{a,\star}_t)\right].$$

Thus identification holds when $m = m_t^{a,\star}$ or $e = e_t^{a,\star}$. The target expansion and its proof are given in Proposition S2. The same target score also supplies the covariance reference for joint aggregation, as detailed next.

S1.3 Covariance for Joint Aggregation

For the primary joint procedure, fix an outer evaluation fold k_1 and use the training counts $n_{r,-k_1}$ from the main text. Suppress $(-k_1)$ on the fitted functions. The target-prediction contribution is $\widehat{h}^a_{t,j,i} = \widehat{m}^a_{s_j}(\boldsymbol{X}_i)$; the target-only and source-residual contributions are

$$\begin{aligned} \widehat{h}^a_{0,i} &= \widehat{m}^a_t(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\widehat{e}^a_t(\boldsymbol{X}_i)}\{Y_i - \widehat{m}^a_t(\boldsymbol{X}_i)\}, \\ \widehat{h}^a_{s_j,j,i} &= I(A_i = a)\widehat{q}^a_{s_j}(\boldsymbol{X}_i)\{Y_i - \widehat{m}^a_{s_j}(\boldsymbol{X}_i)\}. \end{aligned} \tag{S2}$$

These are the contributions underlying the candidate means in Eq. (7) and Eq. (6). Unlike the independent-validation alternative in Section S7.1, the primary procedure evaluates them on the outer training observations used for calibration. Its training-moment requirements are stated in Assumption 4.

Center each contribution within its inner calibration fold. Let $\widehat{B}^{ab}$ be the pooled cross-covariance of the target vectors $(\widehat{h}^a_{0,i}, \widehat{h}^a_{t,1,i}, \ldots, \widehat{h}^a_{t,K,i})^T$ and their arm-b counterparts, with divisor $n_{t,-k_1}$. Let $\widehat{V}^{ab}_{s_j}$ be the corresponding pooled covariance between $\widehat{h}^a_{s_j,j,i}$ and $\widehat{h}^b_{s_j,j,i}$, with divisor $n_{s_j,-k_1}$. For candidate order $(t, s_1, \ldots, s_K)$, the covariance block is

$$\widehat{\Sigma}^{ab} = \frac{\widehat{B}^{ab}}{n_{t,-k_1}} + \operatorname{diag}\left(0, \frac{\widehat{V}^{ab}_{s_1}}{n_{s_1,-k_1}}, \ldots, \frac{\widehat{V}^{ab}_{s_K}}{n_{s_K,-k_1}}\right).$$

The diagonal source term follows from independence across sources; the target block retains covariance between all candidate predictions. Writing $b(\eta^a) = (1 - \sum_j \eta^a_{s_j}, (\eta^a)^T)^T$, the variance in Eq. (10) is equivalently

$$\widehat{V}^{(-k_1)}_\tau(\eta) = b(\eta^1)^T\widehat{\Sigma}^{11}b(\eta^1) + b(\eta^0)^T\widehat{\Sigma}^{00}b(\eta^0) - 2b(\eta^1)^T\widehat{\Sigma}^{10}b(\eta^0).$$

In particular, the discrepancy standard error satisfies $(\widehat{s}^a_{s_j})^2 = \widehat{\Sigma}^{aa}_{jj} + \widehat{\Sigma}^{aa}_{00} - 2\widehat{\Sigma}^{aa}_{0j}$. Candidate means use the pooled site averages before discrepancies and penalties are computed. The

full stacked two-arm covariance matrix is positive semidefinite when assembled from these common sitewise score matrices.

Disjoint arm indicators do not imply zero covariance after centering. At a source, the two uncentered residual contributions have product zero, so their covariance is minus the product of their means; it vanishes only when at least one residual mean is zero. At the target, even the true AIPW scores have cross-arm covariance $\mathrm{Cov}_t\{m_t^{1,\star}(\boldsymbol{X}), m_t^{0,\star}(\boldsymbol{X})\}$. Both facts explain why the joint objective retains cross-arm moments.

An equivalent quadratic implementation stacks the treated-arm discrepancies and the negatives of the control-arm discrepancies:

$$(\widehat{d}^1_{s_1}, \ldots, \widehat{d}^1_{s_K}, -\widehat{d}^0_{s_1}, \ldots, -\widehat{d}^0_{s_K})^T.$$

Its joint variance is the target-only contrast variance, plus twice the covariance of that contrast with the weighted increments, plus the variance of the weighted increments. Multiplication by $n = n_{-k_1}$ gives the quadratic coefficients in Assumption 5; the minus signs on control-arm increments are essential.

S1.4 Outer Evaluation and Batched Communication

The source and target fits above supply the candidate means in Eq. (6) and Eq. (7). For each outer fold k_1, use only its complement to compute those means and the covariance blocks in Section S1.3. Pool the training moments within each site before forming discrepancies and solving the joint weight objective in Eq. (11). The calibrated models and learned weights then determine the held-out site contributions in Eq. (9). Pooling those contributions with each site's full sample size gives Eq. (12). This order separates final evaluation from fitting and weight learning.

The inner folds separate each initial fit from the observations contributing its calibration loss. The final calibrated fits nevertheless use the outer training sample on which aggregation

weights are learned. The training-score control stated in Assumption 4 is verified by Proposition S3. The independent-inner-validation alternative is developed separately in Section S7.1.

With target outcome initialization, the fitting tasks for all folds can be declared in advance and batched into one target broadcast and one source return. The target sends initial outcome coefficients, initial-training feature means, and calibration-fold derivative moments. Each source returns calibrated outcome coefficients, residual means, second moments, cross-arm products, and counts for the specified training and evaluation tasks. The target computes prediction covariances and joint aggregation weights locally; evaluation summaries do not enter weight learning. This is the protocol in figure 1. Its first- and second-moment messages support point estimation and the leading variance. Additional products for the optional local weight-sensitivity diagnostic, which is not included in the primary standard errors, are described in Section S3.6.

S1.5 Tuning and Numerical Stabilization

For the sufficient theoretical regime below, use a common feature basis, intercept convention, and clipping radius across folds and sites. The intercept is unpenalized; all slope coefficients receive an ℓ_1 penalty. Validation, if used, is confined to the corresponding training sample and matches the fitted loss. The inference theorem additionally requires the high-probability gradient event and restricted curvature below; it does not assert that a validation-selected penalty automatically satisfies them.

We use the clipping convention from Section 3.2: $\mathcal{T}_M(u) = \operatorname{sign}(u)\min(|u|, 2M)$ and $\sigma_M(u) = \operatorname{expit}\{\mathcal{T}_M(u)\}$. The population bound is M and the numerical predictor radius is $2M$, with $M > 0$ fixed. The following prescriptions specify where clipping enters the fitting sequence.

(a) In the source joint-weight loss, use the identical factor $g'\{\mathcal{T}_M(\phi^T\widehat{\alpha}_{t,a,\text{init}})\}$ for the target linear moment and the source exponential term.

(b) In the source outcome loss, use $I(A=a)\exp\{-\mathcal{T}_M(\phi^T\widehat{\gamma}_{s_j,a,\text{init}})\}$. In the target outcome loss, use $I(A=a)\exp\{-\mathcal{T}_M(\phi^T\widehat{\beta}_{t,a,\text{init}})\}$.

(c) In the target propensity loss, the single factor $g'\{\mathcal{T}_M(\phi^T\widehat{\alpha}_{t,a,\text{init}})\}$ multiplies the entire bracket $(1-I(A=a))\phi^T\beta + I(A=a)\exp(-\phi^T\beta)$.

(d) Evaluate final outcome means as $g(\phi^T\widehat{\alpha})$, final source weights as $\exp(-\phi^T\widehat{\gamma})$, and calibrated target propensities as $\sigma_M(\phi^T\widehat{\beta}_t^a)$. Thus target probabilities lie in $[\text{expit}(-2M), \text{expit}(2M)]$. No additional probability clipping is imposed.

The final target clipping is asymptotically inactive under the margin and ℓ_1 consistency conditions below, including when the final propensity model is misspecified. Initial clipping need not be inactive for a misspecified model. This is the role of fixed predictor clipping in the casewise proof; an arbitrarily selected radius alone does not establish overlap or inference. The construction uses the clipping principle of Hou et al. (2025, eqs. (11)–(13), Assumption 4); the transported source score requires its own two-sample balance argument.

For a sample of total size n used to learn aggregation weights, $c_n = \sqrt{\log(n \vee 3)}$ is one choice satisfying the theorem's cutoff conditions and gives penalty $(\widehat{t}_j/c_n - 1)_+$. Here n excludes the final evaluation fold. The joint covariance is assembled from target and source empirical covariance blocks; a ridge, if used, must vanish asymptotically. If an unscaled discrepancy variance is floored, $n\epsilon_{V,n} \to 0$ is sufficient to preserve its stated limiting scale. The numerical analyses use a fixed Wald cutoff of 2 and the application-specific predictor radii reported in the main text, which can differ across sites. These settings, including the fixed variance floor in Section S3.6, are distinct from the common-radius, growing-cutoff sufficient regime of the theory.

S2 Calibrated Candidate Expansions

This section proves source identification and calibration from Section 3.1 and Section 3.2, followed by the source expansion in Section 4.1. The argument proceeds from population targets and orthogonality to nuisance rates, second-order remainders, and cross-fitted inference.

Main-text results are restated under their original numbers before their proofs. Only additional supplementary results receive S-numbers. The table gives the proof locations and ingredients.

Main result	Proof ingredients	Proof page
Lemma 1	Direct identification argument in Section S2.3	54
Lemma 2	Population stationarity in Section S2.5	57
Lemma 3	Sitewise covariance decomposition in Section S1.3	73
Lemma 4	Lemmas S3 and S4: nuisance rates and second-order bounds	67
Theorem 1	Source and target expansions, stable biased candidates, and Proposition S3	79

S2.1 Setup and Notation

Fix a source s, target t, and arm $a \in \{0, 1\}$. As in the main text, n_r is the site sample size, $N = \sum_r n_r$ is the full network size, and $n_r/N \to \rho_r^{\text{all}} > 0$. The pair size $n_{t,s} = n_t + n_s$ and all fixed-fold fitting sample sizes are therefore proportional to N.

For coefficient vectors $\alpha, \gamma \in \mathbb{R}^d$, the two working functions are

$$m(\boldsymbol{x}; \alpha) = g\{\phi(\boldsymbol{x})^T \alpha\}, \qquad q_s^a(\boldsymbol{x}; \gamma) = \exp\{-\phi(\boldsymbol{x})^T \gamma\}.$$

The first is an outcome mean; the second is a joint weight for source observations in arm a. The common feature vector $\phi(\boldsymbol{x})$ includes an intercept, while p denotes the raw covariate dimension. The ideal joint weight is

$$q_s^{a,\star}(\boldsymbol{x}) = \frac{f_t^\star(\boldsymbol{x})}{f_s^\star(\boldsymbol{x}) e_s^{a,\star}(\boldsymbol{x})}.$$

Stack the coefficients as $\vartheta = (\alpha^T, \gamma^T)^T \in \mathbb{R}^{2d}$. For a single record $D = (R, \boldsymbol{X}, A, Y)$, the target prediction is $h_t(D; \vartheta) = m(\boldsymbol{X}; \alpha)$ and the source residual contribution is

$$h_s(D; \vartheta) = I(A = a) q_s^a(\boldsymbol{X}; \gamma)\{Y - m(\boldsymbol{X}; \alpha)\}.$$

The subscript indicates the site at which each contribution is averaged. Thus h_t contains only the outcome prediction; the target-only AIPW contribution is h_0 from Section S1.2.

Their population mean is the same source-assisted mean defined in Eq. (2), with the source and arm fixed here:

$$\mu_s^a(\vartheta) = \mu_s^a(\alpha, \gamma) = \mathbb{E}_t\{h_t(D; \vartheta)\} + \mathbb{E}_s\{h_s(D; \vartheta)\}.$$

The stacked argument is used for joint Taylor expansions; the population mean retains the source and arm indices of the main text.

Let $\bar{\vartheta} = (\bar{\alpha}^T, \bar{\gamma}^T)^T$ collect the calibrated population coefficients. The corresponding functions are $\bar{m}_s^a(\boldsymbol{x}) = m(\boldsymbol{x}; \bar{\alpha})$ and $\bar{q}_s^a(\boldsymbol{x}) = q_s^a(\boldsymbol{x}; \bar{\gamma})$, as in Section 4.1. Their source-assisted population mean is $\mu_s^a(\bar{\vartheta})$. The subscript init also identifies initial population minimizers and support sizes. The two correct-model cases defining the calibrated limits are specified in Assumption S3.

For the asymptotic expansion, center each contribution under its own site's distribution:

$$\zeta_{t,s}^a = h_t(D; \bar{\vartheta}) - \mathbb{E}_t\{h_t(D; \bar{\vartheta})\}, \qquad \xi_{t,s}^a = h_s(D; \bar{\vartheta}) - \mathbb{E}_s\{h_s(D; \bar{\vartheta})\}.$$

These are the scores in Section 4.1. We write $\widehat{\mu}_s^a$ for the cross-fitted source-assisted mean. Its variance on the main text's $\sqrt{N}$ scale is $V_{t,s}^{a,*} = (\rho_t^{\text{all}})^{-1} \operatorname{Var}_t(\zeta_{t,s}^a) + (\rho_s^{\text{all}})^{-1} \operatorname{Var}_s(\xi_{t,s}^a)$. Finally, write s_c, s_w for the support sizes of the correct and other calibrated coefficients, including the intercept even when it is zero, and $s_{c,\text{init}}$ for the correct initial support size. Put $s_{\text{eff}} = s_c + s_w + s_{c,\text{init}}$. When both models are correct, include both initial support sizes; this only changes a fixed factor when the initial and calibrated models share the same basis. Coefficient norms are denoted by $\|\cdot\|_q$ and function norms by $\|\cdot\|_{L_2(P_r)}$.

S2.2 Assumptions

We retain the causal identification conditions in Assumption 1. The first two assumptions below give the population-design, tail, overlap, and predictor-margin conditions used in Assumption 2. The third specifies the correct-model branches and population supports underlying Assumption 3; its sparsity rate is imposed when the inference result is applied.

The later oracle-bound event in Assumption S4 is verified from these primitive conditions by Lemma S3. Training moments and source separation are treated in Section S3.

Assumption S1 (Sampling and population design)**.** Observations are i.i.d. within sites and independent across sites. Site fractions have positive limits, and the numbers of sites and all cross-fitting folds are fixed. Fold memberships are balanced and chosen independently of the observed records, as in Section 3.4. At each site used for fitting, $\|\phi\|_\infty \le C_\phi$. For each fitted arm, let $G_{r,a} = \mathbb{E}_r\{I(A=a)\phi\phi^T\}$ and assume

$$\lambda_{\min}(G_{r,a}) \ge \kappa_G > 0, \qquad \lambda_{\max}\{\mathbb{E}_r(\phi\phi^T)\} \le C_\Sigma. \tag{S3}$$

Constants do not depend on d, N. The common, prespecified feature basis includes an intercept and has no exact redundant columns. The lower bound is a condition on population matrices; empirical matrices may be singular when d exceeds the sample size.

Assumption S2 (Link, tails, overlap, and predictor margins)**.** Use the observed conditional means $m_r^{a,\star}(\boldsymbol{x}) = \mathbb{E}_r(Y \mid \boldsymbol{X} = \boldsymbol{x}, A = a)$ from Section 3.1.

(a) The outcome link $g = \Psi'$ is twice continuously differentiable on $\mathbb{R}$, with $g' > 0$. For a fixed $M > 0$, on $[-2M, 2M]$,

$$0 < m_g \le g'(u) \le C_g, \qquad |g(u)| + |g''(u)| \le C_g.$$

Treatment propensities use $\sigma(u) = \mathrm{expit}(u)$ regardless of g. Use $\mathcal{T}_M$ and σ_M as specified in Section 3.2.

(b) For some fixed $B < \infty$, $g(-B) \le m_r^{a,\star}(\boldsymbol{X}) \le g(B)$ almost surely. For $\epsilon_{r,a} = Y - m_r^{a,\star}(\boldsymbol{X})$, there are fixed $v_Y, b_Y > 0$ such that, for $|z| < 1/b_Y$,

$$\mathbb{E}_r\{e^{z\epsilon_{r,a}} \mid \boldsymbol{X}, A = a\} \le \exp\left\{\frac{v_Y z^2}{2(1 - b_Y|z|)}\right\}. \tag{S4}$$

This is a conditional tail condition, not a requirement that Y be bounded or Gaussian.

(c) Target propensities satisfy $\varepsilon \le e_t^{a,\star}(\boldsymbol{X}) \le 1 - \varepsilon$. On the common target–source support the ideal joint weights satisfy $0 < q_- \le q_s^{a,\star}(\boldsymbol{X}) \le q_+ < \infty$. This quantitative overlap condition does not require $q_s^{a,\star}$ to belong to the exponential working family.

(d) Calibrated population predictors and the true predictors of the correctly specified initial models are bounded in absolute value by M on the relevant supports. No bound or convergence requirement is imposed on a misspecified initial fit for the correct-branch inference result: it enters the updates through bounded clipped functions. Sitewise limiting score covariances converge, with positive variances for the contrasts to be studentized.

Section S2.10 discusses the covered outcome links and response distributions.

The conditions imply dimension-free local analytic bounds without bounding realized outcomes. In a convex neighborhood $\mathcal{N}$ in which fitted linear predictors have magnitude at most $2M$, $q_s^a(\boldsymbol{X}; \gamma) \le e^{2M}$, the conditional means and residual second moments are bounded, and

$$\begin{aligned} \sup_{\vartheta \in \mathcal{N}} \|\nabla^2 \mu_s^a(\vartheta)\|_{\mathrm{op}} &\le C, \\ \sum_{r \in \{t,s\}} \mathbb{E}_r\{h_r(D;\vartheta) - h_r(D;\bar{\vartheta})\}^2 &\le C\|\vartheta - \bar{\vartheta}\|_2^2. \end{aligned} \tag{S5}$$

For the Hessian, condition on $(\boldsymbol{X}, A)$ to replace Y by $m_r^{a,\star}(\boldsymbol{X})$; every block is then an expectation of $\phi\phi^T$ times a bounded scalar. For score stability, squared residual factors are controlled by their conditional second moments, and $\mathbb{E}_r(\phi^T\Delta)^2 \le C_\Sigma\|\Delta\|_2^2$. The same argument applies to the target score using $1/e_t^a(\boldsymbol{X}; \beta) = 1 + e^{-\phi^T\beta}$. The conditional

tail bound supplies uniformly bounded score moments of order $2+\delta$. The empirical fits' membership in $\mathcal{N}$ is proved in Lemma S3, rather than assumed here.

Assumption S3 (Correct-model branches and population supports)**.** The source is conditionally mean-transportable and either $m(\boldsymbol{X};\alpha^\star) = m_t^{a,\star}(\boldsymbol{X})$ or $q_s^a(\boldsymbol{X};\gamma^\star) = q_s^{a,\star}(\boldsymbol{X}) = f_t^\star/(f_s^\star e_s^{a,\star})$. In the outcome-correct branch, set $\bar{\alpha} = \alpha^\star$ and define $\bar{\gamma}$ by the population source joint-weight loss with initial outcome $\alpha^\star$. In the weight-correct branch, set $\bar{\gamma} = \gamma^\star$ and define $\bar{\alpha}$ by the population source outcome loss with initial joint-weight fit $\gamma^\star$. The supports of these two calibrated coefficients and the correct initial coefficient have total size s_{eff}, counting intercepts. When both models are correct, include both initial supports. All fitting is confined to the appropriate training sample, using penalties specified below. For the target, replace the source weight branch by correctness of its logistic propensity model and use the corresponding target population losses.

S2.3 Source Identification

Lemma 1 (restated)**.** *For a transportable source and coefficient vectors α, γ with finite expectations, Assumption 1 gives*

$$\mu_{s_j}^a(\alpha,\gamma) - \mu_t^a = \mathbb{E}_t\left[\left(1 - \frac{q_{s_j}^a(\boldsymbol{X};\gamma)}{q_{s_j}^{a,\star}(\boldsymbol{X})}\right)\left(m(\boldsymbol{X};\alpha) - m_t^{a,\star}(\boldsymbol{X})\right)\right].$$

Thus $\mu_{s_j}^a(\alpha,\gamma) = \mu_t^a$ if $m(\boldsymbol{X};\alpha) = m_t^{a,\star}(\boldsymbol{X})$ or $q_{s_j}^a(\boldsymbol{X};\gamma) = q_{s_j}^{a,\star}(\boldsymbol{X})$.

Proof of Lemma 1. Conditional mean transportability gives $\mathbb{E}_s(Y \mid \boldsymbol{X}, A = a) = m_t^{a,\star}(\boldsymbol{X})$. Conditioning the source residual in Eq. (2) therefore replaces Y by this mean. The identity $f_s^\star e_s^{a,\star} q_s^{a,\star} = f_t^\star$ then converts the source expectation to a target expectation with multiplier $q_s^a(\boldsymbol{X};\gamma)/q_s^{a,\star}(\boldsymbol{X})$. Combining it with the target prediction and subtracting $\mu_t^a = \mathbb{E}_t\{m_t^{a,\star}(\boldsymbol{X})\}$ gives the stated identity. Either correct working model makes the resulting bias zero. □

S2.4 Population Targets and Sparsity

To use stationarity at the calibrated limits, we first establish finite, unique population minimizers. We then give a sufficient sparse class under misspecification.

Proposition S1 (Finite population targets and a sparse sufficient class)**.** *Under Assumptions S1 and S2(a)–(c), every initial population loss and every calibrated population loss with a fixed clipped plug-in has a unique finite minimizer for each d. The same holds for untruncated population plug-ins whose predictors have a finite bound.*

For a sufficient sparsity construction, partition $\phi = (Z^T, W^T)^T$, with the intercept in $Z \in \mathbb{R}^q$. Suppose, at each site and for both arms,

$$\mathbb{E}_r(W \mid Z, A = a) = 0, \qquad m_r^{a,\star}(\boldsymbol{X}) = h_{r,a}(Z). \tag{S6}$$

Then all initial population coefficients, and the calibrated population coefficients using those initial population minimizers as plug-ins, have zero W coordinates, even when the working outcome or weighting model is misspecified. If q and the joint law of (Z, A, Y, R) are fixed as inactive coordinates are added, their nonzero coefficients remain fixed. A common finite M can then be chosen to bound the population predictors of all initial and calibrated fits. Thus this class verifies both sparse population targets and the predictor-margin requirement while allowing $d \gg N$.

Proof of Proposition S1. For the outcome loss, conditioning on $(\boldsymbol{X}, A = a)$ gives $\Psi(u) - m_r^{a,\star}(\boldsymbol{X})u$. The interior mean bound in Assumption S2(b) and strict monotonicity of g imply

$$\Psi(u) - \nu u \geq c|u| - C \quad \text{uniformly for } \nu \in [g(-B), g(B)].$$

For example, for $u \geq B+1$ its derivative is at least $g(B+1) - g(B) > 0$, and the negative tail is analogous. Positive bounded calibration weights preserve this lower bound. For a source joint-weight with multiplier $c(\boldsymbol{X}) > 0$, change measure to obtain

$$L_\gamma(\gamma) = \mathbb{E}_s[I(A=a)c(\boldsymbol{X})\{q_s^{a,\star}(\boldsymbol{X})\phi^T\gamma + e^{-\phi^T\gamma}\}].$$

Uniformly for $w \in [q_-, q_+]$, $wu + e^{-u} \geq c|u| - C$. Target propensity calibration instead has conditional loss $c(\boldsymbol{X})\{(1-e_t^{a,\star})u + e_t^{a,\star}e^{-u}\}$, with the same bound by overlap; the initial logistic likelihood has the corresponding interior-mean bound. All clipped multipliers have fixed positive lower and finite upper bounds. Moreover,

$$\mathbb{E}_r\{I(A=a)|\phi^T v|\} \geq \frac{\mathbb{E}_r\{I(A=a)(\phi^T v)^2\}}{C_\phi \|v\|_1} \geq \frac{\kappa_G}{C_\phi\sqrt{d}}\|v\|_2$$

for $v \neq 0$. Hence each loss is coercive for each finite d and attains its minimum. Strictly positive scalar Hessian multipliers and positive population Gram matrices make each loss strictly convex, giving uniqueness and population stationarity. These existence arguments do not claim a dimension-free coefficient bound on their own.

For sparsity, first minimize each initial loss over coefficients of Z only and set the W coefficients to zero. The Z gradient vanishes by restricted minimization. Outcome-loss W gradients vanish because the conditional mean is $h_{r,a}(Z)$ and Eq. (S6) holds. Source joint-weight gradients in W vanish in both target and source terms separately. For the target propensity likelihood, $\mathbb{E}_t(W \mid Z) = 0$ and $\mathbb{E}_t(I(A=a)W \mid Z) = 0$ give the same conclusion. Uniqueness identifies these as the full population minima.

The resulting initial predictors are functions of Z. Therefore clipped outcome derivatives and clipped exponential initial weights are also functions of Z, and the identical gradient argument applies to both calibrated losses at the source and target. This proves sparsity for the entire population fitting sequence without equating initial and calibrated coefficients.

If the active law and q are fixed, the reduced initial coefficients are finite constants. Construct the reduced calibrated population fits using these untruncated initial predictors;

the predictors are bounded because Z is bounded, so the preceding existence proof still applies. There are finitely many resulting coefficients. Choose M larger than all their predictor bounds. Initial population clipping is then inactive and these are also the targets of the clipped procedure with that M. This supplies a nonempty model class with a common margin; it is not a data-driven estimator of the unknown bound. □

For example, bounded inactive features independent of (Z, A, Y, R) and centered at zero satisfy Eq. (S6). The outcome mean may be a nonlinear function such as $h_{r,a}(Z) = g(b_0 + b_1 Z_1 + b_2 Z_1^2)$ while the working active basis contains only $(1, Z_1)$, so its projection remains sparse despite misspecification. Site and treatment mechanisms depending on Z can likewise have misspecified working joint-weight models. Correctness of one nuisance, when required for inference, is imposed separately. With growing active dimension, one must still impose the stated sparsity rate and a uniform predictor bound; this proposition does not make every misspecified projection sparse.

S2.5 Calibrated Population Orthogonality

Stationarity at the minimizers in Proposition S1 gives the calibration result.

Lemma 2 (restated)**.** *For a transportable source and one arm, let $(\bar{\alpha}, \bar{\gamma})$ be the identifiable calibrated population coefficients satisfying the predictor bound in Section 3.2. If the outcome model is correct at $\alpha^\star$, $\bar{\alpha} = \alpha^\star$; if the joint weight model is correct at $\gamma^\star$, $\bar{\gamma} = \gamma^\star$. In either case, the other coefficient minimizes its paired population calibration loss with the correct initial coefficient. Then $\mu^a_{s_j}(\bar{\alpha}, \bar{\gamma}) = \mu^a_t$, and both gradients in Eq. (3) vanish at $(\bar{\alpha}, \bar{\gamma})$.*

Proof of Lemma 2. We verify that both derivatives in Eq. (3) vanish at the calibrated population coefficients.

If the outcome model is correct, conditional mean transportability identifies $\bar{\alpha}_{\text{init}} = \alpha^\star$ for target initialization (one round) or source initialization (two rounds). Any positive clipped initial weight preserves this outcome population minimizer, so $\bar{\alpha} = \bar{\alpha}_{\text{init}} = \alpha^\star$. Conditional residual centering gives $\nabla_\gamma \mu_s^a = 0$ for every calibrated joint weight. Because $|\phi^T \alpha^\star| \le M$, the derivative multiplier in the limiting calibrated joint-weight loss is $g'(\phi^T \alpha^\star)$, so its stationarity gives $\nabla_\alpha \mu_s^a = 0$.

If instead the joint weight model is correct, the initial balance identity identifies $\bar{\gamma}_{\text{init}} = \gamma^\star$. The same identity multiplied by any common clipped outcome derivative identifies $\bar{\gamma} = \gamma^\star$ in the calibrated joint-weight loss. True weighting gives $\nabla_\alpha \mu_s^a = 0$ for every calibrated outcome coefficient. The initial true joint weight is inside the predictor margin, so the limiting outcome calibration is weighted by $q_s^a(\boldsymbol{X}; \gamma^\star)$; its stationarity gives $\nabla_\gamma \mu_s^a = 0$. The misspecified coefficients may differ across stages in either branch. In either branch, Lemma 1 also gives $\mu_s^a(\bar{\alpha}, \bar{\gamma}) = \mu_t^a$, completing the calibration result. □

S2.6 A Sufficient Nuisance Oracle Bound

Following the cross-fitted argument of Hou et al. (2025, Lemma A21), we first bound nuisance error in terms of empirical fluctuations and propagated initialization error. Lemma S3 verifies the required event and gives the final rates.

Fix an outer fold k_1. Let L_N be the unpenalized part of one pooled calibration objective in Sections S1.1 and S1.2, with one common coefficient vector and the prescribed sitewise fold weights $w_{r,k_2} = n_{r,k_2}/n_{r,-k_1}$. Each site's weights sum to one. For a population coefficient $\bar{v}$, let S contain its support and the intercept, $s_v = |S|$, and $\widehat{v} \in \arg\min_v \{L_N(v) + \lambda \|v\|_{1,-0}\}$. Write $r_{c,\text{init}}$ for the largest ℓ_2 error of the correct initial fit across the fixed inner folds.

Assumption S4 (Oracle-bound event)**.** Write $\nabla L_N(\bar{v}) = A_N + B_N$. For each inner block, condition only on its own initial-training data, center its empirical gradient, and then sum

using the original sitewise weights. This gives A_N; the weighted sum of the corresponding conditional expectations gives B_N. On the event under consideration, $\|A_N\|_\infty \le \lambda/2$ and $\|B_N\|_2 \le b_N$. On the expanded cone

$$\mathcal{C}_N = \left\{ \Delta : \|\Delta_{S^c}\|_1 \le 3\|\Delta_S\|_1 + \frac{2b_N}{\lambda}\|\Delta\|_2 \right\},$$

the loss has restricted curvature

$$L_N(\bar{v}+\Delta) - L_N(\bar{v}) - \nabla L_N(\bar{v})^T\Delta \ge \frac{\kappa}{2}\|\Delta\|_2^2$$

for a fixed $\kappa > 0$. If this inequality is local, it must hold on the relevant segments and the minimizer must be localized in that region. For a correct initial fit, the same conditions apply with $B_N = 0$.

Lemma S1 (Nuisance rate with initial-error propagation)**.** *On the preceding event,*

$$\kappa\|\widehat{v} - \bar{v}\|_2 \le 3\sqrt{s_v}\lambda + 2b_N,$$
$$\kappa\|\widehat{v} - \bar{v}\|_1 \le (4\sqrt{s_v} + 2b_N/\lambda)(3\sqrt{s_v}\lambda + 2b_N).$$

Under Assumptions S1 to S3, the pooled calibration updates admit $b_N \le C/N$ for a correctly specified model and $b_N = O_p(r_{c,\text{init}})$ for the other model.

Proof of Lemma S1. Write $\Delta = \widehat{v} - \bar{v}$. The basic inequality for an unpenalized intercept and penalized slopes gives

$$L_N(\bar{v}+\Delta) - L_N(\bar{v}) \le \lambda(\|\Delta_S\|_1 - \|\Delta_{S^c}\|_1).$$

Convexity and the gradient decomposition lower-bound the left side by $-\lambda\|\Delta\|_1/2 - b_N\|\Delta\|_2$. This gives the expanded cone. Restricted curvature then yields

$$\begin{aligned}\frac{\kappa}{2}\|\Delta\|_2^2 &\le \frac{3\lambda}{2}\|\Delta_S\|_1 - \frac{\lambda}{2}\|\Delta_{S^c}\|_1 + b_N\|\Delta\|_2 \\ &\le \left(\frac{3\lambda\sqrt{s_v}}{2} + b_N\right)\|\Delta\|_2.\end{aligned}$$

Division gives the ℓ_2 bound; the expanded cone gives the ℓ_1 bound. Including the intercept in S makes the displayed upper bounds valid without penalizing it. This is the basic-inequality argument of Negahban et al. (2012) with an explicit propagated gradient.

To control B_N, let $\mathcal{F}_{k_2}$ contain the initial-training data outside k_1, k_2. The observations in calibration fold k_2 are independent of this sigma-field. A correct outcome gives a conditionally mean-zero residual for every fixed clipped initial weight, so every source or target outcome block has zero conditional gradient. At the target, a correct propensity gives

$$\mathbb{E}_t\Big[d^a_{k_2}\{I(A \neq a) - I(A = a)e^{-\phi^T\beta^\star}\}\phi \mid \mathcal{F}_{k_2}\Big] = 0.$$

Both target terms use the same sitewise fold fraction, so their pooled conditional gradient is also zero.

For a correct source joint weight, the balance identity applies to the same derivative multiplier $d^a_{k_2}$ on both populations. Its pooled conditional gradient is

$$B_N = \sum_{k_2 \neq k_1} (w_{t,k_2} - w_{s,k_2}) m_{k_2}, \qquad m_{k_2} = \mathbb{E}_t(d^a_{k_2}\phi \mid \mathcal{F}_{k_2}). \tag{S7}$$

The bounded derivative multiplier and population Gram upper bound give $\|m_{k_2}\|_2 \leq C$ by Cauchy–Schwarz, uniformly in d. Balanced sitewise folds have fractions differing by $O(N^{-1})$. With fixed folds and positive site fractions, Eq. (S7) therefore has norm at most C/N. Exact equality of target and source fractions makes this term zero, but is not required for the rates.

For the misspecified update, compare its conditional population gradient with the stationary gradient obtained from the true correct initial coefficient. Both $g' \circ \mathcal{T}_M$ and $\exp(-\mathcal{T}_M)$ are bounded and Lipschitz. For any unit vector v and initial error Δ_{init}, each gradient difference is bounded by sums of terms of the form

$$C\mathbb{E}_r\{|v^T\phi|\,|\phi^T\Delta_{\text{init}}|\} \leq C\{\mathbb{E}_r(v^T\phi)^2\}^{1/2}\{\mathbb{E}_r(\phi^T\Delta_{\text{init}})^2\}^{1/2} \leq C\|\Delta_{\text{init}}\|_2.$$

For outcome-gradient propagation, conditioning on $(\boldsymbol{X}, A)$ replaces the residual by its bounded conditional mean; the final exponential factors are bounded by the population

margin. The stationary comparison uses one common population plug-in. Since each site's weights sum to one, its target and source terms cancel after pooling even if their fold fractions differ. Taking the supremum over v and summing the fixed number of weighted blocks gives $b_N = O_p(r_{c,\mathrm{init}})$. The correct source joint-weight remainder is absorbed by $\sqrt{s_c}\lambda$ when $\lambda \gtrsim N^{-1/2}$, since $s_c \geq 1$. These bounds supply the initialization and calibration steps in Lemma S3. □

S2.7 Verification of Curvature, Localization, and Rates

We verify the event in Lemma S1 by establishing finite initial fits, local curvature, and localization of calibrated minimizers. Let F be the fixed number of site–arm–fold empirical blocks used by the fitting procedure, including fits internal to the aggregation training samples, and let n be their smallest sample size. All such block sizes are proportional to the relevant total sample size. For $\delta \in (0,1)$, put

$$t_\delta = \log(64Fd^2/\delta), \qquad a_n = \sqrt{t_\delta/n} + t_\delta/n, \qquad \lambda = La_n. \tag{S8}$$

Different penalties may be used within a fixed constant factor of λ, with their lower multiplier sufficiently large. Define $\sigma(u) = \mathrm{expit}(u)$ and

$$h_* = e^{-2M}\min\{1, m_g, \sigma'(2M)\} > 0, \qquad \kappa_* = h_*\kappa_G/2, \qquad r_0 = M/C_\phi.$$

Lemma S2 (Finite empirical initial fits)**.** *Under Assumption S1 and Assumption S2(a)–(c), all initial penalized losses attain finite minima with probability at least* $1 - \delta/3$ *when* L *is sufficiently large and* a_n *sufficiently small.*

Proof of Lemma S2. For an initial outcome loss, let P denote its empirical average, $a_i = I(A_i = a)$, $u_i = \phi_i^T v$, $S_0 = Pa_i$, and $\nu_i = m_r^{a,\star}(\boldsymbol{X}_i)$. Write $v = (v_0, v_-^T)^T$, $\xi = P\{a_i(Y_i - \nu_i)\phi_i\}$. The scalar coercivity bound in Proposition S1 gives

$$Q(v) \geq cP(a_i|u_i|) - CS_0 - |\xi_0||v_0| - \|\xi_-\|_\infty \|v_-\|_1 + \lambda\|v_-\|_1.$$

Since $|v_0| \leq S_0^{-1}P(a_i|u_i|) + C_\phi\|v_-\|_1$, the events $|\xi_0|/S_0 \leq c/2$ and $\|\xi_-\|_\infty + C_\phi|\xi_0| \leq \lambda/2$ imply

$$Q(v) \geq (c/2)P(a_i|u_i|) + (\lambda/2)\|v_-\|_1 - CS_0.$$

This is coercive in (v_0, v_-) when $S_0 > 0$. Conditional Bernstein concentration and the positive arm probability give these events at the stated penalty and sample size. The same argument applies to the initial target logistic likelihood with response $I(A = a)$, conditional mean $e_t^{a,\star}(\boldsymbol{X})$, and $a_i = 1$.

For an initial source joint-weight, set $m = P_t\phi$ and $a_i = I(A_i = a)$ at the source. The bounded ideal weights satisfy $\mathbb{E}_s(a_i q_s^{a,\star}(\boldsymbol{X})\phi) = \mathbb{E}_t\phi$. Normalize their empirical version by $\widetilde{q}_i = q_s^{a,\star}(\boldsymbol{X}_i)/(P_s a_i q_s^{a,\star}(\boldsymbol{X}))$, so $P_s a_i \widetilde{q}_i = m_0 = 1$. On the bounded-moment concentration event the normalization is bounded above and away from zero. The identity $P_s a_i \widetilde{q}_i = 1$ and the bound $|\phi_{i,k}| \leq C_\phi$ control the extra normalization error by a constant times the intercept imbalance, so

$$\|m_- - P_s(a_i\widetilde{q}_i\phi_{i,-})\|_\infty \leq \lambda/2.$$

Writing the initial loss as

$$P_s\{a_i(e^{-u_i} + \widetilde{q}_i u_i)\} + \{m_- - P_s(a_i\widetilde{q}_i\phi_{i,-})\}^T v_- + \lambda\|v_-\|_1$$

shows it is at least $cP_s(a_i|u_i|) + (\lambda/2)\|v_-\|_1 - C$, again coercive. Here the ideal weights are used only for the proof and are not supplied to the algorithm. Bounded-coordinate concentration controls the normalization and moment imbalance at order a_n. A finite union bound over all initial-fitting blocks, with a sufficiently large L, proves the stated probability.

Continuity and coercivity give finite minimizers; uniqueness of the empirical minimizer is unnecessary. □

The following lemma verifies the preceding rate bound for the source–target calibration losses, including two-population balance, conditional-tail concentration, and predictor localization. For fixed positive constants C_0, C_1, write the sample-size conditions as

$$C_0 s_{\text{eff}} \sqrt{t_\delta / n} \le \kappa_G / 2, \qquad C_1 s_{\text{eff}} \lambda < r_0. \tag{S9}$$

Lemma S3 (Calibration rates)**.** *Under Assumptions S1 to S3, consider the prescribed pooled losses and penalties with unpenalized intercepts. For sufficiently large fixed L, C_0, C_1, Eq. (S9) gives the following event with probability at least $1 - \delta$: the correct initial fit and both calibrated updates attain finite minima, their predictors are bounded by $2M$, and the oracle bounds in Lemma S1 hold with $\kappa = \kappa_*$. The final fits satisfy*

$$\|\widehat{v}_c - \bar{v}_c\|_2 \le C\sqrt{s_c}\lambda, \qquad \|\widehat{v}_c - \bar{v}_c\|_1 \le C s_c \lambda,$$
$$\|\widehat{v}_w - \bar{v}_w\|_2 \le C\sqrt{s_w + s_{c,\text{init}}}\lambda, \qquad \|\widehat{v}_w - \bar{v}_w\|_1 \le C(s_w + s_{c,\text{init}})\lambda.$$

The target calibrated fits obey the same bounds. For bounded d and $\lambda \asymp n^{-1/2}$, finite minimizers and predictor localization hold with probability tending to one, with the displayed errors in O_p form.

Proof of Lemma S3. Lemma S2 supplies finite initial fits. Only the correctly specified initial fit needs an estimation rate; clipping controls the other fit throughout the conditional concentration argument. We allocate the remaining failure probability to the score and Gram events.

We first control the empirical gradients and Gram matrices. At each population target used in this lemma, its predictor has magnitude at most M. Each empirical gradient coordinate is either bounded or is a bounded multiple of Y plus bounded terms. The conditional Bernstein bound in Eq. (S4) and the bounded conditional mean therefore give a

uniform scalar Bernstein inequality. Apply this bound separately on each calibration block, conditional only on that block's own initial-training data; its clipped plug-in is then fixed and bounded. A union bound over d coordinates and F blocks, followed by summation with the prescribed sitewise weights, gives $\|A_N\|_\infty \le \lambda/2$ for L sufficiently large, except on an event of probability at most $\delta/3$. Initial correct-model gradients have zero population mean.

For each initial or calibration block, the entries of $I(A=a)\phi\phi^T$ are bounded by C_ϕ^2. Hoeffding concentration and a union bound over Fd^2 entries yield, with failure probability at most $\delta/3$,

$$\max_{r,a,\text{block}} \|\widehat{G}_{r,a} - G_{r,a}\|_{\max} \le CC_\phi^2\sqrt{t_\delta/n} =: e_n. \tag{S10}$$

Neither event assumes independence of the overlapping training sets.

These concentration events also give local curvature. For a fitted coefficient v within $\|v-\bar{v}\|_1 \le r_0$, every predictor along the connecting segment has magnitude at most $2M$. The source outcome Hessian is an empirical average of $I(A=a)e^{-\mathcal{T}_M(u_{\widehat{\gamma}_{\text{init}}})}g'(u_\alpha)\phi\phi^T$; the source joint-weight Hessian uses $I(A=a)e^{-u_\gamma}g'\{\mathcal{T}_M(u_{\widehat{\alpha}_{\text{init}}})\}\phi\phi^T$. The target calibrated Hessians have the same respective forms with β in place of γ. Initial outcome and joint-weight losses omit the extra plug-in factor; the initial target logistic likelihood uses $\sigma'(u_\beta)\phi\phi^T$. Each Hessian therefore dominates h_* times its empirical arm Gram matrix (or its full-target Gram matrix, which dominates an arm Gram matrix).

For any perturbation, Eq. (S10) gives

$$\Delta^T\widehat{G}_{r,a}\Delta \ge \kappa_G\|\Delta\|_2^2 - e_n\|\Delta\|_1^2.$$

The basic inequality implies $\|\Delta\|_1 \le (4\sqrt{s_v} + 2b_N/\lambda)\|\Delta\|_2$. For a correct initial fit $b_N = 0$. Once its rate is established, the propagation bound gives $b_N \le C\sqrt{s_{c,\text{init}}}\lambda$ for the opposite update, and $b_N \le C/N$ for a correct source joint-weight update, with zero conditional gradient for the other correct updates. Since $N^{-1}/\lambda = O(N^{-1/2})$, these cones all satisfy $\|\Delta\|_1 \le C\sqrt{s_{\text{eff}}}\|\Delta\|_2$. The first condition in Eq. (S9) makes their empirical Gram curvature

at least $\kappa_G/2$. Integrating the Hessian along the segment yields the loss remainder at least $h_*\kappa_G\|\Delta\|_2^2/4 = \kappa_*\|\Delta\|_2^2/2$.

The curvature bound now allows us to localize the fits. Consider first the correct initial fit, then the two calibrated fits in that order. Let $Q(v) = L_N(v) + \lambda\|v\|_{1,-0}$. If a point on the sphere $\|v - \bar{v}\|_1 = r_0$ had $Q(v) \le Q(\bar{v})$, its basic inequality would imply the applicable cone. The local curvature and the basic-inequality calculation in Lemma S1, applied to this boundary point, would then give $\|v - \bar{v}\|_1 \le C_1 s_{\text{eff}}\lambda < r_0$, a contradiction. Continuity ensures a minimizer over the closed ℓ_1 ball, and the boundary argument puts it in the interior. Convexity makes it a global minimizer. Any global minimizer outside the ball would, by convexity along its segment from $\bar{v}$, produce a boundary point with $Q(v) \le Q(\bar{v})$, again a contradiction. Thus every minimizer is localized. Applying the deterministic lemma inside the ball gives the stated rates. The correct initial rate supplies the propagated bound before it is used for the opposite update, so the argument is not circular.

For bounded d, the preceding probability bound with a fixed multiplier is replaced by a finite-dimensional argument. The coercivity inequalities in Proposition S1, together with bounded d and the fixed design, mean, and overlap constants, place all initial population minimizers in a common compact set. Their predictors are consequently uniformly bounded, although the misspecified initial bounds need not equal M. On a fixed neighborhood of each minimizer, strict positivity and continuity of g' and the exponential and logistic multipliers give a positive Hessian lower bound. The empirical arm Gram matrices converge, and each initial gradient at its population minimizer is $O_p(n^{-1/2})$ by the second-moment bounds. For a perturbation satisfying the basic inequality, local curvature thus gives

$$\frac{\kappa}{2}\|\Delta\|_2^2 \le \{\|\nabla L_N(\bar{v})\|_2 + \lambda\sqrt{d}\}\|\Delta\|_2.$$

The convex local-ball argument first yields finite initial minimizers and $O_p(n^{-1/2})$ errors. For the calibrated updates, blockwise conditional variances give $A_N = O_p(n^{-1/2})$; the

correct initial rate and Eq. (S7) give the same order for B_N. Their clipped multipliers have uniform positive lower bounds, so the same curvature and local-ball argument applies at the calibrated population coefficients. Hence every calibrated minimizer has ℓ_2 and ℓ_1 errors $O_p(n^{-1/2})$. The population margin then gives predictor localization with probability tending to one. Here stochastic gradient control replaces the high-probability gradient event used for growing d. □

For growing d, taking $\delta = (2d)^{-c}$ with fixed $c > 0$ gives $\lambda \asymp \sqrt{\log(2d)/N}$. The bounded-dimensional choice $\lambda \asymp N^{-1/2}$ has the same order. In both regimes, $s_{\text{eff}} \log(2d) = o(\sqrt{N})$ ensures the sample-size conditions in Eq. (S9).

The lemma yields $\|\widehat{v} - \bar{v}\|_1 = o_p(1)$ and hence

$$\sup_{\boldsymbol{x}} |\phi(\boldsymbol{x})^T(\widehat{v} - \bar{v})| \le C_\phi \|\widehat{v} - \bar{v}\|_1 = o_p(1).$$

The population margin therefore makes final target propensity clipping inactive with probability tending to one, uniformly over the fixed folds.

S2.8 Second-Order Bias and Score Stability

Orthogonality and Eq. (S5) convert coefficient error into second-order population bias and L_2 score error.

Lemma S4 (Second-order bias and score stability)**.** *Under Assumption S2 and Lemma 2, for $\vartheta \in \mathcal{N}$,*

$$|\mu_s^a(\vartheta) - \mu_s^a(\bar{\vartheta})| \le (C/2)\|\vartheta - \bar{\vartheta}\|_2^2,$$

$$\sum_{r \in \{t,s\}} \mathbb{E}_r\{h_r(D;\vartheta) - h_r(D;\bar{\vartheta})\}^2 \le C\|\vartheta - \bar{\vartheta}\|_2^2.$$

Proof of Lemma S4. For $\Delta = \vartheta - \bar{\vartheta}$, Taylor's formula along the segment from $\bar{\vartheta}$ to ϑ gives

$$\mu_s^a(\vartheta) - \mu_s^a(\bar{\vartheta}) = \nabla \mu_s^a(\bar{\vartheta})^T \Delta + \int_0^1 (1-u)\Delta^T \nabla^2 \mu_s^a(\bar{\vartheta} + u\Delta)\Delta \, du.$$

The first term is zero and the Hessian bound controls the integral. The second conclusion follows from the dimension-free bound in Eq. (S5). A general nonlinear GLM remainder includes squared errors of each nuisance; it need not reduce to their product under misspecification. □

S2.9 Candidate Expansion and Variance Consistency

Lemma 4 (restated)**.** *For the fitting procedure and penalty levels in Section 4.1, under Assumptions 1 to 3, each $s_j \in \mathcal{J}^{a,*}$ satisfies*

$$\widehat{\mu}_{s_j}^a - \mu_t^a = n_t^{-1} \sum_{i \in \mathcal{D}_t} \zeta_{t,s_j,i}^a + n_{s_j}^{-1} \sum_{i \in \mathcal{D}_{s_j}} \xi_{t,s_j,i}^a + o_p(N^{-1/2}),$$

and its fitted site-score covariances are consistent.

Proof of Lemma 4. Fix a source $s = s_j \in \mathcal{J}^{a,*}$. The conditions of Assumptions 2 and 3 give the pairwise design, tail, overlap, and population-margin conditions above. With the prescribed penalties, the sparsity condition implies

$$s_{\text{eff}} \lambda_N = o(1), \qquad \sqrt{N}\, s_{\text{eff}} \lambda_N^2 = o(1). \tag{S11}$$

Lemma S3 therefore gives $\|\widehat{\vartheta}^{(-k)} - \bar{\vartheta}\|_2 = O_p(\sqrt{s_{\text{eff}}}\lambda_N)$ and predictor localization on every fold. By Lemmas 1 and 2, $\mu_s^a(\bar{\vartheta}) = \mu_t^a$ and the population gradient vanishes.

For each evaluation fold k, set $\Delta h_{r,k} = h_r(D; \widehat{\vartheta}^{(-k)}) - h_r(D; \bar{\vartheta})$. Add and subtract the two population means to write the remainder as $K_f^{-1} \sum_k (R_{1,k} + R_{2,k}) + O_p(N^{-1})$, where the last term accounts for balanced fold sizes and

$$R_{1,k} = \sum_{r \in \{t,s\}} (\widehat{\mathbb{E}}_{r,k} - \mathbb{E}_r)\Delta h_{r,k}, \qquad R_{2,k} = \mu_s^a(\widehat{\vartheta}^{(-k)}) - \mu_s^a(\bar{\vartheta}).$$

Conditional on the training data for fold k, the evaluation terms are independent and centered. Since both site fold sizes are proportional to N,

$$\mathrm{Var}(R_{1,k} \mid \mathcal{D}^{(-k)}) \leq \frac{C}{N} \|\widehat{\vartheta}^{(-k)} - \bar{\vartheta}\|_2^2.$$

Conditional Chebyshev's inequality and consistency yield $R_{1,k} = o_p(N^{-1/2})$. The folds' training samples overlap, but no independence across the fixed number of fold remainders is needed to sum these bounds. By the bias lemma, $|R_{2,k}| = O_p(s_{\mathrm{eff}}\lambda_N^2) = o_p(N^{-1/2})$. This proves the expansion. The leading target and source sums are independent, site-centered averages. Uniform $(2+\delta)$ moments and positive limiting variance give the Lyapunov central limit theorem for their sum. Thus $\sqrt{N}(\widehat{\mu}_s^a - \mu_t^a) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{a,*})$.

To establish covariance consistency, center the cross-fitted scores within their sites and form

$$\widehat{V}_{t,s}^a = \frac{N}{n_t}\widehat{\mathbb{E}}_t((\widehat{\zeta}_{t,s}^a)^2) + \frac{N}{n_s}\widehat{\mathbb{E}}_s((\widehat{\xi}_{t,s}^a)^2) \xrightarrow{p} V_{t,s}^{a,*}.$$

Conditionally on each training sample, L_2 stability and Markov's inequality give $\widehat{\mathbb{E}}_{r,k}(\Delta h_{r,k})^2 = o_p(1)$. Summing over finitely many folds gives the same conclusion sitewise. Cauchy–Schwarz then shows that empirical first and second moments of fitted scores differ by $o_p(1)$ from their population-limit-score counterparts. The ordinary law of large numbers applies to the latter using their uniform integrability. Centering and the limiting site fractions complete the proof. The same argument applied to products gives consistency of covariances between two score components evaluated on the same observations.

The numbers of sources, arms, and folds are fixed, so the expansion and covariance conclusions hold simultaneously for all valid candidate components. □

S2.10 Outcome Links and Continuous Responses

For continuous outcomes with $g(u) = u$, $g'(u) = 1$ and $\Psi(u) = u^2/2$. The joint-weight calibration loss becomes the initial exponential-balancing loss, and the outcome calibration loss is weighted least squares with an ℓ_1 penalty. No initial outcome coefficient is required for the target derivative moment, which is simply $\widehat{\mathbb{E}}_t\phi$. The target regression average is $\widehat{\mathbb{E}}_t(\phi)^T\widehat{\alpha}^a_{t,s_j}$. With $\widehat{\Sigma}_{\phi,t} = \widehat{\mathrm{Cov}}_t\{\phi(\boldsymbol{X})\}$, the covariance between two source-specific target predictions is $(\widehat{\alpha}^a_{t,s_j})^T\widehat{\Sigma}_{\phi,t}\widehat{\alpha}^a_{t,s_h}$. Thus the same covariance structure applies, with particularly simple moment summaries. At the target, the propensity calibration also drops its initial-outcome multiplier, and the outcome update is weighted least squares with propensity-odds weights. The following corollary extends the inference result to unbounded continuous responses under conditional tail conditions.

Corollary S1 (Continuous-outcome inference)**.** *Take $g(u) = u$ and $\Psi(u) = u^2/2$. Under the design, conditional-tail, overlap, population-margin, and sparsity conditions of Lemma S3, together with Eq. (S11), the source-assisted mean has the expansion and consistent variance in Lemma 4 if either the source outcome mean is a correct linear model or the joint exponential weight is correct, with conditional-mean transportability in either case. At the target, impose the corresponding rate conditions with n_t and $s_{\mathrm{eff},0,a}$; either a correct linear outcome mean or a correct logistic propensity then yields the expansion and consistent variance in Proposition S2. These conclusions allow unbounded continuous Y. They also yield valid Wald inference for the treatment contrast, retaining cross-arm covariances.*

Proof of Corollary S1. The identity link has $g' = 1$ and $g'' = 0$, satisfying the link conditions on every bounded predictor interval. Source joint-weight calibration is the ordinary exponential-balancing loss and target propensity calibration has multiplier one. Outcome calibration is weighted least squares, with the same clipped initial source weights or target propensity-odds weights. Conditional residual centering and the respective balance equa-

tions give the casewise derivatives exactly as in Lemma 2 and Proposition S2. No step uses $Y \in \{0, 1\}$. Conditional Bernstein tails establish gradient concentration, and conditional second moments establish score stability and variance consistency. The verified nuisance rates and second-order argument then prove the stated expansions and normal limits. □

In particular, bounded conditional means with conditionally Gaussian errors of uniformly bounded variance satisfy the continuous-outcome conditions, without requiring equal variances across sites or treatment arms. The same general theorem covers Bernoulli outcomes with a logistic working mean and Poisson outcomes with an exponential working mean under the stated mean bounds. A working GLM specifies the nuisance mean and convex loss; arbitrary continuous distributions, heavy tails, and restricted-domain links are not automatically covered by that notation.

S2.11 From Arm Means to the Treatment Contrast

Apply the preceding expansion for $a = 1$ and $a = 0$, using their respective calibrated coefficients. Joint moment bounds give a bivariate central limit theorem by the Cramér–Wold argument. The pairwise treatment contrast consequently has variance

$$\frac{\mathrm{Var}_t(\zeta^1_{t,s} - \zeta^0_{t,s})}{n_t} + \frac{\mathrm{Var}_s(\xi^1_{t,s} - \xi^0_{t,s})}{n_s}$$

on the unscaled estimator scale. This expression includes both cross-arm covariances. Independence of the two potential-outcome mean estimators is not assumed.

S3 Joint Aggregation and Inference

We combine the source expansions with a target-only reference to prove Theorem 1. The steps below establish the target expansion, joint covariance, and stability of potentially biased candidates before source selection and oracle equivalence.

S3.1 Target-Only Limit Theory

Use the score, calibrated population coefficients, and losses defined in Section S1.2. The target-only reference has the following expansion.

Proposition S2 (Model-doubly-robust target expansion)**.** *Assume target causal identification, either a correct outcome GLM or a correct logistic propensity model, and the target versions of Assumptions S1 to S3. Use Eq. (S8) with $\delta_N \to 0$ when $d \to \infty$, or $\lambda_N \asymp n_t^{-1/2}$ when d is bounded, as covered by Lemma S3. Require $s_{\mathrm{eff},0,a}\lambda_N = o(1)$ and $\sqrt{n_t}s_{\mathrm{eff},0,a}\lambda_N^2 = o(1)$. For growing d and $\delta_N = (2d)^{-c}$, $s_{\mathrm{eff},0,a}\log(2d) = o(\sqrt{n_t})$ suffices. With $\varphi_t^a = h_0(D;\bar{\alpha}_0,\bar{\beta}_0) - \mu_t^a$, the target estimator satisfies*

$$\widehat{\mu}_t^a - \mu_t^a = n_t^{-1}\sum_{i\in\mathcal{D}_t}\varphi_{t,i}^a + o_p(n_t^{-1/2}).$$

It is asymptotically normal at rate $\sqrt{n_t}$, and the empirical variance of its centered cross-fitted scores is consistent for $\mathrm{Var}_t(\varphi_t^a)$ when this limit is positive. The corresponding bivariate result holds for both arms and their contrast.

Proof of Proposition S2. The target doubly robust identity gives $\mu_0(\bar{\alpha}_0,\bar{\beta}_0) = \mu_t^a$. Differentiation of the untruncated smooth functional gives

$$\nabla_\alpha\mu_0(\alpha,\beta) = \mathbb{E}_t[\{(1-I(A=a)) - I(A=a)e^{-\phi^T\beta}\}g'(\phi^T\alpha)\phi],$$

$$\nabla_\beta\mu_0(\alpha,\beta) = -\mathbb{E}_t\{I(A=a)e^{-\phi^T\beta}(Y-m(\boldsymbol{X};\alpha))\phi\}.$$

If outcome is correct, any positive clipped initial propensity-odds weight preserves the outcome minimizer $\alpha^\star$, and residual centering makes the propensity gradient zero for any β.

The outcome gradient is the stationarity equation of the target propensity loss with the correct outcome derivative; its clipping is inactive at truth.

If propensity is correct, $\mathbb{E}_t\{(1 - I(A = a)) - I(A = a)e^{-\phi^T\beta^\star} \mid \boldsymbol{X}\} = 0$. This both identifies the calibrated propensity under any clipped initial outcome multiplier and makes the outcome gradient vanish at any α. Because the true initial propensity predictor lies within the margin, the target outcome loss has weight $I(A = a)e^{-\phi^T\beta^\star}$; its stationarity makes the propensity gradient vanish at $\bar{\alpha}_0$. Thus both derivatives are zero in either branch, without equality of the misspecified initial and calibrated coefficients.

The conditional-gradient calculation and the verified rates of Lemma S3 apply to both target losses. The local target Hessian and L_2 bounds follow from Eq. (S5). Population bias is therefore $O_p(s_{\text{eff},0,a}\lambda_N^2) = o_p(n_t^{-1/2})$. On the event of uniform predictor localization, final propensity clipping is inactive; its complement has probability tending to zero and does not affect the expansion in probability. Cross-fitting bounds the conditional empirical-process remainder exactly as in the source proof. The ordinary central limit theorem, the cross-fitted second-moment argument, and Cramér–Wold prove the remaining claims. □

The limit score is evaluated at $(\bar{\alpha}_0, \bar{\beta}_0)$, not at a hypothetical pair of true functions when one working model is misspecified. Only when both models are correct does it reduce to the usual efficient target AIPW score. Model-doubly-robust inference here requires the specified calibrated fitting procedure and regularity; the same conclusion is not claimed for ordinary predictive nuisance fits.

S3.2 Covariance Decomposition

The covariance decomposition in Lemma 3 accounts for shared target observations and dependence between treatment arms.

Lemma 3 (restated)**.** *Independent site samples give additive variances for the first-order site contributions. The joint treatment-effect variance is estimated by*

$$\widehat{V}_{\tau}^{(-k_1)}(\eta) = \sum_{r} \frac{\widehat{\mathrm{Var}}_{r,-k_1}\{\widehat{G}^1_{r,i}(\eta^1) - \widehat{G}^0_{r,i}(\eta^0)\}}{n_{r,-k_1}}.$$

Here each empirical variance centers contributions within the inner calibration folds and pools their second central moments within the site. Under the candidate linear expansions and fitted-score covariance consistency, this expression consistently estimates the treatment-effect variance on the valid candidate subspace.

Proof of Lemma 3. Fix a pair of weight vectors supported on the valid candidate arms. The source expansion in Lemma 4 and the target expansion in Proposition S2 express their treatment contrast as a sum of centered site contributions, up to a negligible remainder. Samples from different sites are independent, so the variance of this sum is the sum of the site variances divided by their respective sample sizes.

Within a site, taking the variance of the difference between the two arm contributions preserves their cross-arm covariance. At the target, all candidate predictions are evaluated on the same observations, giving the full target covariance block. At each source, only that source's residual scores contribute, giving the diagonal source blocks across candidates. Expanding these within-site variances yields exactly the block formula in Section S1.3 and the expression in Eq. (10).

Under the assumed fitted-score covariance consistency, every entry of these blocks converges to its population counterpart on the valid candidate subspace. There are finitely many entries, and the variance is quadratic in the weights, so the corresponding scaled

variance converges uniformly on each compact set of valid weights. This establishes the claimed variance consistency. □

S3.3 Verification of Training Moments

To justify the covariance estimator and aggregation penalties, we verify Assumption 4 on the reused training sample. The proof applies population orthogonality and nuisance rates to an empirical Taylor expansion, then controls fitted-score second moments.

Proposition S3 (Training means and score covariances)**.** *For the primary pooled calibration procedure, assume the conditions of Lemma 4 for valid source arms and of Proposition S2 for the target. Let $n = n_{-k_1}$, let λ_N denote their common penalty order, and let $s_{\max}$ be the largest relevant effective support size. If $s_{\max}\lambda_N = o(1)$ and $\sqrt{n}\, s_{\max}\lambda_N^2 = o(1)$, the conditions in Assumption 4 hold uniformly over the fixed outer folds. Consequently, valid training discrepancies are $O_p(n^{-1/2})$ and their fitted-score covariance blocks are consistent, including shared-target and cross-arm products.*

Proof of Proposition S3. Fix one outer training sample and a valid source arm, suppressing $(-k_1)$ on fitted coefficients. Define its empirical functional by $\widehat{\mu}^a_{s,\mathrm{tr}}(\vartheta) = \widehat{\mathbb{E}}_{t,-k_1} h_t(D;\vartheta) + \widehat{\mathbb{E}}_{s,-k_1} h_s(D;\vartheta)$, and put $\Delta = \widehat{\vartheta}^{(-k_1)} - \bar{\vartheta}$. The rates in Lemma S3 give $\|\Delta\|_1 = O_p(s_{\max}\lambda_N)$ and $\|\Delta\|_2 = O_p(\sqrt{s_{\max}}\lambda_N)$, with predictors bounded by $2M$ along the connecting segment. The population gradient at $\bar{\vartheta}$ is zero by Lemma 2. Each empirical gradient coordinate at this deterministic coefficient is a sum of bounded terms and bounded multiples of $I(A = a)Y$. The conditional-tail bound and a coordinate union bound therefore give

$$\|\nabla \widehat{\mu}^a_{s,\mathrm{tr}}(\bar{\vartheta})\|_\infty = O_p(\lambda_N).$$

The gradient is evaluated at the population coefficient, so the bound remains valid when Δ depends on the training records.

To control the empirical Taylor remainder, use the target Gram matrix and the source matrix weighted by $I(A = a)(1 + |Y|)$. Their population operator norms are bounded: condition on $(\boldsymbol{X}, A)$ and apply the conditional first-moment and design bounds. The conditional-tail bound also gives uniformly subexponential tails for $I(A = a)|Y|$, so their maximum entrywise deviations are $O_p(\lambda_N)$ by Bernstein concentration. Every local Hessian block of $\widehat{\mu}^a_{s,\mathrm{tr}}$ is dominated in quadratic form by these two matrices, up to a fixed constant; mixed blocks are bounded using $2|xy| \le x^2 + y^2$. Uniformly along the segment, this yields

$$|\Delta^T \nabla^2 \widehat{\mu}^a_{s,\mathrm{tr}}(\vartheta)\Delta| \le C\|\Delta\|_2^2 + O_p(\lambda_N)\|\Delta\|_1^2.$$

Taylor's formula consequently gives

$$\begin{aligned} |\widehat{\mu}^a_{s,\mathrm{tr}}(\widehat{\vartheta}) - \widehat{\mu}^a_{s,\mathrm{tr}}(\bar{\vartheta})| &= O_p\{\lambda_N\|\Delta\|_1 + \|\Delta\|_2^2 + \lambda_N\|\Delta\|_1^2\} \\ &= O_p(s_{\max}\lambda_N^2 + s_{\max}^2\lambda_N^3) = O_p(s_{\max}\lambda_N^2). \end{aligned}$$

The last equality uses $s_{\max}\lambda_N = o(1)$. For the target-only functional, write the inverse propensity as $1 + e^{-\phi^T\beta_0}$. Its population gradient vanishes by the stationarity argument in Proposition S2, and its Hessian is controlled by the target Gram matrix weighted by $1 + I(A = a)|Y|$. The same Taylor bound therefore applies. Final propensity clipping is inactive on the localization event. Since $\sqrt{n}\, s_{\max}\lambda_N^2 = o(1)$, these bounds prove part (a) of Assumption 4.

For part (b), local Lipschitz bounds and bounded features give, for each applicable raw site contribution,

$$|h_r(D; \widehat{\vartheta}) - h_r(D; \bar{\vartheta})|^2 \le C\|\Delta\|_1^2\{1 + I(A = a)Y^2\}.$$

The empirical arm outcome second moments are $O_p(1)$, and $\|\Delta\|_1 = o_p(1)$, so the training averages of these squared differences are $o_p(1)$. The analogous bound holds for the target-only score. Centering fitted contributions within calibration folds preserves this conclusion: Jensen's inequality controls their mean differences, and each fold's population-coefficient

mean converges to its site expectation. Cauchy–Schwarz and the laws of large numbers for population-coefficient scores then give covariance consistency, including all shared-target and cross-arm entries. Finally, the population-nuisance training means have expectation μ_t^a and variance $O(n^{-1})$. The replacement bound makes their fitted discrepancies $O_p(n^{-1/2})$. A finite union over folds and valid arms completes the proof. □

S3.4 Candidate Consistency under Stable Initial Fits

Selection also uses moments from potentially biased candidates. These require consistency and a positive discrepancy scale. The following result supplies them under stability of both initial fits.

For a fixed source s_j and arm a, write $\widehat{\mu}_j = \widehat{\mu}_{s_j}^a$ and define $\bar{\alpha}_{\text{init}}, \bar{\gamma}_{\text{init}}$ by their initial population losses; the one-round $\bar{\alpha}_{\text{init}}$ is the target initial outcome minimizer. Define $\bar{\gamma}$ by the calibrated population joint-weight loss with plug-in $\bar{\alpha}_{\text{init}}$ and $\bar{\alpha}$ by the calibrated population outcome loss with plug-in $\bar{\gamma}_{\text{init}}$. Set $\bar{\vartheta} = (\bar{\alpha}^T, \bar{\gamma}^T)^T$. The corresponding functions are $\bar{m}_{s_j}^a(\boldsymbol{x}) = m(\boldsymbol{x}; \bar{\alpha})$ and $\bar{q}_{s_j}^a(\boldsymbol{x}) = q_{s_j}^a(\boldsymbol{x}; \bar{\gamma})$. These definitions retain the stated clipping, whether either model is correct or not. Let s_{stab} be the sum of all four support sizes, counting intercepts, and write

$$\mu_{j,N}^* = \mathbb{E}_t \bar{m}_{s_j}^a(\boldsymbol{X}) + \mathbb{E}_{s_j}\{I(A = a)\bar{q}_{s_j}^a(\boldsymbol{X})(Y - \bar{m}_{s_j}^a(\boldsymbol{X}))\}.$$

Proposition S4 (Stable candidate means and score covariances)**.** *Suppose the population design, GLM, conditional-tail, and quantitative-overlap conditions in Assumptions S1 and S2(a)–(c) hold, without assuming transportability or either correct nuisance model. Assume all four initial and calibrated population predictors are bounded by M, and $s_{\text{stab}}\lambda_N = o(1)$ with the penalty choice in Eq. (S8), $\delta_N \to 0$, for growing d, or $\lambda_N \asymp N^{-1/2}$ for bounded d. The complete outer-evaluation estimate satisfies*

$$\widehat{\mu}_j - \mu_{j,N}^* = O_p\{\sqrt{s_{\text{stab}}}\lambda_N + N^{-1/2}\} = o_p(1).$$

On both training and evaluation samples, fitted sitewise score means, variances, and cross-products differ by $o_p(1)$ from their population-coefficient counterparts. The reused training candidate means satisfy $\widehat{\mu}_j - \mu^*_{j,N} = o_p(1)$. Cross-products require the corresponding conditions for the other candidate, using Proposition S2 for the target-only reference. When the population means and score covariances have limits, these empirical moments converge to them. If additionally $\mathrm{Var}_{s_j}(Y \mid \boldsymbol{X}, A = a) \geq v_- > 0$, the discrepancy standard error satisfies $\sqrt{n_{-k_1}}\, \widehat{s}^a_{s_j} \to_p s^{a,*}_{s_j} \in (0, \infty)$.

Proof of Proposition S4. The population targets exist uniquely by Proposition S1. At each initial population minimizer the gradient is zero, even under misspecification, and the predictor bound gives the same Bernstein concentration as in Lemma S3. Its Gram and local-ball arguments with $b_N = 0$ give initial ℓ_2 errors $O_p(\sqrt{s_{\mathrm{stab}}}\lambda_N)$ and ℓ_1 errors $O_p(s_{\mathrm{stab}}\lambda_N)$, since each initial support is included in s_{stab}.

For either calibrated update, compare its conditional population gradient with the gradient using its own limiting initial plug-in. With that common population plug-in, each site's fold weights sum to one, so pooling recovers the population calibration loss and its zero gradient. The Lipschitz propagation calculation in Lemma S1 therefore gives $b_N = O_p(\sqrt{s_{\mathrm{stab}}}\lambda_N)$ without requiring model correctness. The expanded cone has $\|\Delta\|_1 \leq C\sqrt{s_{\mathrm{stab}}}\|\Delta\|_2$. Since $s_{\mathrm{stab}}\lambda_N = o(1)$, the Gram and localization conditions hold eventually, and the same local-ball argument gives final ℓ_2 errors $O_p(\sqrt{s_{\mathrm{stab}}}\lambda_N)$ and ℓ_1 errors $O_p(s_{\mathrm{stab}}\lambda_N)$. For bounded d, the finite-dimensional argument in Lemma S3 applies at all four population minimizers and gives the same rates.

On the resulting local event, the population functional is Lipschitz in coefficient ℓ_2 norm. Each gradient is an expectation of ϕ times a scalar with bounded conditional mean;

Cauchy–Schwarz and the population Gram upper bound control every unit directional derivative. Thus, uniformly in the local neighborhood,

$$\|\nabla \mu^a_{s_j}(\vartheta)\|_2 \le C, \qquad |\mu^a_{s_j}(\widehat{\vartheta}) - \mu^*_{j,N}| \le C\|\widehat{\vartheta} - \bar{\vartheta}\|_2.$$

This population error is $O_p(\sqrt{s_{\text{stab}}}\lambda_N)$; orthogonality is not required. Conditional on each outer training sample, the held-out site averages have bounded variance of order N^{-1}. As in the proof of Lemma 4, replacing balanced sitewise fold fractions by $1/K_f$ costs $O_p(N^{-1})$. Summing the fixed number of fold terms gives the stated outer-evaluation rate.

On each outer evaluation fold, the L_2 score bound in Eq. (S5), conditional Markov's inequality, and Cauchy–Schwarz make empirical fitted-score first and second moments differ by $o_p(1)$ from their population-coefficient counterparts. The population-predictor bound supplies the local neighborhood used in that inequality. Uniform conditional outcome moments give the corresponding laws of large numbers. The same argument applies to products with another stable score, including the target-only score.

On reused training observations, the ℓ_1 rates and bounded features give uniform $o_p(1)$ predictor errors. Local Lipschitz bounds then control each squared score error by a common $o_p(1)$ factor times $1 + Y_i^2$. The sitewise empirical outcome second moments are $O_p(1)$, so the same moment consistency holds on these training samples without conditioning on the fitted model. The fixed number of internal training constructions preserves these bounds.

Finally, for $U_j = I(A = a)\bar{q}^a_{s_j}(\boldsymbol{X})(Y - \bar{m}^a_{s_j}(\boldsymbol{X}))$, the law of total variance gives

$$\mathrm{Var}_{s_j}(U_j) \ge \mathbb{E}_{s_j}\{I(A = a)\bar{q}^a_{s_j}(\boldsymbol{X})^2 \,\mathrm{Var}(Y \mid \boldsymbol{X}, A = a)\} \ge e^{-2M} v_- P_{s_j}(A = a).$$

Because the intercept is in the design, Eq. (S3) implies $P_{s_j}(A = a) \ge \kappa_G$. Under the stated covariance limits, the limit of $n_{-k_1}(\widehat{s}^a_{s_j})^2$ is the source variance divided by $\rho^{\text{all}}_{s_j}$, plus a nonnegative target-score variance. The uniform second-moment bounds make this limit finite. Balanced folds preserve these site fractions, so

$$(s^{a,*}_{s_j})^2 \ge \frac{e^{-2M} v_- \kappa_G}{\rho^{\text{all}}_{s_j}} > 0. \qquad \square$$

The fixed-active-law construction in Proposition S1 supplies stable sparse targets even when site outcome means differ. We now combine this candidate stability with the valid-candidate training moments to prove selection and oracle equivalence.

S3.5 Proof of the Joint Aggregation Theorem

Work on an outer training sample of size $n = n_{-k_1}$, suppressing $(-k_1)$ until evaluation. All statements hold uniformly over the fixed number of outer folds.

Theorem 1 (restated)**.** *For the fitting procedure and penalty levels in Section 4.1, under Assumptions 1 to 5 and the target-only conditions of Proposition S2, the outer-fold weights satisfy* $\widehat{\eta}^{(-k_1)} \to_p \eta^*$ *and*

$$P\{\widehat{\eta}_{s_j}^{a,(-k_1)} = 0 \text{ for all } a = 0, 1,\ s_j \notin \mathcal{J}^{a,*}\} \to 1.$$

Moreover,

$$\sqrt{N}(\widehat{\tau}_t - \tau_t) \xrightarrow{d} \mathcal{N}(0, V_\tau^*), \qquad V_\tau^* \le L_\tau^*(0).$$

If $\mathcal{J}^{0,*} = \mathcal{J}^{1,*} = \varnothing$*, the aggregate equals the target-only contrast with probability tending to one.*

Proof of Theorem 1. The quadratic representation in Assumption 5 has $\widehat{Q}_n \to_p Q^*$ with $\lambda_{\min}(Q^*) > 0$ and $\widehat{\boldsymbol{l}}_n = O_p(1)$. On an event of probability tending to one, $\lambda_{\min}(\widehat{Q}_n) \ge \kappa$ for some fixed $\kappa > 0$. Comparing the penalized objective at $\widehat{\eta}$ with its value at zero, and using nonnegative penalty coefficients, gives

$$\kappa\|\widehat{\eta}\|_2^2 \le 2\|\widehat{\boldsymbol{l}}_n\|_2\|\widehat{\eta}\|_2, \qquad \|\widehat{\eta}\|_2 = O_p(1).$$

This argument controls the unconstrained optimizer without imposing a compact weight set.

We next examine the penalty on each coordinate. For a valid source arm, Proposition S3 gives a training-mean nuisance replacement error $o_p(n^{-1/2})$. The centered population contributions have finite second moments by Assumption 2, and their expectation equals μ_t^a. Hence $\widehat{d}_{s_j}^a = O_p(n^{-1/2})$ and $\widehat{t}_{s_j}^a = O_p(1)$. Its penalty coefficient $(\widehat{t}_{s_j}^a/c_n - 1)_+$ is therefore zero with probability tending to one.

For a biased arm, $\widehat{d}_{s_j}^a \to_p \bar{d}_{s_j}^a \neq 0$ and $\sqrt{n}\widehat{s}_{s_j}^a \to_p s_{s_j}^{a,*} > 0$. Consequently $\widehat{t}_{s_j}^a = \Theta_p(\sqrt{n})$ and, since $c_n/\sqrt{n} \to 0$, its penalty coefficient diverges. The quadratic gradient at the optimizer is $2\widehat{\boldsymbol{l}}_n + 2\widehat{Q}_n\widehat{\eta} = O_p(1)$. A nonzero biased coordinate would require its gradient component to equal the diverging penalty coefficient in absolute value, contradicting this stochastic bound. There are finitely many source arms, so all biased coordinates are zero with probability tending to one. On the remaining valid coordinates the penalty is zero, and quadratic coefficient convergence with positive curvature gives the unique restricted minimizer $\widehat{\eta} \to_p \eta^*$.

Restore the outer fold index k_1. Form a signed discrepancy vector from held-out valid source-minus-target means, with negative signs in the control-arm coordinates and zeros on biased coordinates. The calibrated source expansion in Lemma 4 and the target-anchor expansion in Proposition S2 give $\widehat{\boldsymbol{D}}_{k_1} = O_p(N^{-1/2})$. The foldwise difference between the fitted-weight and oracle-weight contrasts is therefore bounded by

$$|(\widehat{\eta}^{(-k_1)} - \eta^*)^T \widehat{\boldsymbol{D}}_{k_1}| \leq \|\widehat{\eta}^{(-k_1)} - \eta^*\|_2 \|\widehat{\boldsymbol{D}}_{k_1}\|_2 = o_p(N^{-1/2}).$$

The balanced folds satisfy $n_{r,k_1}/n_r = K_f^{-1} + O(n_r^{-1})$. Their site fractions therefore differ by $O(N^{-1})$, which contributes $o_p(N^{-1/2})$ to the pooled fitted-weight difference. The finite number of folds preserves the preceding product order. Independence between the learned weights and held-out candidate differences is unnecessary for this bound.

At the oracle weights, the candidate expansions yield site-centered population contributions G_r^a as defined in the main text. Pooling folds assigns each observation coefficient $1/n_r$ at site r. Site independence and the $(2+\delta)$ moment bounds give

$$\sqrt{N}(\widehat{\tau}_t - \tau_t) \xrightarrow{d} \mathcal{N}(0, V_\tau^*), \qquad V_\tau^* = \sum_r \frac{\mathrm{Var}_r(G_r^1 - G_r^0)}{\rho_r^{\mathrm{all}}}.$$

The fitted-score covariance argument in the proof of Lemma 4 and the target score consistency in Proposition S2, together with $\widehat{\eta} \to_p \eta^*$, imply empirical L_2 convergence of the outer contributions to these population contributions. Their pooled second moments then give $N\widehat{V}_\tau^{\mathrm{out}} \to_p V_\tau^*$. Finally, zero weights are feasible for the oracle objective, so $V_\tau^* \le L_\tau^*(0)$. If both valid source sets are empty, selection sets every source weight to zero with probability tending to one, leaving the target-only contrast. □

S3.6 Reported Standard Errors

The preceding proof establishes consistency of the leading variance in Eq. (13). The simulation and RHC analyses use its square root as the primary RoCE standard error, with the learned weights plugged into the site-centered outer contributions. Their Wald intervals do not include an additional weight-learning correction. The All of Us figure retains its supplied intervals pending a matching export.

Optional local sensitivity diagnostic. The implementation also retains a local delta-method calculation for estimating the aggregation weights. It is reported separately from the primary standard error. We derive it by perturbing training moments and differentiating the weight optimality equations.

The calculation holds nuisance fits, feature transformations, folds, tuning values, and the selected active set fixed. It describes the reported local sensitivity calculation; inference at selection boundaries or under full refitting requires additional analysis.

Fix outer fold k and write $n_{r,-k}$ for its training count at site r and $n_{-k} = \sum_r n_{r,-k}$. Stack the uncentered training records as $W_{r,i,k} = (B_{r,i,k}, U_{r,i,k}^T)^T$, with $U_{r,i,k} \in \mathbb{R}^{2K}$. At the target, $B_{t,i,k} = \widehat{h}^1_{0,i} - \widehat{h}^0_{0,i}$ and

$$U_{t,i,k} = (\widehat{h}^1_{t,1,i} - \widehat{h}^1_{0,i}, \ldots, \widehat{h}^1_{t,K,i} - \widehat{h}^1_{0,i}, -\widehat{h}^0_{t,1,i} + \widehat{h}^0_{0,i}, \ldots, -\widehat{h}^0_{t,K,i} + \widehat{h}^0_{0,i})^T.$$

At source s_j, $B_{s_j,i,k} = 0$; $U_{s_j,i,k}$ has only two potentially nonzero entries: $\widehat{h}^1_{s_j,j,i}$ in coordinate j and $-\widehat{h}^0_{s_j,j,i}$ in coordinate $K + j$. All fitted functions exclude outer fold k. These are the scores of Eq. (S2), expressed as a target-only contrast plus signed source increments.

Let $\bar{W}_{r,k}$ be the pooled site mean and $\widetilde{W}_{r,i,k}$ the record centered within its inner calibration fold. Define

$$\widehat{C}_{r,k} = \frac{1}{n_{r,-k}} \sum_{i \in \mathcal{I}_{r,-k}} \widetilde{W}_{r,i,k} \widetilde{W}^T_{r,i,k}, \qquad \begin{pmatrix} \widehat{v}_k & \widehat{l}_k^T \\ \widehat{l}_k & \widehat{H}_k \end{pmatrix} = \sum_r \frac{\widehat{C}_{r,k}}{n_{r,-k}}.$$

Thus $\widehat{v}_k + 2\widehat{l}_k^T \eta + \eta^T \widehat{H}_k \eta$ is the unscaled variance objective. In the main theorem's scaled notation, $\widehat{Q}_n = n_{-k} \widehat{H}_k$. The signed discrepancy vector is $\widehat{\delta}_k = \sum_r \bar{U}_{r,k}$. The numerical implementation uses the variance floor $\epsilon_V = 10^{-6}$. Set

$$\widehat{s}_{k,j} = \{\max(\widehat{H}_{k,jj}, \epsilon_V)\}^{1/2}, \quad \widehat{t}_{k,j} = |\widehat{\delta}_{k,j}|/\widehat{s}_{k,j}, \quad p_{k,j} = (\widehat{t}_{k,j}/c - 1)_+.$$

The reported analyses use $c = 2$. The sign convention for control-arm discrepancies does not affect their absolute Wald statistics.

To differentiate the moments, perturb the empirical mass of one training observation i at site r from one to $1 + \varepsilon$, retaining the original sample counts in the variance scaling. Means and within-fold centering use the perturbed masses. A dot denotes differentiation at $\varepsilon = 0$. For the covariance blocks $\widehat{C}^{UU}_{r,k}$ and $\widehat{C}^{UB}_{r,k}$, the moment derivatives satisfy

$$n_{r,-k} \dot{\delta}_k = U_{r,i,k} - \bar{U}_{r,k},$$

$$n^2_{r,-k} \dot{H}_k = \widetilde{U}_{r,i,k} \widetilde{U}^T_{r,i,k} - \widehat{C}^{UU}_{r,k},$$

$$n^2_{r,-k} \dot{l}_k = \widetilde{U}_{r,i,k} \widetilde{B}_{r,i,k} - \widehat{C}^{UB}_{r,k}.$$

Only coordinates with $\widehat{t}_{k,j} > c$ contribute a penalty derivative. Away from the variance-floor kink, the chain rule gives

$$\dot{s}_{k,j} = \frac{I(\widehat{H}_{k,jj} > \epsilon_V)\dot{H}_{k,jj}}{2\widehat{s}_{k,j}}, \qquad \dot{t}_{k,j} = \frac{\operatorname{sign}(\widehat{\delta}_{k,j})\dot{\delta}_{k,j} - \widehat{t}_{k,j}\dot{s}_{k,j}}{\widehat{s}_{k,j}}.$$

Hence $\dot{p}_{k,j} = \dot{t}_{k,j}/c$ in this region, whereas $\dot{p}_{k,j} = 0$ when $\widehat{t}_{k,j} < c$. Below the variance floor, $\dot{s}_{k,j} = 0$.

Let J_k be the selected nonzero coordinates, with their signs held fixed. Numerically, coordinates with $|\widehat{\eta}_{k,j}| > 10^{-12}$ are treated as active, and $|\widehat{t}_{k,j}/c - 1| < 10^{-6}$ flags proximity to a penalty kink. Differentiating the stationarity equations of the soft-penalized objective yields

$$\widehat{H}_{k,J_kJ_k}\dot{\eta}_{k,J_k} = -(\dot{H}_k\widehat{\eta}_k)_{J_k} - \dot{l}_{k,J_k} - (2n_{-k})^{-1}\operatorname{diag}\{\operatorname{sign}(\widehat{\eta}_{k,J_k})\}\dot{p}_{k,J_k}.$$

Inactive coordinates have $\dot{\eta}_{k,J_k^c} = 0$; for an empty active set, the entire derivative is zero. The calculation requires nonsingular active curvature and does not differentiate a change in the selected coordinates.

The effect of each weight perturbation on the final estimate depends on the actual outer-fold increments. For $a \in \{0, 1\}$ define

$$v_{k,j}^a = \frac{1}{n_t}\sum_{i\in\mathcal{I}_{t,k}}(\widehat{h}_{t,j,i}^a - \widehat{h}_{0,i}^a) + \frac{1}{n_{s_j}}\sum_{i\in\mathcal{I}_{s_j,k}}\widehat{h}_{s_j,j,i}^a.$$

Stack these as $v_k = ((v_k^1)^T, -(v_k^0)^T)^T$. The functions are evaluated on outer fold k, and the denominators are the full site counts, as in Eq. (12). Accumulate an observation's indirect contribution over every outer training set containing it:

$$b_{r,i} = \sum_{k:\, i\in\mathcal{I}_{r,-k}} v_k^T\dot{\eta}_{k;r,i}.$$

Let $T_{r,i} = \widehat{G}_{r,i}^1 - \widehat{G}_{r,i}^0$ be its held-out contrast contribution and set $a_{r,i} = (T_{r,i} - \bar{T}_r)/n_r$. The leading variance and the separate local diagnostic variance are

$$\widehat{V}_\tau^{\text{out}} = \sum_r\sum_{i\in\mathcal{D}_r} a_{r,i}^2, \qquad \widehat{V}_{\text{diag}} = \sum_r\sum_{i\in\mathcal{D}_r}(a_{r,i} + b_{r,i})^2. \tag{S12}$$

Using empirical second moments with divisor n_r, $\sum_{r,i} a_{r,i}^2$ equals the leading fixed-weight variance. The diagnostic adds both $\sum_{r,i} b_{r,i}^2$ and $2\sum_{r,i} a_{r,i} b_{r,i}$, and hence need not increase the standard error. Primary intervals use $\widehat{V}_\tau^{\text{out}}$, not $\widehat{V}_{\text{diag}}$. Accumulating by original observation before squaring retains dependence across overlapping training folds. This local calculation neither refits nuisance models nor accounts for discrete selection changes; it does not resolve the weak-bias coverage limitation discussed in the main paper.

The first- and second-moment messages in figure 1 suffice for point estimation and the leading variance. The optional diagnostic above uses observation-level score records in the numerical implementation. Computing it through federated summaries would also require the higher-order and cross-fold products entering Eq. (S12); those summaries are beyond the message list in that figure.

S4 Simulation Models and Comparators

This section supplies the model and comparator details for Section 5. The simulations vary nuisance misspecification and source incompatibility separately. Every fitted model uses an intercept and the $p = 200$ covariate main effects. The generated interaction X_1X_2 enters through $z_\kappa = X_1(1 + \kappa X_2)/\sqrt{1 + \kappa^2}$ whenever the corresponding κ is nonzero. The outcome interaction is controlled by κ_Y, while κ_A controls interactions in treatment assignment and population shift. The statuses in table S1 concern $\rho = 0$; for $\rho > 0$, source 1's treated outcome additionally violates transportability. A correct outcome functional form at a site does not remove that violation.

Table S1: Generated and fitted nuisance models in the three simulation scenarios at $\rho = 0$. All fitted models omit X_1X_2.

Scenario	κ_Y	κ_A	Outcome model	Treatment and joint-weight models
1	0	0	Correct	Correct
2	0.75	0	Misspecified: omitted interaction	Correct
3	0	0.75	Correct	Misspecified: omitted interaction

The benchmark names refer to the implementations compared in this manuscript. They combine familiar AIPW and federated aggregation ideas (Bang and Robins, 2005; Xiong et al., 2023), with the following distinctions:

Calibrated target-only.

Calibrated target outcome and propensity models produce an AIPW treatment contrast using target observations alone. This is the reference for assessing the gain from source borrowing.

Sample-size aggregation (SS).

Local arm-mean estimates are combined according to site sample sizes. The procedure does not transport the source arm means to the target covariate distribution.

Inverse-variance aggregation (IVW).

Local arm-mean estimates are combined using inverse-variance weights, including a nonnegative between-site variance adjustment. As with SS, the source means are not transported to the target.

Federated-DR.

Density-ratio-weighted source AIPW estimates are combined with the target estimate

by inverse variance. Treatment and population adjustments use separate nuisance components, and there is no RoCE pivotal penalty.

Pooled-DR.

Density-ratio-weighted AIPW uses common nuisance fits obtained from pooled records. This comparator provides a pooled-data reference and does not satisfy the same record-sharing restriction as the federated procedure.

RoCE combines paired calibration with arm-specific pivotal penalties and joint TATE variance minimization. The comparisons evaluate these complete procedures; differences cannot be attributed to a single component in the absence of a corresponding ablation. All methods use the same generated dataset in each replicate. RMSE measures the combined effect of bias and variability, while coverage assesses the reported intervals. Pointwise Monte Carlo intervals describe simulation uncertainty and are not simultaneous bands across settings.

S5 All of Us Cohort Definition and Site Composition

This section supplements Section 6 with the cohort rules and site composition used in the All of Us analysis.

The analysis defines an initial-exposure contrast in the evaluable one-year cohort. Index is the earlier first recorded exposure to GLP-1 or DPP-4; simultaneous first exposures are assigned to DPP-4. Participants must have a type 2 diabetes diagnosis by index and no prior diagnosis in the composite outcome set. Subsequent exposure to the other class does not change the initial assignment. The design imposes no separate washout or background-therapy restriction.

[To be completed: list the diseases or diagnostic categories in the composite dermatologic outcome, the corresponding diagnosis codes or concept identifiers, and the rule used to combine them.]

The composite outcome included any of the nine dermatologic conditions listed in table S2. These conditions were identified using their SNOMED CT concepts, represented by the OMOP concept IDs in table S2, together with all descendant concepts in the All of Us condition-concept hierarchy.

Table S2: Starting concepts for the composite dermatologic outcome. Each concept and its descendants are included.

Disease or diagnostic category	OMOP concept ID
Hidradenitis suppurativa	4241223
Inflammatory dermatosis	45766714
Psoriasis	140168
Cutaneous lupus erythematosus	4324123
Eczema	133835
Urticaria	139900
Bullous pemphigoid	4298692
Pemphigus vulgaris	4170723
Alopecia areata	141933

For each participant, the earliest recorded diagnosis matching any of these concepts or their descendants defined composite-outcome onset, with participants whose onset preceded index excluded from the cohort. Among those retained, a diagnosis of any component condition between index and one calendar year later, including both endpoints, constituted

a composite event. For participants without such an event, a negative outcome was assigned only when the last condition-occurrence date was on or after the one-year endpoint.

Of 15,339 baseline participants with geographic information, 2,074 have unclassifiable outcomes and are excluded, leaving 13,265 participants. No censoring weights are used. This rule defines the evaluable population; it does not establish continuous observation throughout the year.

Covariates comprise age, ethnicity, gender, race, sex at birth, and diagnosis-derived PheCode counts for the year before index and earlier history. Diabetes and outcome-defining concepts are excluded. The 5% prevalence screen precedes splitting and yields 154 count features among 181 model features. Scaling uses outer training samples. Northeast is the largest evaluable geographic group and is designated as the target across all methods. Table S3 describes the four groups; the marginal event counts are descriptive and are not tests of conditional transportability.

Table S3: All of Us evaluable one-year cohort by geographic site. Treatment-arm entries give participants and recorded composite-outcome events in parentheses. These unadjusted counts are distinct from the adjusted risk differences in figure 4.

Site	Region	Total n	GLP-1 n (events)	DPP-4 n (events)
t	Northeast	5,040	1,082 (90)	3,958 (277)
s_1	Midwest	3,433	874 (84)	2,559 (205)
s_2	South	2,429	567 (42)	1,862 (136)
s_3	West	2,363	588 (39)	1,775 (118)

S6 Right Heart Catheterization Analysis

This second application complements the All of Us analysis in Section 6 using an insurance-defined target population.

We analyze 30-day mortality after right heart catheterization (RHC) in 5,039 SUPPORT patients (Connors et al., 1996; Hirano and Imbens, 2001). Private insurance defines the target group ($n_t = 1{,}698$); Medicare, private insurance with Medicare, and Medicaid define three candidate sources. Uninsured patients and those with both Medicare and Medicaid are excluded. All methods use the same cohort and 61 covariates plus an intercept. RoCE and calibrated target-only use two-layer ten-fold cross-fitting; median/mode imputation and standardization are fitted on each pooled outer training sample. After this pooled preprocessing, the implemented joint-TATE analysis uses one batched source-model exchange, a fixed cutoff of 2, and numerical predictor radii of 3 at sources and 5 at the target. Pooled medians and modes may require additional communication in a genuinely distributed implementation. SS and IVW use site-specific training-fold preprocessing, while the two DR benchmarks use full-sample fitting. These application-specific radii differ from the common-radius sufficient regime in the main paper.

Figure S1 shows six mortality risk-difference estimates. RoCE gives 3.02 percentage points (95% CI $[-1.13, 7.16]$), versus 2.40 points (95% CI $[-2.42, 7.22]$) for calibrated target-only estimation. Their standard errors are 2.12 and 2.46 points, respectively, for an estimated relative efficiency of 1.35. The RoCE interval uses the leading variance; reaggregation of the saved fit leaves the point estimate and source weights unchanged and requires no nuisance-model refit. The insurance groups emulate sites within one study, not independently recruited hospitals. Target-anchor weights averaged across outer folds are approximately 0.671 for the treated mean and 0.729 for the control mean (table S4).

Table S4: RHC insurance groups and RoCE source weights averaged over ten outer folds. RHC and mortality are observed proportions.

Insurance	n	RHC	Mortality	$\widehat{\eta}^1_{s_j}$	$\widehat{\eta}^0_{s_j}$
Private (target)	1698	0.431	0.309	–	–
Medicare	1458	0.350	0.362	0.092	0.111
Private & Medicare	1236	0.396	0.367	0.087	0.067
Medicaid	647	0.298	0.269	0.149	0.093

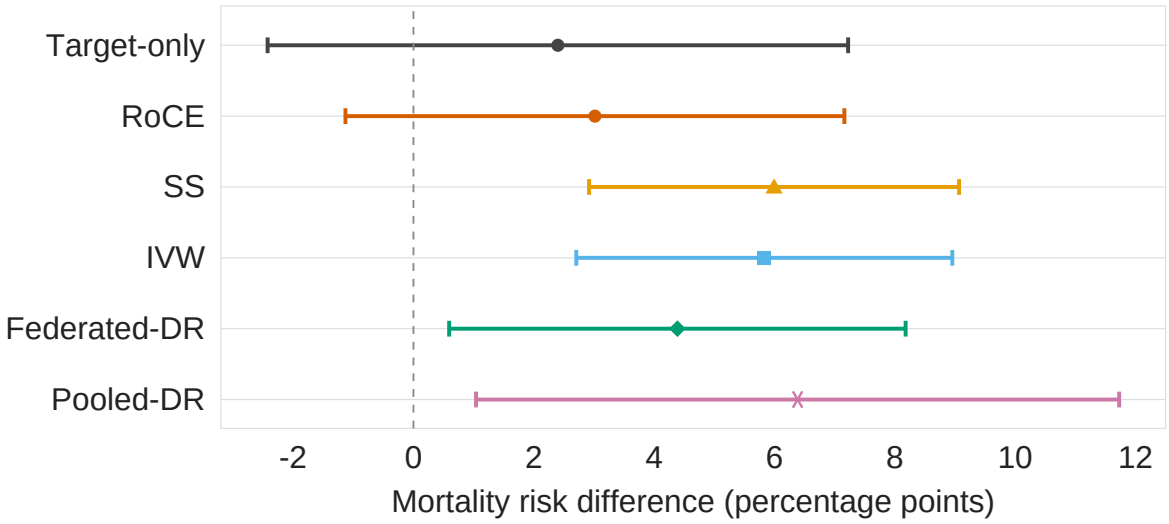


Figure S1: Estimated target-population effects of RHC on 30-day mortality with 95% confidence intervals. RoCE uses the leading site-centered variance; baseline intervals retain their original inference procedures. Positive risk differences indicate higher mortality under RHC; the dashed line marks zero.

S7 Alternative Validation and Armwise Aggregation

We consider two alternatives to the primary procedure: independent inner validation for weight learning, and armwise variance minimization on a bounded weight set. Unless both arms are displayed, fix a, write $\mu = \mu_t^a$, and suppress the arm index.

S7.1 Independent Inner Validation

Partition each site into $K_f \geq 4$ balanced folds $\mathcal{I}_{r,h}$. Reserve k for outer evaluation and $\ell \neq k$ for inner validation. The remaining index h denotes a calibration fold. For an excluded set B, define $\mathcal{T}_r(B) = \bigcup_{h \notin B} \mathcal{I}_{r,h}$ and $n_r(B) = |\mathcal{T}_r(B)|$. Outer refits use $B = \{k\}$; inner validation fits use $B = \{k, \ell\}$. Initial fits for block $h \notin B$ use only $\mathcal{T}_r(B \cup \{h\})$. The same exclusion applies to target summaries, preprocessing, and tuning.

Write $L_{\gamma,h}$ and $L_{\alpha,h}$ for the unpenalized empirical loss contributions in Eq. (4) and Eq. (5), evaluated on calibration fold h. The complete calibration fit is

$$\begin{aligned} \widehat{\gamma}^{(-B)} &\in \arg\min_{\gamma} \left\{ \frac{1}{K_f - |B|} \sum_{h \notin B} L_{\gamma,h}(\gamma; \widehat{\alpha}_{\text{init}}^{(-B,-h)}) + \lambda_\gamma(B) \|\gamma\|_{1,-0} \right\}, \\ \widehat{\alpha}^{(-B)} &\in \arg\min_{\alpha} \left\{ \frac{1}{K_f - |B|} \sum_{h \notin B} L_{\alpha,h}(\alpha; \widehat{\gamma}_{\text{init}}^{(-B,-h)}) + \lambda_\alpha(B) \|\alpha\|_{1,-0} \right\}. \end{aligned} \tag{S13}$$

Each site average uses all observations in its calibration fold; $I(A = a)$ selects the treatment arm without changing the denominator. For unequal folds, weight each site's contribution by its own fraction $|\mathcal{I}_{r,h}|/n_r(B)$ before combining the target and source terms. Add each penalty once. The target derivative moment uses target fold h and the initial outcome fit excluding $B \cup \{h\}$; the feature mean for an initial joint-weight fit uses only the initial-training target sample.

Apply the same construction to the target losses in Eq. (S1), stated with their primary fold updates in Section S1.2. Initial target outcome fits use the corresponding treatment arm;

initial logistic propensity fits use all target training observations with response $I(A = a)$. Calibrated target propensities are $\sigma_M(\phi^T \widehat{\beta}_{0,a})$.

For inner validation fold ℓ, evaluate only the calibrated models with $B = \{k, \ell\}$. Denote the resulting uncentered target-only contribution by $h_{0,i}^{(-k,-\ell)}$, target prediction for source j by $h_{t,j,i}^{(-k,-\ell)}$, and source residual contribution by $h_{s_j,j,i}^{(-k,-\ell)}$. Put $n_{r,-k} = n_r(\{k\})$ and $n_{-k} = \sum_r n_{r,-k}$. Pool the validation contributions as

$$\begin{aligned} \widehat{\mu}_{0,\text{in}}^{(-k)} &= \frac{1}{n_{t,-k}} \sum_{\ell \neq k} \sum_{i \in \mathcal{I}_{t,\ell}} h_{0,i}^{(-k,-\ell)}, \\ \widehat{\mu}_{j,\text{in}}^{(-k)} &= \frac{1}{n_{t,-k}} \sum_{\ell \neq k} \sum_{i \in \mathcal{I}_{t,\ell}} h_{t,j,i}^{(-k,-\ell)} + \frac{1}{n_{s_j,-k}} \sum_{\ell \neq k} \sum_{i \in \mathcal{I}_{s_j,\ell}} h_{s_j,j,i}^{(-k,-\ell)}. \end{aligned} \tag{S14}$$

For each site, center its pooled validation contributions by their pooled site mean and compute their covariance matrix. Divide the target covariance by $n_{t,-k}$ and the source residual variances by $n_{s_j,-k}$ to obtain $\widehat{\Sigma}_{\text{in}}^{(-k)}$. Its off-diagonal entries retain the shared target observations. Form the discrepancies from Eq. (S14), use n_{-k} in the variance objective and $c_{n_{-k}}$ in the Wald penalty, and solve for the aggregation weights after pooling all validation contributions. For the joint TATE objective, stack both arms and use the covariance construction in Section S1.3 on these independent validation records; the alternative armwise objective in Section S7.4 instead solves separately for each arm.

Outer refits use $B = \{k\}$ in Eq. (S13). Evaluate them on fold k and apply $\widehat{\eta}^{a,(-k)}$. For balanced folds, average the outer estimates with weights $1/K_f$. Estimate final variance from the pooled sitewise outer contributions, including the two-arm covariance. Do not treat outer estimates as independent: their training samples overlap. The proof in Lemma S5 replaces the fitted contributions by common population contributions before applying the central limit theorem.

S7.2 One-Round and Two-Round Communication

In the one-round version, the target prepares messages for every $B \in \{\{k\}, \{k, \ell\} : k \neq \ell\}$ and $h \notin B$. Each message contains the initial target outcome coefficient trained outside $B \cup \{h\}$, the feature mean on that initial-training sample, and the clipped derivative moment on target calibration fold h. The source uses these quantities in Eq. (S13). It returns calibrated outcome coefficients indexed by B, plus residual means, second moments, cross-arm products, and counts on each requested validation or evaluation fold. All tasks are fixed in advance and batched into one broadcast and one return. Target predictions and their covariances are computed locally after the coefficients arrive. Joint-weight coefficients remain at the source.

In the two-round version, the first broadcast supplies all initial-training target feature means and task labels, and the first return supplies source-specific initial outcome coefficients. The second broadcast supplies the corresponding target calibration-fold derivative moments; the second return supplies the calibrated outcome coefficients and residual summaries. The dependencies are the same for inner validation and outer refits. Aggregation weights are computed locally at the target and do not require a further source refit. These round counts require a predeclared fitting protocol whose tuning and retries do not request new data-dependent target summaries.

For treatment-effect inference, each source must return the covariance of its two arm residual scores in addition to their marginal variances. If $\widehat{\xi}^{a}_{t,s_j,i}$ is the centered residual score for arm a, the required source contribution is the empirical variance of $\widehat{\eta}^{1}_{s_j}\widehat{\xi}^{1}_{t,s_j,i} - \widehat{\eta}^{0}_{s_j}\widehat{\xi}^{0}_{t,s_j,i}$. Although the uncentered arm residuals have disjoint treatment indicators, their centered covariance need not be zero. The target likewise uses the difference of its two aggregated target scores.

S7.3 Connecting Inner Validation to Outer Evaluation

Lemma S5 (Common population limits and validation moments)**.** *Use the complete calibration fits and pooled validation moments in Eqs. (S13) and (S14), with fixed $K_f \geq 4$, balanced sitewise folds, and no evaluation observations entering fitting or tuning. Use the same prespecified feature map, clipping radius, losses, and sitewise normalization for every excluded set B.*

For the target, assume the conditions of Proposition S2. For every unbiased source-assisted estimator, assume transportability and the correct-model, design, tail, and sparsity conditions of Lemma 4. Choose penalties of the stated order using each fit's own training counts and require the stated remainder conditions. For every other source-assisted estimator, assume Proposition S4 and a fixed nonzero limiting bias. Then the following statements hold simultaneously over the finitely many inner validation fits and outer refits.

(a) The calibrated coefficient limits are independent of B. The rates of Lemma S3 hold with the corresponding training sample sizes. Initial and calibrated limits are equated only for a correctly specified model.

(b) With site-centered population contributions $\varphi_t, \zeta_{t,s_j}, \xi_{t,s_j}$ as defined above, inner validation gives

$$\widehat{\mu}_{0,\mathrm{in}}^{(-k)} - \mu = \frac{1}{n_{t,-k}} \sum_{i \in \mathcal{T}_t(\{k\})} \varphi_{t,i} + o_p(n_{-k}^{-1/2}),$$
$$\widehat{\mu}_{j,\mathrm{in}}^{(-k)} - \mu = \frac{1}{n_{t,-k}} \sum_{i \in \mathcal{T}_t(\{k\})} \zeta_{t,s_j,i} + \frac{1}{n_{s_j,-k}} \sum_{i \in \mathcal{T}_{s_j}(\{k\})} \xi_{t,s_j,i} + o_p(n_{-k}^{-1/2})$$

for every unbiased source-assisted estimator. Each biased estimator instead converges to its own fixed limit.

(c) Inner and outer empirical sitewise score covariances converge to the same population covariances. On the unbiased coordinates, $n_{-k}\widehat{\Sigma}_{\mathrm{in}}^{(-k)}$ converges to the covariance used

by the full-sample oracle objective. The same statement holds jointly for the two arms. For biased coordinates, only convergence of the score-based covariance is asserted.

(d) Each outer-fold estimator has the analogous expansion on its evaluation fold. Averaging these expansions over k gives the full sitewise expansion, with each observation contributing once.

Proof of Lemma S5. Fold labels are fixed independently of the observations. Thus every sitewise empirical mean in an initial or calibration loss estimates the same site expectation, regardless of B. Its normalization is by that site's own sample size. In a correct-model branch, the correct update has the same true coefficient for every bounded clipped opposite initial fit. The other calibrated update converges to the population minimizer obtained using that true initial coefficient. For a biased candidate, both initial population minimizers are stable by Proposition S4, and their calibrated population minimizers are consequently common across B. This identifies common limiting population targets without equating misspecified initial and calibrated coefficients. Unequal sitewise fold fractions can leave the $O(N^{-1})$ conditional-gradient term in Eq. (S7); the revised rate bound absorbs it for each excluded set B.

Every initial or calibration block contains a fixed positive fraction of its site's observations. There are finitely many triples (k, ℓ, h), sources, and arms. Apply the gradient concentration, Gram concentration, and localization arguments of Lemma S3 to each block with its own initial-training sample. Allocate the failure probability over these finitely many applications and use a union bound. The penalty order and sparsity requirements are unchanged because all training sizes are proportional to the site sizes. This proves (a); independence of overlapping training sets is not needed.

For (b), condition separately on the training data for each fit excluding k, ℓ. Those data determine the calibrated coefficients, while fold ℓ remains independent of them. Write its

fitted contribution as its population-limit contribution plus their difference. The centered empirical average of this difference is $o_p(n_{-k}^{-1/2})$ by the conditional variance bound and the L_2 consistency in Eq. (S5). For an unbiased candidate, calibration bounds the uncentered population difference by $O_p(s_{\text{eff}}\lambda^2) = o_p(n_{-k}^{-1/2})$. The target uses Proposition S2. Sum the resulting bounds over the fixed number of validation folds using Eq. (S14). For biased candidates, Proposition S4 supplies consistency for their common limits; no quadratic bias bound is required.

For (c), condition on each validation fit's training data to bound the empirical squared difference between fitted and limiting contributions. Conditional Markov's inequality and L_2 consistency make this difference $o_p(1)$. Cauchy–Schwarz gives the same result for cross-products, including shared-target and cross-arm products. Laws of large numbers then apply to the common population contributions pooled within each site. Balanced folds preserve the site fractions asymptotically: $n_{r,-k}/n_{-k} = n_r/N + O(N^{-1})$. Centering and dividing by the site counts therefore gives the asserted covariance limit. Biased estimators need not have these contributions as influence functions; their convergent covariance components are used only as discrepancy scales and as a finite extension of the quadratic objective.

For (d), repeat the preceding conditional argument for $B = \{k\}$ on fold k. Each remainder is $o_p(N^{-1/2})$. For exactly equal folds, averaging assigns each observation coefficient $1/n_r$ at site r. With balanced integer fold sizes, replacing the fold weights by $1/K_f$ changes the expansion by $O_p(N^{-1})$; the same rounding bound applies when pooling inner validation folds. Using the actual site counts gives the sitewise expansion directly. The central limit theorem applies to these sitewise sums, not to independent outer-fold estimates. At no step do we condition on all overlapping fitted models and claim that all their evaluation samples remain independent. □

This lemma verifies the validation inputs needed by the aggregation theorem. It does not derive correct-model assumptions from marginal agreement: each unbiased source-assisted estimator is required to satisfy the source inference conditions. A nontransportable source whose marginal bias happens to cancel, but whose fitted model has a nonzero first derivative, is not covered by this sufficient regime.

S7.4 Armwise Covariance and Objective

For armwise aggregation, index the target-only estimator by $j = 0$ and sources by $j = 1, \ldots, K$, with $\widehat{\mu}_0 = \widehat{\mu}_t^a$, $\widehat{\mu}_j = \widehat{\mu}_{s_j}^a$, and $\eta_j = \eta_{s_j}^a$. Let μ_j^* be each candidate's probability limit, with $\mu_0^* = \mu$. Let $Z_t = (\varphi_t, \zeta_{t,s_1}, \ldots, \zeta_{t,s_K})^T$ contain the site-centered target scores. With $B = \mathrm{Cov}_t(Z_t)$, $v_j = \mathrm{Var}_{s_j}(\xi_{t,s_j})$, and $b(\eta) = (1 - \mathbf{1}^T\eta, \eta^T)^T$, the limiting quadratic variance is

$$L^*(\eta) = \frac{b(\eta)^T B b(\eta)}{\rho_t^{\mathrm{all}}} + \sum_j \frac{\eta_j^2 v_j}{\rho_{s_j}^{\mathrm{all}}}.$$

On the unbiased candidate subspace this is the asymptotic variance. For a biased candidate, its limiting fitted-score covariance may merely define a finite quadratic extension used in screening; it need not be the covariance of a valid linear expansion for that candidate.

Lemma S6 (Shared-target covariance)**.** *If all estimated score covariance components converge to finite limits, and on the unbiased subspace these are the covariances in the candidate linear expansions, the empirical version $\widehat{L}$ converges uniformly to L^* on every fixed compact weight set. Its gradient is uniformly $O_p(1)$ there.*

Proof of Lemma S6. The target sample is the only sample appearing in more than one candidate estimator, so it supplies all off-diagonal covariances. Independent source samples supply $\sum_j \eta_j^2 v_j / n_{s_j}$. With fixed K, consistency of finitely many variance and covariance components implies coefficientwise convergence of the quadratic polynomial and hence uniform convergence of it and its gradient on a compact set. The estimated quadratic

is positive semidefinite when assembled from one target empirical covariance matrix and nonnegative source variances. □

Use the main-text discrepancy and standard-error conventions, suppressing the arm index: $\widehat{d}_j = \widehat{\mu}_j - \widehat{\mu}_0$ and $\widehat{s}_j = \widehat{s}^a_{s_j}$. The variance $\widehat{s}_j^2$ includes shared-target covariance. Its scaled version $N\widehat{s}_j^2$ consistently estimates the limiting discrepancy variance for unbiased candidates; for biased candidates only a positive finite limit is required. On an outer training sample, use its size n in place of N and set $c_n = \sqrt{\log(n \vee 3)}$. Define $\widehat{t}_j = |\widehat{d}_j|/\widehat{s}_j$ and $a_{j,N} = (\widehat{t}_j/c_n - 1)_+$. The adaptive criterion is

$$\widehat{\eta} \in \arg\min_{\eta \in \mathcal{B}_\eta} \left\{ \widehat{L}(\eta) + \sum_j a_{j,N} |\eta_j| \right\}.$$

The bounded ball $\mathcal{B}_\eta = \{\eta : \|\eta\|_2 \le C_\eta\}$ contains zero and is closed under setting any coordinate to zero. These two properties will be used in the selection argument.

S7.5 Unbiased Candidates and Separation

The assumption below applies to this bounded, armwise criterion. It is distinct from the conditions for the primary joint procedure in Assumption 5. Let $\mathcal{J}^* = \{j : \mu_j^* = \mu\}$ and $b_{j,N} = |\mu_j^* - \mu|$. This is the unbiased candidate set, which need not coincide with a set of conditionally transportable populations. Its definition permits $\mathcal{J}^* = \varnothing$. Define η^* as the minimizer of L^* over the oracle subspace $\{\eta \in \mathcal{B}_\eta : \eta_j = 0 \text{ for } j \notin \mathcal{J}^*\}$.

Assumption S5 (Conditions for bounded armwise aggregation)**.** (a) The target-only estimator is asymptotically linear. Each biased candidate satisfies $\widehat{\mu}_j - \mu_j^* = o_p(1)$ for a fixed limit $\mu_j^* \neq \mu$, including on all training samples used to construct weights. Each unbiased candidate has the source-assisted linear expansion. All estimated covariance components converge, and $\sqrt{N}\widehat{s}_j \to_p s_j \in (0, \infty)$.

(b) If $\mathcal{J}^* \neq \varnothing$, the oracle quadratic is strictly convex on its subspace and its unique minimizer is interior to the ball. If $\mathcal{J}^* = \varnothing$, the oracle subspace is $\{0\}$ and $\eta^* = 0$ without a curvature requirement.

(c) If there are biased candidates, put $b_{0,N} = \min_{j \notin \mathcal{J}^*} b_{j,N}$. Require $\inf_N b_{0,N} > 0$, $c_n \to \infty$, and $c_n/\sqrt{n} \to 0$ on each outer training sample. If all candidates are unbiased, only $c_n \to \infty$ is needed.

(d) The number of sites and folds is fixed, all site fractions are positive, and the stated moment bounds and positive limiting variances hold for the final oracle aggregate.

The requirement on biased candidates is only consistency and a positive finite limiting discrepancy scale. Proposition S4 gives a sufficient route from stable sparse initial fits to these properties for the specified GLM class. Orthogonality or root-N inference for those candidates is unnecessary. For a fixed nonzero bias, consistency implies $|\widehat{d}_j|/b_{j,N} \to_p 1$, whereas the unbiased-candidate expansions imply $\sqrt{N}\widehat{d}_j = O_p(1)$. Consequently,

$$\widehat{t}_j = O_p(1) \quad (j \in \mathcal{J}^*), \qquad \widehat{t}_j = \Theta_p(\sqrt{N} b_{j,N}) \quad (j \notin \mathcal{J}^*).$$

The proposed growing cutoff satisfies the threshold window for these fixed biases. For a drifting bias, a separate sufficient extension requires candidate errors $o_p(b_{0,N})$ and $c_n/(\sqrt{n} b_{0,N}) \to 0$, in addition to the other uniform conditions; consistency alone is insufficient in that regime. The retained fixed cutoff 2 lies outside the theorem's growing-cutoff regime.

S7.6 Oracle Aggregation Result

Theorem S1 (Adaptive aggregation limit)**.** *Under Assumption S5 and the target and source expansion conditions above,* $\|\widehat{\eta} - \eta^*\|_2 = o_p(1)$ *and*

$$P\{\widehat{\eta}_j = 0 \text{ for all } j \notin \mathcal{J}^*\} \longrightarrow 1.$$

The outer-cross-fitted aggregate has the same first-order distribution as the oracle:

$$\sqrt{N}(\widehat{\mu}_{\text{agg}} - \mu) \xrightarrow{d} \mathcal{N}\{0, L^*(\eta^*)\}, \qquad \widehat{L}(\widehat{\eta}) \to_p L^*(\eta^*).$$

If no candidate is unbiased, selection returns the target-only estimator with probability tending to one.

S7.7 Selection and Oracle Equivalence

Proof of Theorem S1. For an unbiased candidate, $\widehat{t}_j/c_n \to_p 0$, so its penalty is exactly zero with probability tending to one. For a biased candidate, $a_{j,N} \to_p \infty$ by separation. Fixed K makes these statements simultaneous over all coordinates.

Write $G_N = \sup_{\eta \in \mathcal{B}_\eta} \|\nabla \widehat{L}(\eta)\|_\infty = O_p(1)$. For any η with $\eta_j \neq 0$, let $\eta^{(j=0)}$ be obtained by setting that coordinate to zero. It remains in the ball, and the mean-value inequality gives

$$\widehat{L}(\eta) - \widehat{L}(\eta^{(j=0)}) \geq -G_N|\eta_j|.$$

All other penalty terms are unchanged. Thus the full objective at η exceeds its value at $\eta^{(j=0)}$ whenever $a_{j,N} > G_N$. This event holds for every biased coordinate with probability tending to one, so every minimizer excludes them. This argument also covers $\mathcal{J}^* = \varnothing$ directly.

On the oracle subspace, all remaining penalties vanish with probability tending to one. Uniform convergence of $\widehat{L}$ and uniqueness of its population minimizer imply $\widehat{\eta} \to_p \eta^*$ by the

compact argmin argument. The same reasoning applies to each outer-training fit; fixed fold count makes the conclusion simultaneous.

For outer fold k, the difference between the fitted-weight and oracle-weight estimates is

$$\sum_{j \in \mathcal{J}^*} (\widehat{\eta}_j^{(-k)} - \eta_j^*)(\widehat{\mu}_{j,k} - \widehat{\mu}_{0,k}) = o_p(N^{-1/2}),$$

because each coefficient difference is $o_p(1)$ and each unbiased candidate difference is $O_p(N^{-1/2})$. Averaging over finitely many folds gives oracle equivalence. Replacing the candidate estimates by their linear expansions yields a sum of independent, site-centered oracle scores. The sitewise central limit theorem proves the normal limit. Finally, uniform covariance consistency and $\widehat{\eta} \to_p \eta^*$ imply $\widehat{L}(\widehat{\eta}) \to_p L^*(\eta^*)$, and Slutsky's theorem gives studentized inference. □

The oracle arm-mean variance is bounded above by $L^*(0)$ because the target-only weight vector is feasible. A strict gain requires a nonzero source direction that lowers the quadratic. No assumption that at least one source exists or receives positive weight is required for validity. Separate minimization in the two arms does not minimize the variance of their difference; its covariance terms must be retained as described in the main text.

S7.8 Squared Discrepancies and Local Bias

The raw-discrepancy version of the FACE criterion (Han et al., 2025) replaces $a_{j,N}$ by $\lambda_{\mathrm{sq},N}\widehat{d}_j^2$. On unbiased coordinates its penalty is $O_p(\lambda_{\mathrm{sq},N}/N)$; on separated biased coordinates it has order $\lambda_{\mathrm{sq},N}b_{j,N}^2$. Sufficient conditions for the same compact-set selection and consistency argument are

$$\lambda_{\mathrm{sq},N}/N \to 0, \qquad \lambda_{\mathrm{sq},N}b_{0,N}^2 \to \infty.$$

The more restrictive choice $\lambda_{\mathrm{sq},N} = o(\sqrt{N})$ also makes the unbiased-coordinate penalty $o_p(N^{-1/2})$. That stronger restriction is not needed merely for consistency of weights in

the cross-fitted oracle-equivalence proof above. The Wald form is used here because it places discrepancies from differently variable sources on a comparable scale, not because an intrinsically different detection boundary follows without specifying the tuning regime.

If $b_{j,N} = O(N^{-1/2})$, the biased Wald statistic need not diverge. Exact selection is then not guaranteed. For a fixed weight vector the first-order mean shift is $\sqrt{N}\sum_j \eta_j(\mu_j^* - \mu)$. Biases of smaller order are negligible, but biases on the exact root-N scale may shift the limit. If data-adaptive weights fail to converge to a deterministic limit, even this fixed-weight normal approximation does not automatically justify inference. Addressing such local alternatives requires an additional analysis.